HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
School of Communication and Information Technology

Final Report
Escrow Payment System (IPS)

Course: System Analysis and Design – IT3120E
Class Code: 175881
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Group Members:

Group 1

Chapter 1

Introduction

1.1 Problem Definition

Today, online transactions are part of almost every digital service – from e-commerce and freelancing platforms to cross-border trade and marketplace payments. However, many transactions still face a fundamental problem: the lack of trust between buyers and sellers who do not know each other. In a typical online transaction, the buyer must pay before receiving the goods or services, while the seller must deliver before receiving payment. This creates a high risk of fraud, non-delivery, late delivery, or payment reversal.

In addition, traditional payment systems are designed to move money quickly, but they do not support conditional release. Once a payment is authorized, funds are transferred immediately to the seller, leaving the buyer unprotected if the goods are not delivered as agreed. Similarly, sellers may be exposed to non-payment risk or chargeback fraud. When disputes arise, resolving them often requires manual intervention, chargebacks, or legal action, which is slow, expensive, and damaging to trust.

These challenges are even more serious for high-value and cross-border transactions, where currency conversion, different regulations, and extra transaction steps increase complexity and cost. In markets like Vietnam and Southeast Asia, existing escrow solutions are often limited to large institutions, law firms, or specific marketplaces, making them inaccessible to small and medium-sized enterprises (SMEs), freelancers, and individual buyers.

To solve these problems, a more efficient and trustworthy payment structure is needed – one that can hold funds securely, release them only when predefined conditions are met, and resolve disputes fairly. The proposed **Escrow Payment System (EPS)** is designed to meet this need. It acts as a trusted third party between the payer and the receiver, helping to verify, hold, and release payments in a fast, secure, and transparent way.

1.2 Proposed Solution

The Escrow Payment System offers a modern and flexible solution for managing conditional payments. Instead of transferring funds directly from the buyer to the seller, it creates a single platform that holds the funds in a secure escrow account and automates the escrow lifecycle. The goal is to make high-value and cross-border transactions safer, more reliable, and easier to track.

Key features of the proposed system include:

- **Conditional Release:** A rules engine that releases funds only when predefined conditions are met, such as buyer confirmation, logistics delivery verification, or milestone approval.
- **Dispute Resolution:** A structured workflow for evidence submission, mediation, and arbitration to resolve conflicts fairly without litigation.
- **Centralized Escrow Management:** A single dashboard to create, fund, and monitor transactions, with the ability to handle complex flows involving multiple parties (buyers, sellers, and brokers).
- **Interoperability & Verification:** Seamless integration with banks, e-wallets, KYC/AML providers, and third-party logistics via APIs to automatically confirm identity and delivery.
- **Cross-Border Support:** Multi-currency handling, automated currency conversion, and compliance with regional financial regulations.
- **Enhanced Security:** Strong encryption, tokenization, multi-factor authentication, and end-to-end traceability for every transaction.
- **Analytics Dashboard:** Tools to view escrow volume, dispute rates, and release times to help administrators monitor system health.

By holding funds securely, releasing them conditionally, and providing a fair dispute resolution mechanism, the Escrow Payment System helps organizations and users reduce fraud, build trust, and gain confidence in their online financial operations. their online financial operations.

Chapter 2

Planning

2.1 Motivation

With the rise of online transactions, users expect fast, secure, and reliable payments. However, existing systems still suffer from delays, high fees, and poor coordination between buyers, sellers, and platforms. More importantly, they do not address the trust gap: buyers risk paying for goods that are never delivered, while sellers risk delivering goods that are never paid for. These problems slow down transactions, increase fraud, and make high-value or cross-border deals especially difficult.

The proposed Escrow Payment System aims to solve these issues by acting as a trusted bridge between payers and receivers. It holds funds securely in an escrow account, releases them only when predefined conditions are met, and provides a fair dispute resolution mechanism. The system automates verification, ensures regulatory compliance, and provides faster, safer, and more transparent transactions, especially in high-value and cross-border scenarios

2.2 Initial System Request

2.2.1 Business Need

Businesses and individuals turn to escrow payment systems to fulfill a range of essential operational requirements:

- **Building Trust Between Strangers:** Protect buyers and sellers who do not know each other by holding funds in a neutral account until conditions are met.
- **Supporting High-Value Transactions:** Provide a safeguard for large purchases such as real estate, vehicles, or bulk orders where both parties need extra protection.

- **Enabling Cross-Border Trade:** Provide a neutral ground for international transactions involving different currencies, regulations, and legal systems.
- **Reducing Disputes and Chargebacks:** Automate verification and provide a structured dispute workflow to reduce the cost of chargebacks and legal action.
- **Supporting Marketplaces and Platforms:** Allow e-commerce and freelancing platforms to embed escrow functionality and reduce platform liability.
- **Automating Financial Workflows:** Automate the escrow lifecycle from funding to conditional release, dispute handling, and reporting.

2.2.2 Business Requirements

When choosing an escrow provider, businesses must evaluate a number of key requirements:

- **Security and Compliance:** Compliance with escrow licensing, PCI DSS, KYC/AML, and GDPR regulations.
- **Segregated Accounts and Auditability:** Escrow funds held in segregated accounts with immutable audit logs for every state change.
- **Conditional Release Engine:** Flexible release conditions such as buyer confirmation, delivery verification, milestone approval, or time-based release.
- **Dispute Resolution Workflow:** A structured process for evidence submission, mediation, and arbitration.
- **Integration Capabilities:** Seamless integration with e-commerce platforms, accounting software, logistics providers, and banks via APIs.
- **Scalability and Reliability:** Handle growing escrow volume without compromising performance or uptime.
- **Reporting and Analytics:** Detailed reporting on escrow volume, release times, dispute rates, and compliance metrics.
- **Cost and Fee Structure:** Transparent and competitive pricing covering escrow, transaction, and currency conversion fees.
- **Customer Support:** Responsive multi-channel support for dispute and payment issues.
- **International Payment Capabilities:** Multi-currency support with clear foreign exchange rates and regional compliance.

- **Dispute Resolution Workflow:** A structured process and communication portal for evidence submission, mediation, and arbitration.

2.2.3 Business Value

The proposed Escrow Payment System will deliver value through:

- **Improved Trust:** A neutral escrow account protects both parties and reduces fraud.
- **Streamlined Processes:** Automated verification and conditional release replace manual paperwork and reduce operational delays.
- **Enhanced Security:** Segregated accounts, encryption, and audit logs protect funds and data.
- **Cost Savings:** Fewer disputes, chargebacks, and legal costs.
- **Better User Experience:** A single platform that simplifies the escrow lifecycle.
- **Market Expansion:** Cross-border and multi-currency support opens new markets.

2.2.4 Special Issues or Constraints

- **Regulatory Compliance:** Must obtain escrow licenses and comply with central bank regulations in all operating regions.
- **Segregated Account Management:** Requires careful banking partnerships and strict internal controls.
- **Scalability:** Must handle high concurrent escrow volume during peak periods.
- **Security:** Strong encryption, MFA, and fraud detection are essential.
- **Integration:** Smooth interoperability with banks, gateways, logistics, and platforms.
- **Dispute Resolution Fairness:** Transparent rules for evidence submission and arbitration, requiring a dedicated operational team or AI-assisted moderation to handle edge cases.
- **Usability:** Simple interface for buyers, sellers, and administrators.
- **Cross-Border Complexity:** Multiple currencies, regulations, and legal systems add significant complexity.

2.3 Feasibility Analysis

2.3.1 Technical Feasibility

Technical Resources: Modern web and cloud technologies provide the necessary infrastructure for secure, high-performance escrow systems. EPS can be implemented using:

- **Frontend:** ReactJS or Angular for responsive web interfaces.
- **Backend:** Node.js, Django, or Spring Boot for API and escrow logic.
- **Database:** PostgreSQL or MongoDB for storing escrow records and audit logs.
- **Cloud Infrastructure:** AWS, Azure, or Google Cloud for scalability, load balancing, and disaster recovery.
- **Security Frameworks:** AES-256 encryption, SSL/TLS, OAuth 2.0, and multi-factor authentication.
- **Escrow-Specific:** Segregated account integration with banking partners, an automated rules engine for conditional release, and logistics/KYC API connectors.

Technical Expertise: Development skills – API integration, database management, encryption, financial data processing, and escrow compliance.

Risk Assessment: Three main types of risk have been identified: time, resource, and functional, with additional escrow-specific risks around regulatory compliance and banking partnerships.

2.3.2 Operational Feasibility

- **User Readiness:** Banks, fintech firms, marketplaces, and users are already familiar with escrow concepts and online payment platforms.
- **Organizational Impact:** Enhances workflow efficiency by automating escrow creation, conditional release, and dispute routing. The system separates technical operations from business operations, ensuring clear ownership of each function.
- **Maintenance and Support:** Ongoing operations are handled by two distinct teams:
  - A **Technical Team** responsible for system uptime, security monitoring, bug fixes, and infrastructure maintenance.
  - An **Operations & Compliance Team** responsible for dispute resolution, KYC/AML reviews, customer support, and regulatory reporting.

2.3.3 Economic Feasibility

Cost of Development:

- **Initial Setup:** $\$120{,}000 - \$150{,}000$, covering backend and frontend development, escrow account integration, API integration, UI design, security implementation, testing, and deployment.
- **Recurring Costs:** $\$4{,}000 - \$7{,}000$ per month, including cloud hosting, segregated account maintenance, compliance monitoring, and support.

Revenue Generation:

- **Escrow Fees:** A commission $(0.5\% - 1.5\%)$ per successful escrow transaction. This is higher than standard payment gateways because escrow provides greater protection, longer fund-holding periods, and dispute handling.
- **Subscription Model:** Monthly or annual plans for marketplaces, SMEs, and financial institutions.
- **Value-Added Services:** Premium features such as dispute arbitration, real-time analytics, and custom API integrations.

Cost Efficiency: Automating escrow lifecycle, verification, and dispute routing significantly reduces manual intervention and legal costs.

Return on Investment (ROI): Break-even point within 2–3 years, supported by escrow fees, subscription revenue, and expanding transaction volume.

2.3.4 Legal & Regulatory Feasibility

Compliance:

- **Laws and Regulations:** In Vietnam, there is no standalone "escrow license." The system must obtain relevant **intermediary payment licenses** from the State Bank of Vietnam (SBV) and strictly maintain **Payment Guarantee Accounts** (Tài khoản đảm bảo thanh toán) with partnered commercial banks to hold user funds legally.
- **Third-party Service:** Contracts with banking partners, logistics providers, and KYC/AML vendors must be carefully managed.

Data Privacy: Robust security measures and compliance with PCI DSS, GDPR, and Vietnamese regulations on Non-Cash Payments and SBV guidance. Includes encryption, strong authentication, segregated accounts, and transparent terms of service.

2.3.5 Market Feasibility

Demand: Rapid growth of digital commerce, cross-border trade, and high-value online transactions has created strong demand for secure escrow solutions.

Competitive Landscape: Many existing escrow providers are tied to specific banks, law firms, or marketplaces, with high fees and limited interoperability. An independent, API-first, and cost-effective escrow platform can fill a significant market gap and establish a strong competitive advantage.

Target Market: SMEs, online marketplaces, freelancers, cross-border traders, and financial institutions seeking flexible escrow integration.

Market Potential: Adoption rates for escrow-backed online payments are expected to rise significantly, driven by government initiatives promoting digital transformation and financial inclusion.

2.4 Initial Project Plan

2.4.1 Project Scope Estimation

Main Objective: Develop a Minimum Viable Product (MVP) for the Escrow Payment System (EPS) that supports basic escrow flows (create → fund → hold → release/refund), integration with multiple payment methods, conditional release rules, and basic dispute resolution. The MVP will also include an administrative dashboard and APIs for merchants and marketplaces.

Functional Scope (MVP):

- User authentication and registration (buyer, seller, admin).
- Escrow creation, funding, and tracking.
- Conditional release engine with configurable rules.
- API connections to at least one bank and one payment gateway (mock or real).
- Basic dispute workflow with evidence submission.
- Logging, audit trail, and basic analytics dashboard.
- Security mechanisms: OAuth2, TLS encryption, MFA, and segregated account integration.

Estimated Timeline (21 weeks total):

- Requirement Analysis: 2 weeks

- System & UI/UX Design: 3 weeks
- Backend, Escrow Logic & API Development: 7 weeks
- Frontend and Dashboard Development: 5 weeks (in parallel with backend)
- Dispute Workflow & Integration: 2 weeks
- Testing, Security Review, Penetration Testing, and Deployment: 2 weeks

The Testing phase is extended to 2 weeks because escrow systems hold user funds for extended periods and enforce complex disbursement rules. Security review and penetration testing are critical to prevent financial loss and regulatory violations. The total timeline is 21 weeks, with any remaining buffer absorbed into the extended testing phase.

Deliverables:

- Requirement Specification Document
- System Architecture and Database Design
- API Specification
- Frontend & Backend Source Code
- Escrow Condition & Dispute Policy Document
- Test Report and Deployment Guide

2.4.2 Staffing Plan

For a small company with fewer members, the team structure must remain lean yet cover all critical roles required for developing, securing, and maintaining the Escrow Payment System (EPS):

- **Project Manager / Business Analyst:** Oversees project planning, requirement gathering, and stakeholder coordination. Ensures deliverables align with business goals and manages the overall project timeline. Works closely with external Legal/Compliance Consultants to define dispute resolution frameworks and ensure central bank compliance.
- **Technical Lead:** Defines system architecture, technology stack, and coding standards. Oversees integration, code reviews, and ensures system scalability, performance, and security compliance.

- **Full-Stack Developer (Backend + Frontend):** Implements backend APIs and frontend dashboards. Handles database operations, escrow logic, and UI functionalities.
- **DevOps & Security Engineer:** Manages CI/CD pipelines, deployment, and infrastructure monitoring. Also responsible for implementing security policies, SSL/TLS setup, and disaster recovery.
- **QA Engineer / UI-UX Designer:** Performs testing (functional, integration, load) and ensures KYC/AML, escrow compliance, and data protection standards are met. Assists with UI/UX improvements.
- **Operations & Compliance Specialist (part-time or shared):** Handles dispute resolution, KYC/AML reviews, customer support, and regulatory reporting. Ensures the platform operates within SBV guidelines and maintains Payment Guarantee Accounts properly.

2.4.3 CASE Tools

Modeling and Design: StarUML / Enterprise Architect for use case, class, and sequence diagrams. Figma for UI wireframes and collaborative design.

Project Management: Jira for backlog and sprint tracking.

Development and CI/CD: Git + GitHub/GitLab for version control. GitHub Actions / GitLab CI for build, test, and deploy automation. Docker for containerization; Kubernetes for orchestration if needed.

Database and API Tools: PostgreSQL. Postman / Swagger (OpenAPI) for API testing and documentation.

Testing and Security: Pytest / Jest for unit and integration testing. SonarQube / Dependabot for code quality and dependency security scanning. k6 / JMeter for performance and load testing.

Monitoring and Logging: ELK Stack (Elasticsearch, Logstash, Kibana) for centralized logging. Prometheus + Grafana for system monitoring and alerts.

Chapter 3

Analysis

3.1 General Use cases Diagram

EPS is an academic escrow simulation for goods and services. Each agreement has one Buyer, one distinct Seller and one full VND amount, funded through a Bank or Payment Gateway mock. The overview is divided into focused views for readability.

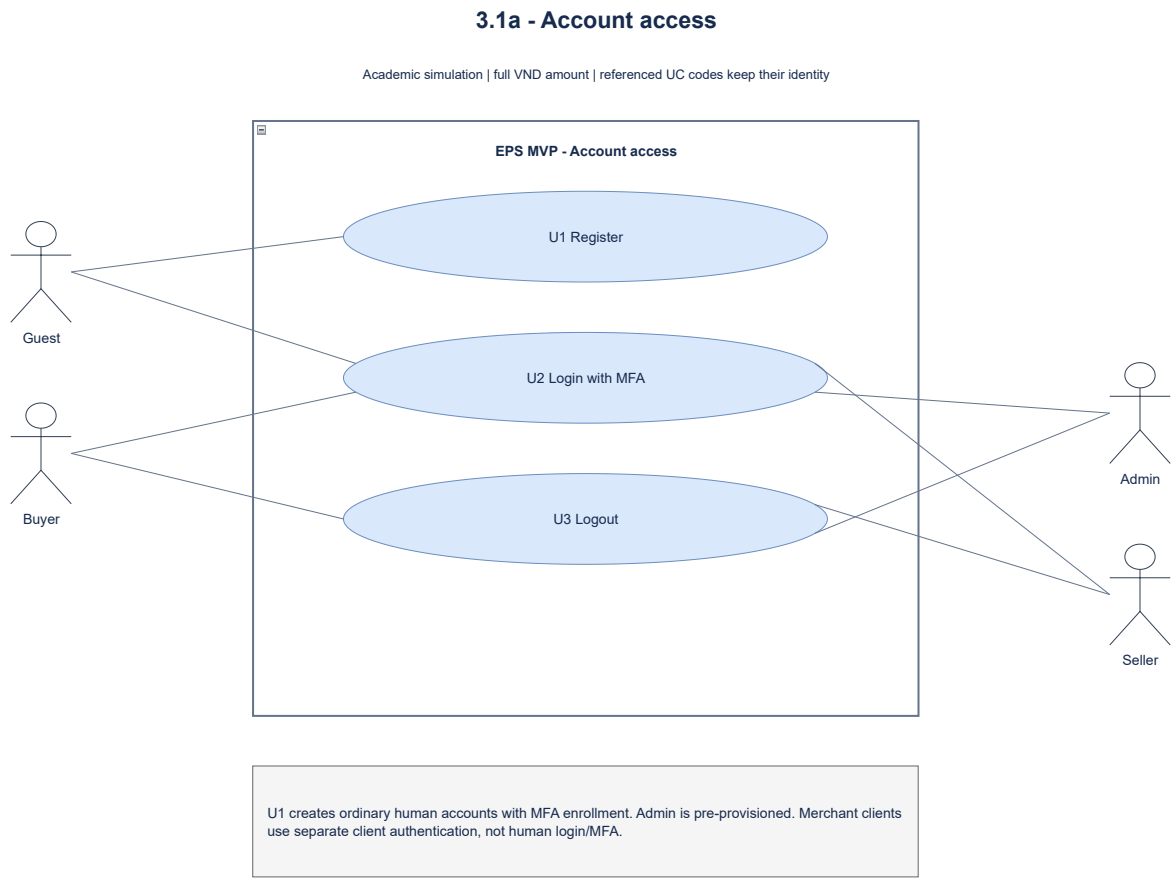


Figure 3.1: Account access

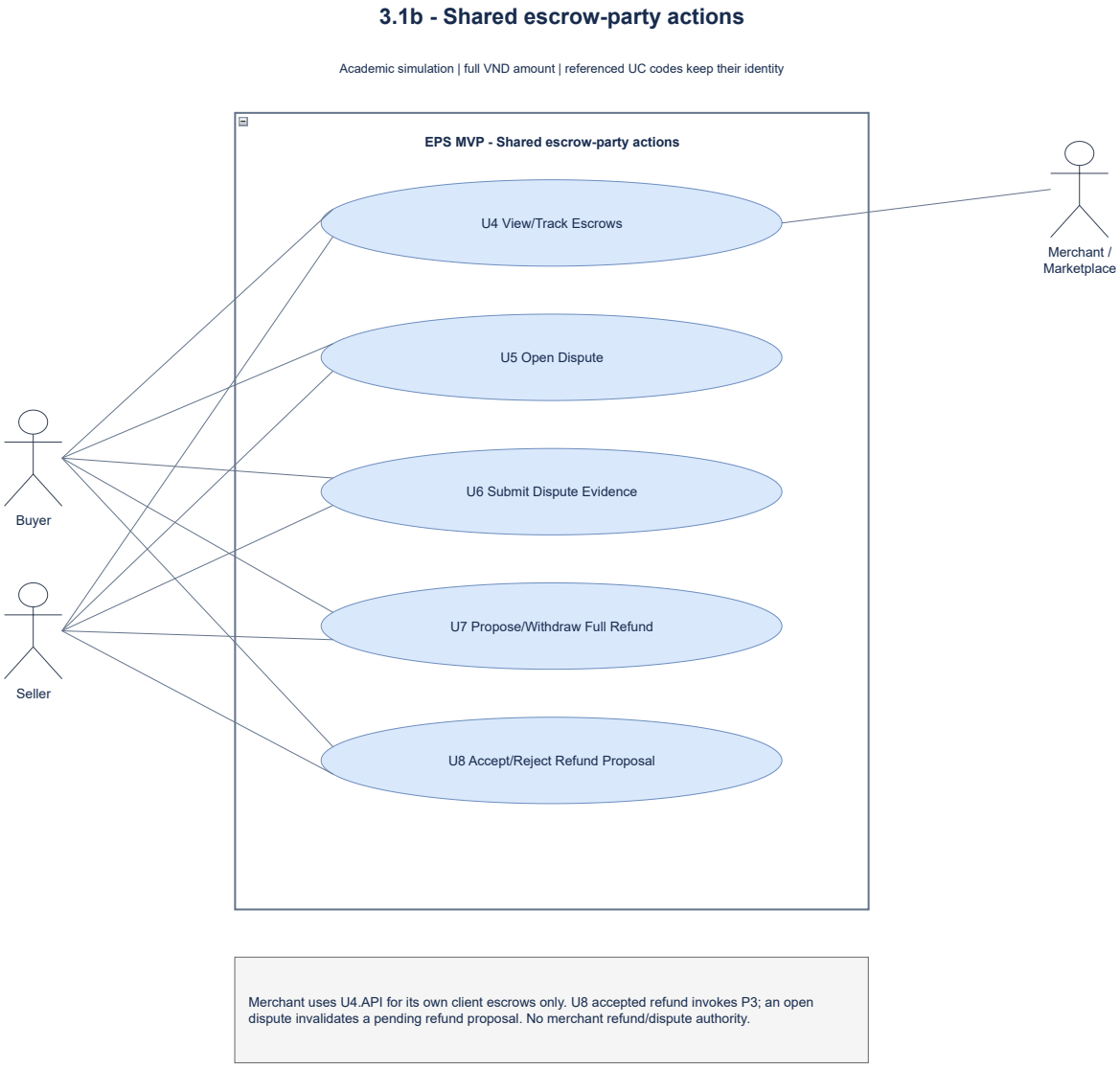


Figure 3.2: Shared escrow-party actions

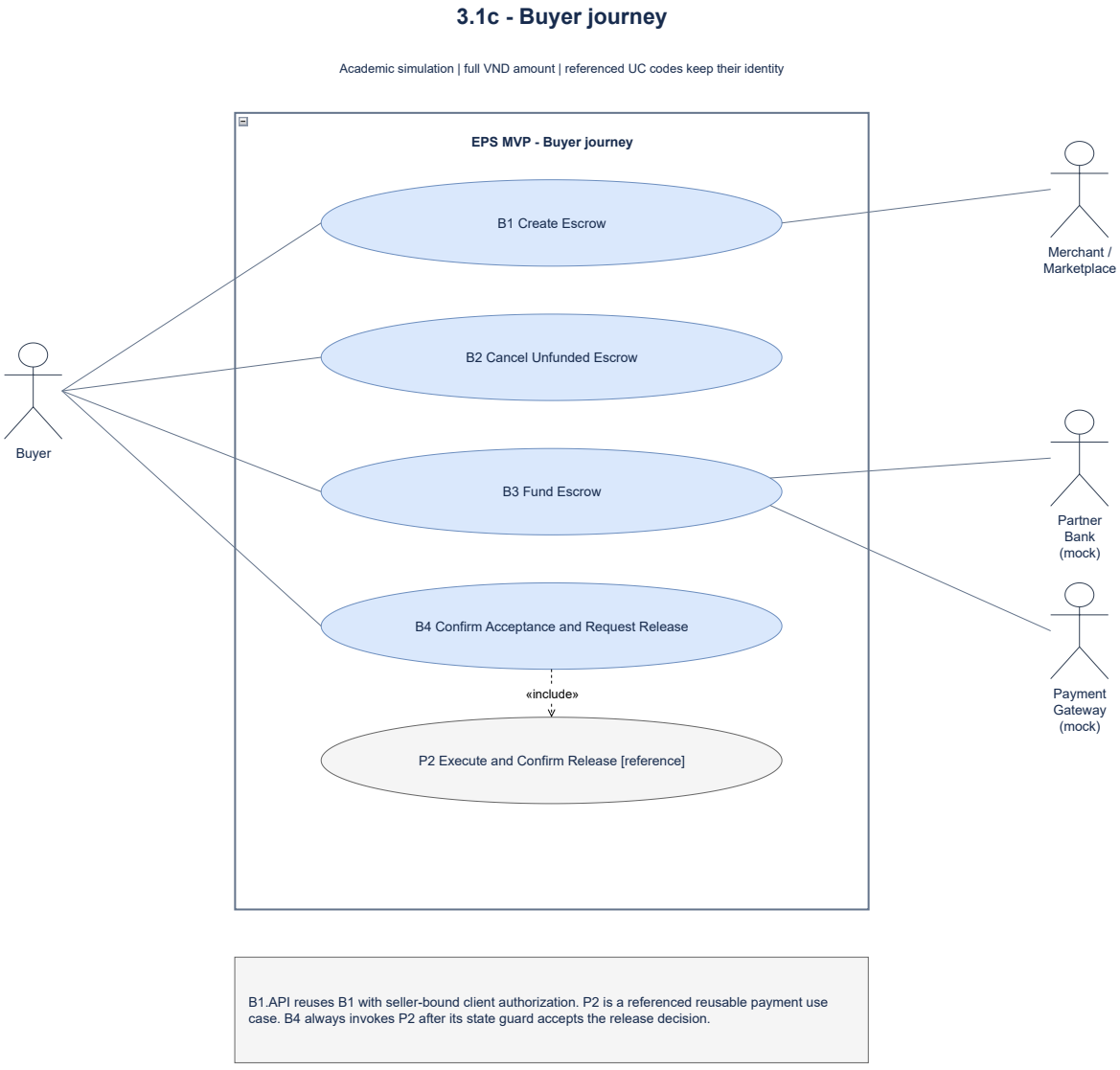


Figure 3.3: Buyer journey

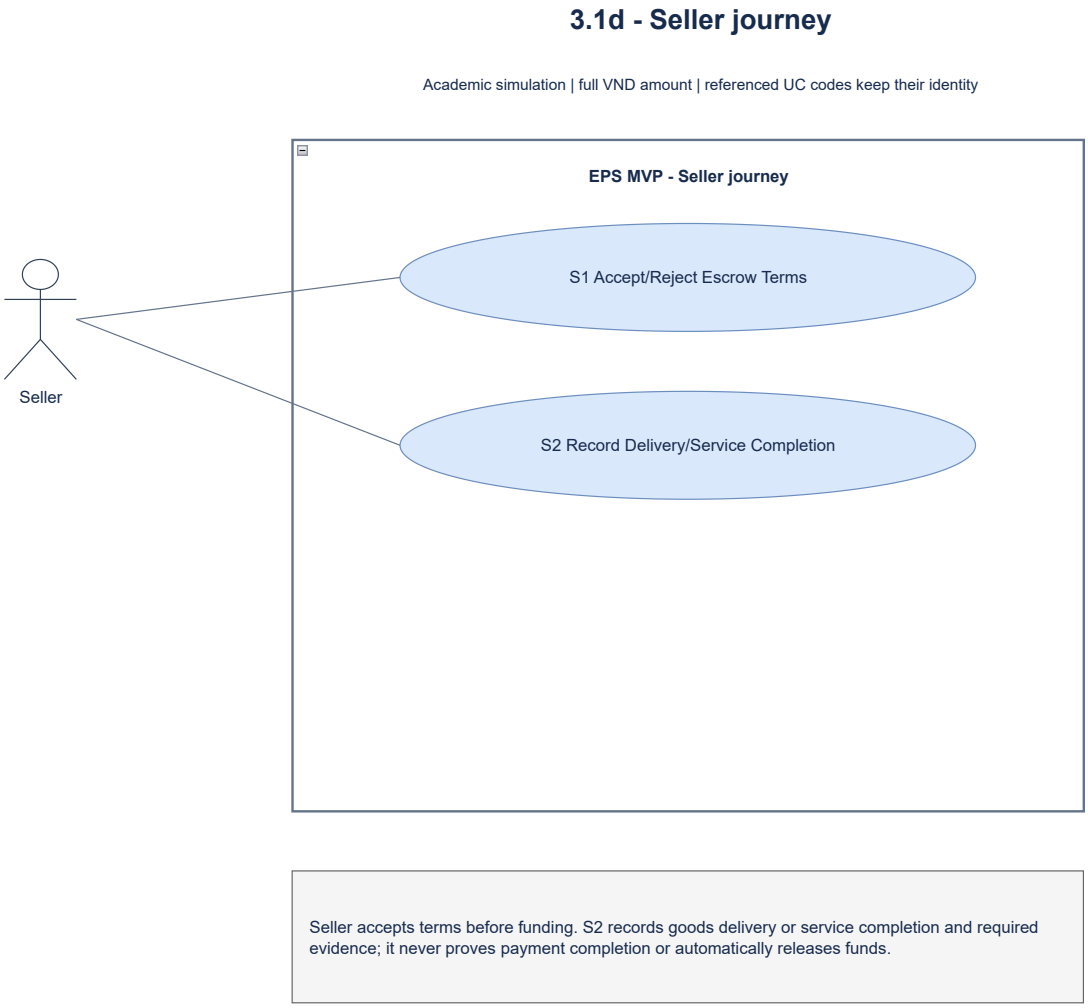


Figure 3.4: Seller journey

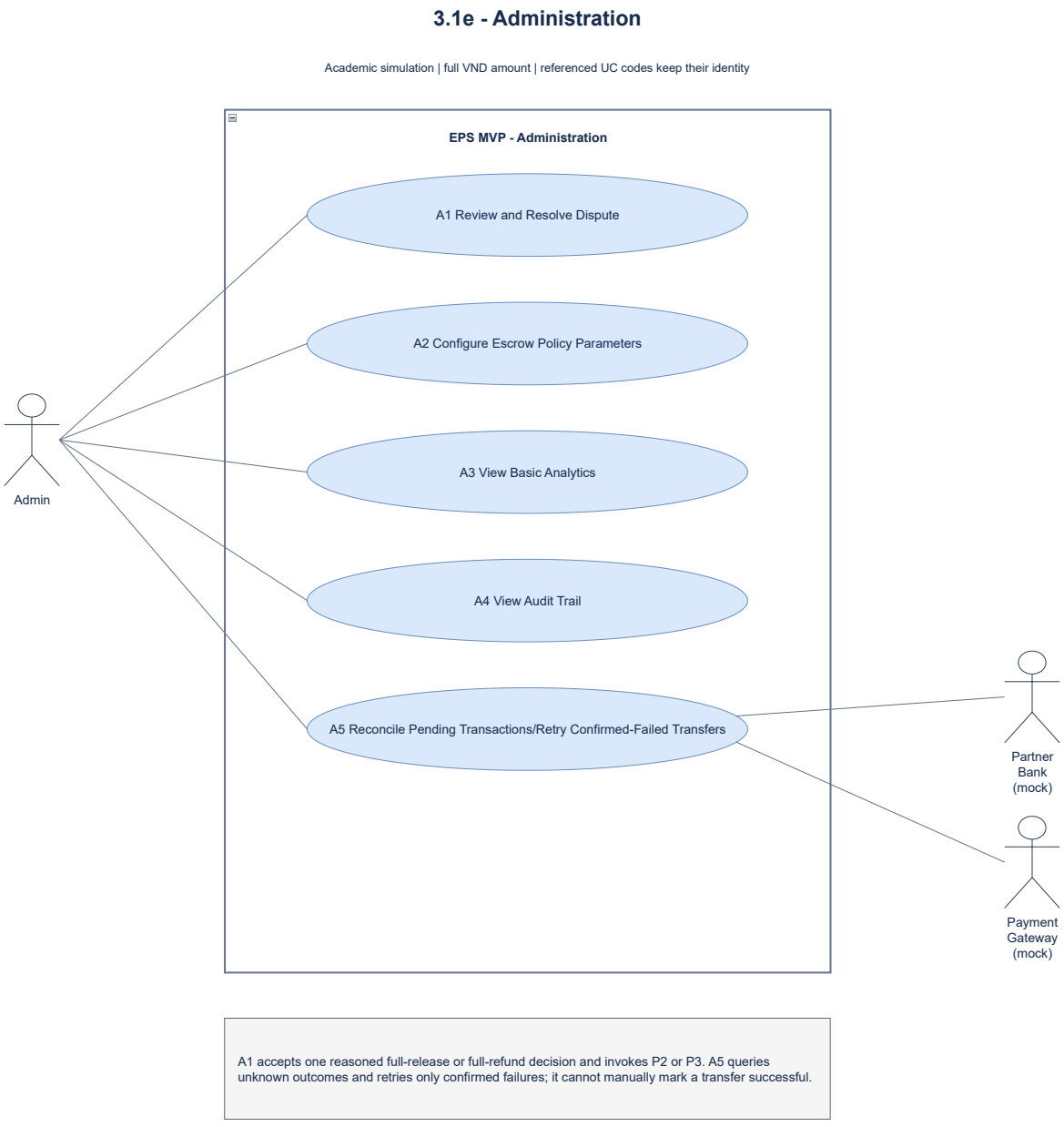


Figure 3.5: Administration

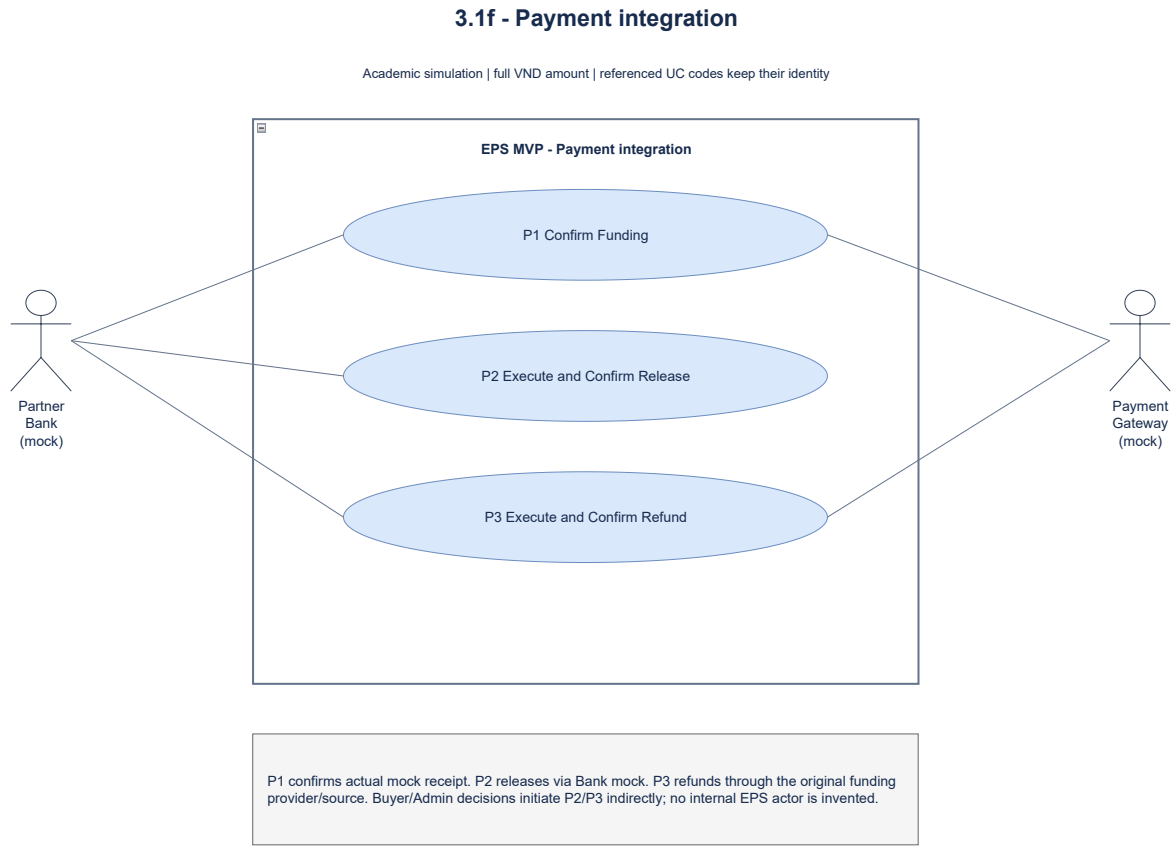


Figure 3.6: Payment integration

- **Guest**: registers an ordinary account and logs in with MFA.
- **Buyer**: creates and funds an escrow, accepts delivery and requests release.
- **Seller**: accepts the agreement and records delivery or service completion. Both participants may request a refund or open a dispute.
- **Admin**: resolves disputes, updates policy, views analytics/audit and reconciles pending transfers.
- **Merchant/Marketplace**: creates and queries only its own escrows through a separately authenticated client.
- **Partner Bank**: simulates funding, Seller release and refunds for Bank-funded escrows.
- **Payment Gateway**: simulates funding and refunds to the original Gateway source.

Buyer and Seller are roles in each transaction. Release uses the Bank mock; refund returns through the original funding provider. Real money, partial payments, milestones and automatic release are outside this MVP.

3.2 Use case Details

3.2.1 User's Use cases Details

U1: Register

Use Case Code	U1	**Use Case Name**	Register
Actor	Guest		
Usage	Create an ordinary user account.		
Activation Event	Guest selects Register.		
Preconditions	The login identifier is not already registered.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Guest	Enter account details and password.
	2	EPS	Validate the details and start MFA enrollment; do not grant Admin or Merchant privileges.
	3	Guest	Complete the MFA enrollment challenge.
	4	EPS	Create the account after successful verification and direct the user to Login.
	5	EPS	For invalid details, a duplicate identifier or failed MFA, reject registration without activating an account.
Postconditions	A valid ordinary account is created; the user must log in separately.		

U2: Login with MFA

Use Case Code	U2	**Use Case Name**	Login with MFA
Actor	Buyer; Seller; Admin		
Usage	Start an authenticated user session.		
Activation Event	User selects Login.		
Preconditions	An active user or pre-provisioned Admin account exists.		
Main Event Flow	**Index**	**Done by**	**Action**

U2 (continued)			
	1	User	Submit the login identifier and password.
	2	EPS	Verify credentials and request the enrolled MFA factor.
	3	User	Submit the MFA response.
	4	EPS	Verify MFA, create the session and show the permitted workspace.
	5	EPS	Reject invalid credentials or MFA without creating a session. Merchant client credentials cannot be used as a human login.
Postconditions	A successful login creates a session with the account's existing permissions.		

U3: Logout

Use Case Code	U3	**Use Case Name**	Logout
Actor	Buyer; Seller; Admin		
Usage	End the current user session.		
Activation Event	User selects Logout.		
Preconditions	A current or previously expired session is available.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	User	Select Logout.
	2	EPS	Invalidate the current session and show Login. An already expired session returns the same result.
Postconditions	The previous session cannot access protected functions.		

U4: View/Track Escrows

Use Case Code	U4	**Use Case Name**	View/Track Escrows
Actor	Buyer; Seller; Merchant/Marketplace (U4.API variant)		
Usage	View escrows and their current progress.		
Activation Event	User opens the escrow list or an escrow detail.		
Preconditions	The user is authenticated and is a participant in the requested escrow.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	User	Request the list or select an escrow.
	2	EPS	Check access and retrieve the authorised record.
	3	EPS	Show terms, lifecycle, payment status, delivery, refund proposal and dispute status.
	4	EPS	Show unresolved operations as pending with a query reference; viewing does not move money.
	5	EPS	Reject access to another user's escrow without disclosing its data.

U4 (continued)			
	API	Merchant	U4.API: Authenticate the Merchant client and return only its owned escrow status. Pending results include a query reference. Reject other clients' escrows and exclude private dispute evidence and Admin audit data.
Postconditions	The authorised current status is displayed; the escrow is unchanged.		

U5: Open Dispute

Use Case Code	U5	**Use Case Name**	Open Dispute
Actor	Buyer; Seller		
Usage	Request Admin resolution of a held escrow.		
Activation Event	Buyer or Seller selects Open Dispute.		
Preconditions	The escrow is HELD or DELIVERED; no dispute has previously been opened and no payout is reserved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Enter the dispute reason.
	2	EPS	Recheck participant access and dispute eligibility.
	3	EPS	Open the dispute and invalidate any pending refund proposal in one state change.
	4	EPS	Save the case and allow both parties to submit evidence; block delivery and ordinary release.
	5	EPS	Reject a stale or concurrent request if another payout or dispute decision has already been accepted.
Postconditions	One OPEN dispute exists; the full amount remains held.		

U6: Submit Dispute Evidence

Use Case Code	U6	**Use Case Name**	Submit Dispute Evidence
Actor	Buyer; Seller		
Usage	Add evidence to an open dispute.		
Activation Event	A participant selects Submit Evidence.		
Preconditions	The participant belongs to the escrow and the dispute remains OPEN.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Submit evidence and a short explanation.
	2	EPS	Validate the submission and recheck that no final decision has been recorded.
	3	EPS	Save the evidence with its author and submission time.
	4	EPS	Reject unauthorised submissions or evidence received after the final decision.
Postconditions	Accepted evidence is available for dispute review.		

U7: Propose/Withdraw Full Refund

Use Case Code	U7	**Use Case Name**	Propose/Withdraw Full Refund
Actor	Buyer; Seller		
Usage	Propose or withdraw a full refund.		
Activation Event	A participant selects Propose Refund or Withdraw Proposal.		
Preconditions	The escrow is HELD or DELIVERED, with no open dispute or reserved payout.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Submit a full-refund reason, or select their pending proposal for withdrawal.
	2	EPS	For a new proposal, check that no other proposal is pending; save it and notify the counterpart.
	3	EPS	While the proposal is pending, block new delivery recording and ordinary Buyer release.
	4	EPS	Allow only the proposer to withdraw before approval. Restore otherwise eligible actions without deleting saved delivery.
	5	EPS	Reject stale requests; repeated requests return the recorded result.
Postconditions	The proposal is PENDING or WITHDRAWN; no money is transferred.		

U8: Accept/Reject Refund Proposal

Use Case Code	U8	**Use Case Name**	Accept/Reject Refund Proposal
Actor	Buyer; Seller		
Usage	Accept or reject the counterpart's full-refund proposal.		
Activation Event	The counterpart responds to a proposal.		
Preconditions	The proposal is PENDING and the escrow has no open dispute or reserved payout.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Review the proposal and choose Accept or Reject.
	2	EPS	Recheck that the responder is the counterpart and the proposal remains eligible.
	3	EPS	On acceptance, approve the proposal and reserve the full refund as REFUND_ PENDING; invoke P3.
	4	EPS	On rejection, mark REJECTED and restore otherwise eligible actions without reverting delivery.
	5	EPS	Reject self-approval or a stale response. A repeated response returns the recorded result.
Postconditions	Approval starts P3; rejection makes no payment. Only confirmed provider success produces REFUNDED.		

3.2.2 Buyer's Use cases Details

B1: Create Escrow

Use Case Code	B1	**Use Case Name**	Create Escrow
Actor	Buyer; Merchant/Marketplace (B1.API variant)		
Usage	Create an escrow agreement.		
Activation Event	Buyer selects Create Escrow.		
Preconditions	The Buyer is authenticated; the Buyer and Seller are distinct existing users.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Enter the Seller, GOODS/SERVICES, full VND amount, deliverable and acceptance criteria.
	2	EPS	Validate the agreement and capture the current evidence policy and Seller destination.
	3	EPS	Save the escrow as AWAITING_SELLER and notify the Seller.
	4	EPS	Reject invalid terms or self-trading. An identical repeated creation returns the existing escrow.
	API	Merchant	B1.API: Authenticate the Merchant client, use its bound Seller and validate the same B1 terms. Save client ownership and return the escrow reference and AWAITING_SELLER. Reject another Seller or conflicting request reuse; identical repeats return the existing escrow.
Postconditions	The agreement awaits Seller acceptance; no funding has started.		

B2: Cancel Unfunded Escrow

Use Case Code	B2	**Use Case Name**	Cancel Unfunded Escrow
Actor	Buyer		
Usage	Cancel an escrow before funding.		
Activation Event	Buyer selects Cancel.		
Preconditions	The escrow is AWAITING_SELLER or AWAITING_FUNDING; no money was received and no funding result is unresolved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Confirm cancellation of their escrow.
	2	EPS	Recheck ownership and funding status.
	3	EPS	Set CANCELLED and record the reason.
	4	EPS	Reject cancellation while funding is active or unknown. A repeated cancellation returns the existing result.
Postconditions	The cancelled escrow cannot accept funding or payout.		

B3: Fund Escrow

Use Case Code	B3	**Use Case Name**	Fund Escrow
Actor	Buyer; Partner Bank; Payment Gateway		
Usage	Fund the full accepted agreement through a mock provider.		
Activation Event	Buyer selects Fund Escrow.		
Preconditions	Seller has accepted; the escrow is AWAITING_FUNDING with no unresolved funding attempt.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Review the terms, accept them and choose Bank or Gateway.
	2	EPS	Save a funding reference and set FUNDING_ PENDING.
	3	Bank/Gateway	Process the full VND funding request and return a result through P1.
	4	EPS	Show the confirmed result, or keep pending if the result is uncertain.
	5	EPS	Only confirmed failure with no receipt returns AWAITING_FUNDING, allowing the Buyer to retry or cancel.
Postconditions	Only P1 confirmation establishes HELD. A timeout does not permit another funding attempt.		

B4: Confirm Acceptance and Request Release

Use Case Code	B4	**Use Case Name**	Confirm Acceptance and Request Release
Actor	Buyer; Partner Bank		
Usage	Accept delivery and request full release to the Seller.		
Activation Event	Buyer selects Confirm Acceptance.		
Preconditions	The escrow is DELIVERED and its evidence policy is satisfied; no refund proposal, open dispute or payout is pending.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Review delivery and confirm acceptance.
	2	EPS	Recheck access, evidence and escrow eligibility.
	3	EPS	Reserve the full amount as RELEASE_ PENDING and invoke P2 through the Bank mock.
	4	EPS	Show the pending or confirmed result.
	5	EPS	Reject a conflicting request; a repeated acceptance returns the saved release decision.
Postconditions	Release is requested once. RELEASED requires confirmed Bank settlement.		

3.2.3 Seller's Use cases Details

S1: Accept/Reject Escrow Terms

Use Case Code	S1	**Use Case Name**	Accept/Reject Escrow Terms
Actor	Seller		
Usage	Accept or reject a proposed escrow agreement.		
Activation Event	Seller opens the proposed terms.		
Preconditions	The escrow is AWAITING_SELLER and the user is its Seller.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Seller	Review the terms and choose Accept or Reject.
	2	EPS	Recheck Seller access and the current state.
	3	EPS	On acceptance, lock the terms and set AWAITING_FUNDING; on rejection, set CANCELLED with the reason.
	4	EPS	Reject a stale change or return the previously recorded response.
Postconditions	Accepted terms cannot be edited; funding is permitted only after acceptance.		

S2: Record Delivery/Service Completion

Use Case Code	S2	**Use Case Name**	Record Delivery/Service Completion
Actor	Seller		
Usage	Record goods delivery or service completion.		
Activation Event	Seller selects Record Delivery/Completion.		
Preconditions	The escrow is HELD, with no pending refund proposal, open dispute or reserved payout.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Seller	Submit delivery information or the completed work and required evidence.
	2	EPS	Check Seller access and the escrow's captured evidence policy.
	3	EPS	Save completion, set DELIVERED and notify the Buyer.
	4	EPS	Reject missing required evidence or an ineligible state; do not release money automatically.
Postconditions	Delivery is recorded; the full amount remains held for Buyer acceptance or another eligible outcome.		

3.2.4 Admin's Use cases Details

A1: Review and Resolve Dispute

Use Case Code	A1	**Use Case Name**	Review and Resolve Dispute
Actor	Admin		
Usage	Resolve a dispute with one full payout decision.		
Activation Event	Admin selects an open dispute.		
Preconditions	Admin is authenticated; the dispute is OPEN, funds are held and no payout is reserved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Review the agreement, evidence and payment status.
	2	Admin	Choose full release or full refund and enter a reason.
	3	EPS	Recheck the case; save RESOLVED and reserve the chosen payout in one state change.
	4	EPS	Invoke P2 for release or P3 for refund. This decision does not require Buyer acceptance.
	5	EPS	Show the pending or confirmed settlement; reject a conflicting second decision.
Postconditions	The decision and payout direction are final; only provider confirmation completes the payment.		

A2: Configure Escrow Policy Parameters

Use Case Code	A2	**Use Case Name**	Configure Escrow Policy Parameters
Actor	Admin		
Usage	Configure delivery-evidence requirements for new escrows.		
Activation Event	Admin opens Escrow Policy.		
Preconditions	Admin is authenticated with policy permission.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Review and edit the evidence-required flag for GOODS and SERVICES.
	2	EPS	Validate the flags and save a new policy version.
	3	EPS	Apply it only to escrows created afterwards; keep existing snapshots unchanged.
	4	EPS	Record the change in the audit trail. Automatic release and milestones are not configurable.
Postconditions	A valid policy version is available for new agreements.		

A3: View Basic Analytics

Use Case Code	A3	**Use Case Name**	View Basic Analytics
Actor	Admin		
Usage	View basic statistics for simulated escrows.		
Activation Event	Admin opens Analytics.		
Preconditions	Admin is authenticated with analytics permission.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Select the period and transaction type.
	2	EPS	Calculate counts and VND amounts by lifecycle state.
	3	EPS	Show the observation time and simulated-data label; separate pending amounts from final outcomes.
Postconditions	Statistics are displayed without changing escrow records.		

A4: View Audit Trail

Use Case Code	A4	**Use Case Name**	View Audit Trail
Actor	Admin		
Usage	Inspect recorded system and escrow events.		
Activation Event	Admin opens Audit Trail.		
Preconditions	Admin is authenticated with audit permission.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Filter events by escrow, period or event type.
	2	EPS	Return attributable events and correlation references, excluding authentication secrets.
	3	EPS	Display the audit trail as read-only; show an empty result when no events match.
Postconditions	Audit records remain unchanged.		

A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers

Use Case Code	A5	**Use Case Name**	Reconcile Pending Transactions/Retry Confirmed-Failed Transfers
Actor	Admin; Partner Bank; Payment Gateway		
Usage	Reconcile uncertain operations and retry confirmed-failed payouts.		
Activation Event	Admin selects an unresolved or failed payment attempt.		
Preconditions	Admin is authenticated; the original provider and operation references are available.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Request the status of the original operation.
	2	EPS	Authenticate and match the provider result, then apply P1, P2 or P3 once.
	3	EPS	For an unknown result, retain pending and allow queries only.

A5 (continued)			
	4	EPS	For confirmed failed funding with no receipt, return AWAITING_FUNDING; only the Buyer may fund again.
	5	Admin	For a provider-confirmed failed payout with no money moved, request a retry.
	6	EPS	Create one linked attempt with the same amount, direction and beneficiary. Never permit manual success or an opposite payout.
Postconditions	The status is reconciled, remains pending, or has one authorised same-direction retry.		

3.2.5 Payment Integration's Use cases Details

P1: Confirm Funding

Use Case Code	P1	**Use Case Name**	Confirm Funding
Actor	Partner Bank; Payment Gateway		
Usage	Confirm whether the full funding amount was received.		
Activation Event	Bank/Gateway sends a callback or answers a status query.		
Preconditions	A saved funding attempt exists.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	EPS	Authenticate the provider and match escrow, operation, attempt, full amount and VND.
	2	EPS	On confirmed receipt, record the held balance and set HELD once.
	3	EPS	On confirmed failure with no receipt, return AWAITING_FUNDING.
	4	EPS	For pending or unknown results, retain FUNDING_PENDING.
	5	EPS	Return the stored result for duplicates. Reject mismatched or contradictory results and flag them for reconciliation.
Postconditions	Funds are held only after a valid receipt; a duplicate never creates another credit.		

P2: Execute and Confirm Release

Use Case Code	P2	**Use Case Name**	Execute and Confirm Release
Actor	Partner Bank		
Usage	Execute and confirm one full release to the Seller.		
Activation Event	B4/A1 authorises release, or A5 authorises an eligible retry.		
Preconditions	A full release decision and fixed Seller destination are saved; a retry requires confirmed failure with no money moved.		
Main Event Flow	**Index**	**Done by**	**Action**

P2 (continued)			
	1	EPS	Send the full VND amount to the Bank mock using the saved release reference.
	2	Bank	Execute the mock transfer and return a callback or query result.
	3	EPS	Authenticate and match the attempt, amount, currency and destination.
	4	EPS	On success, set RELEASED once. On failure, keep RELEASE_PENDING and its reservation for an eligible A5 retry; on uncertainty, query only.
	5	EPS	Reuse duplicate results and reject mismatches. Do not accept a new dispute or opposite payout after reservation.
Postconditions	The Seller is paid once on confirmed success; otherwise the same release remains pending.		

P3: Execute and Confirm Refund

Use Case Code	P3	**Use Case Name**	Execute and Confirm Refund
Actor	Partner Bank; Payment Gateway		
Usage	Execute and confirm one full refund to the original source.		
Activation Event	U8/A1 authorises refund, or A5 authorises an eligible retry.		
Preconditions	A full refund decision and original funding source are saved; a retry requires confirmed failure with no money moved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	EPS	Send the full VND refund through the original funding provider using the saved refund reference.
	2	Bank/Gateway	Execute the mock refund and return a callback or query result.
	3	EPS	Authenticate and match the attempt, amount, currency and original source.
	4	EPS	On success, set REFUNDED once. On failure, keep REFUND_PENDING and its reservation for an eligible A5 retry; on uncertainty, query only.
	5	EPS	Reuse duplicate results and reject mismatches. Do not accept a new dispute or opposite payout after reservation.
Postconditions	The original source is refunded once on confirmed success; otherwise the same refund remains pending.		

3.3 Activity Diagram

3.3.1 User's Use cases

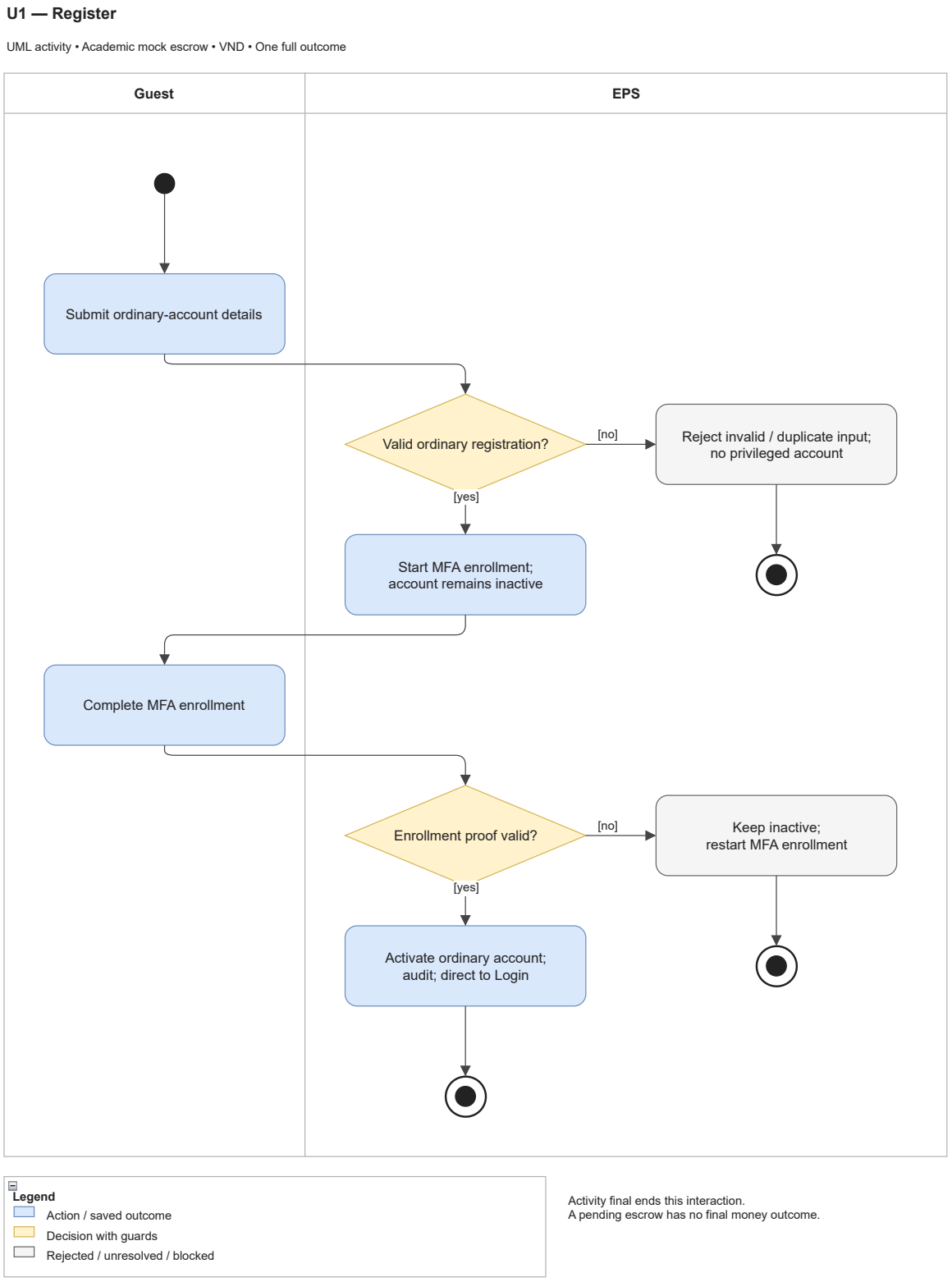


Figure 3.7: U1: Register

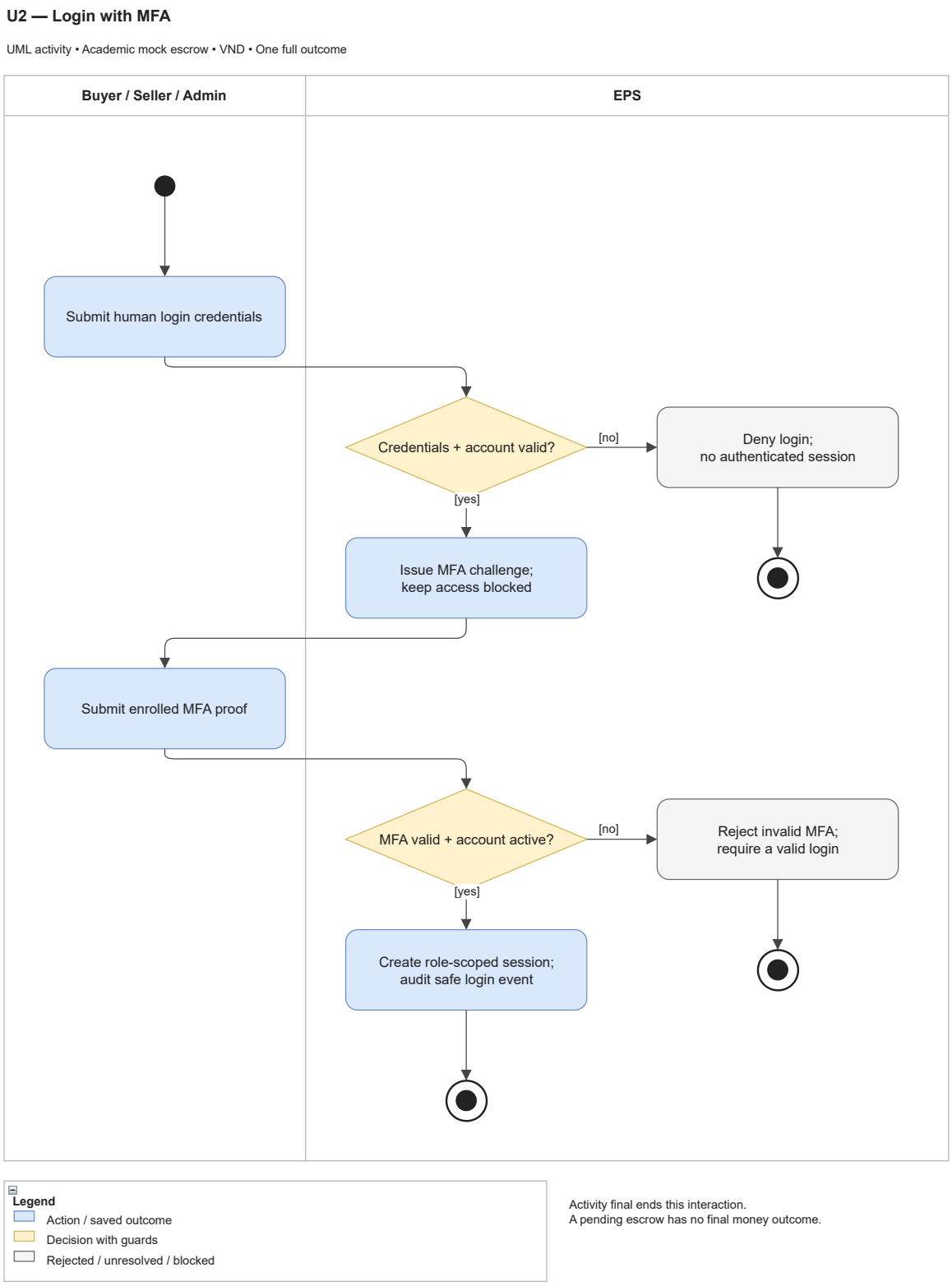


Figure 3.8: U2: Login with MFA

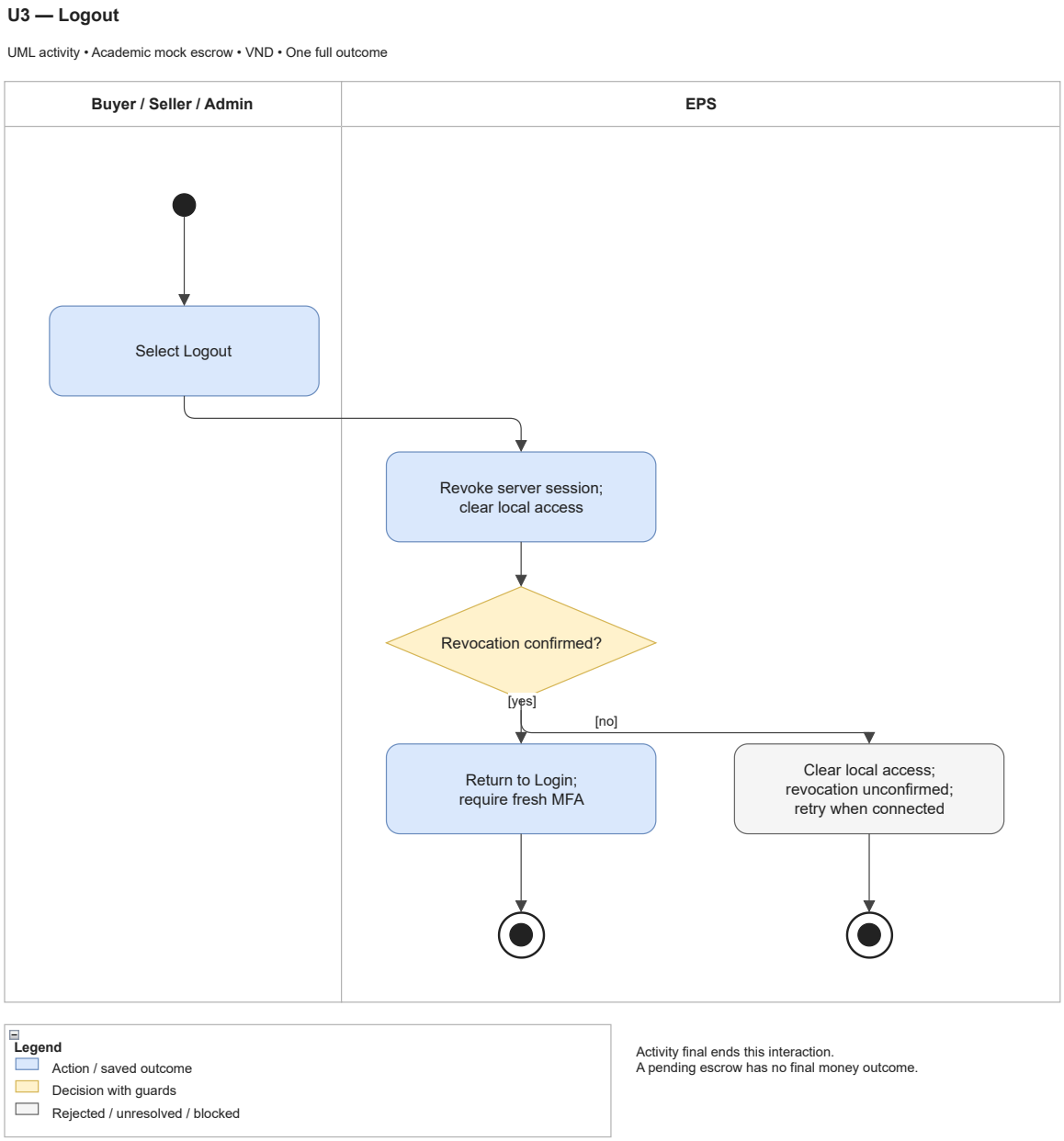


Figure 3.9: U3: Logout

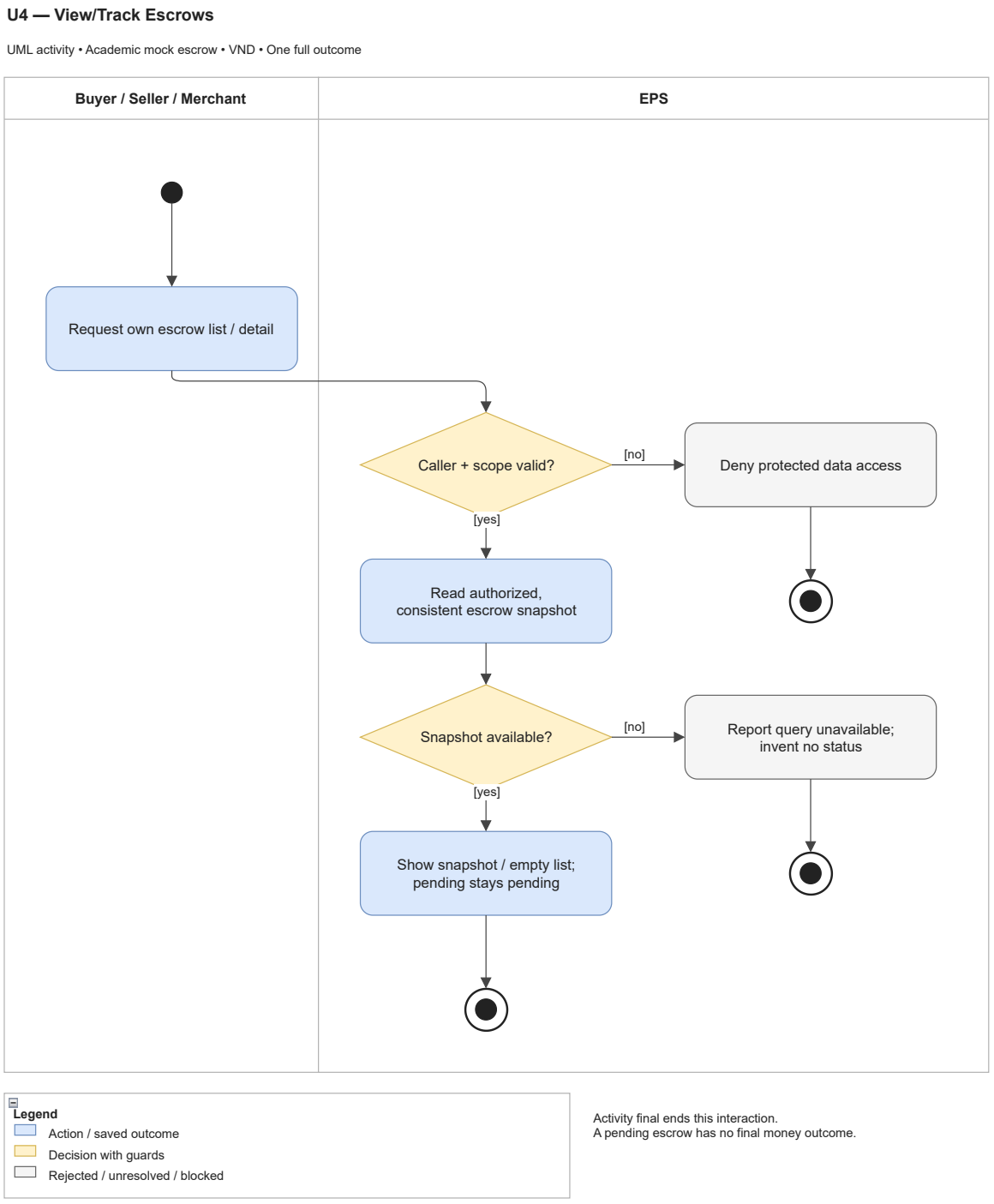


Figure 3.10: U4: View/Track Escrows

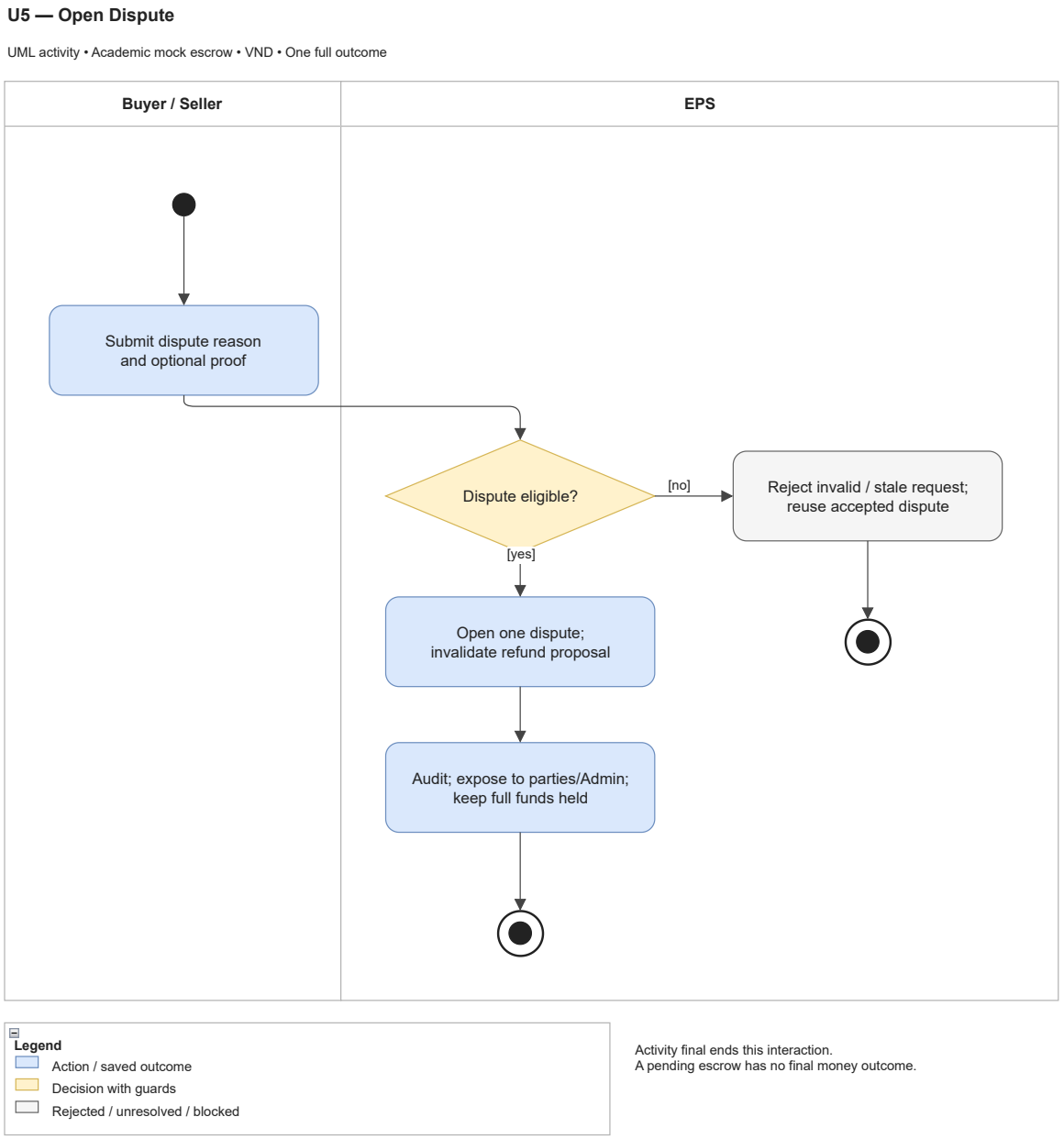


Figure 3.11: U5: Open Dispute

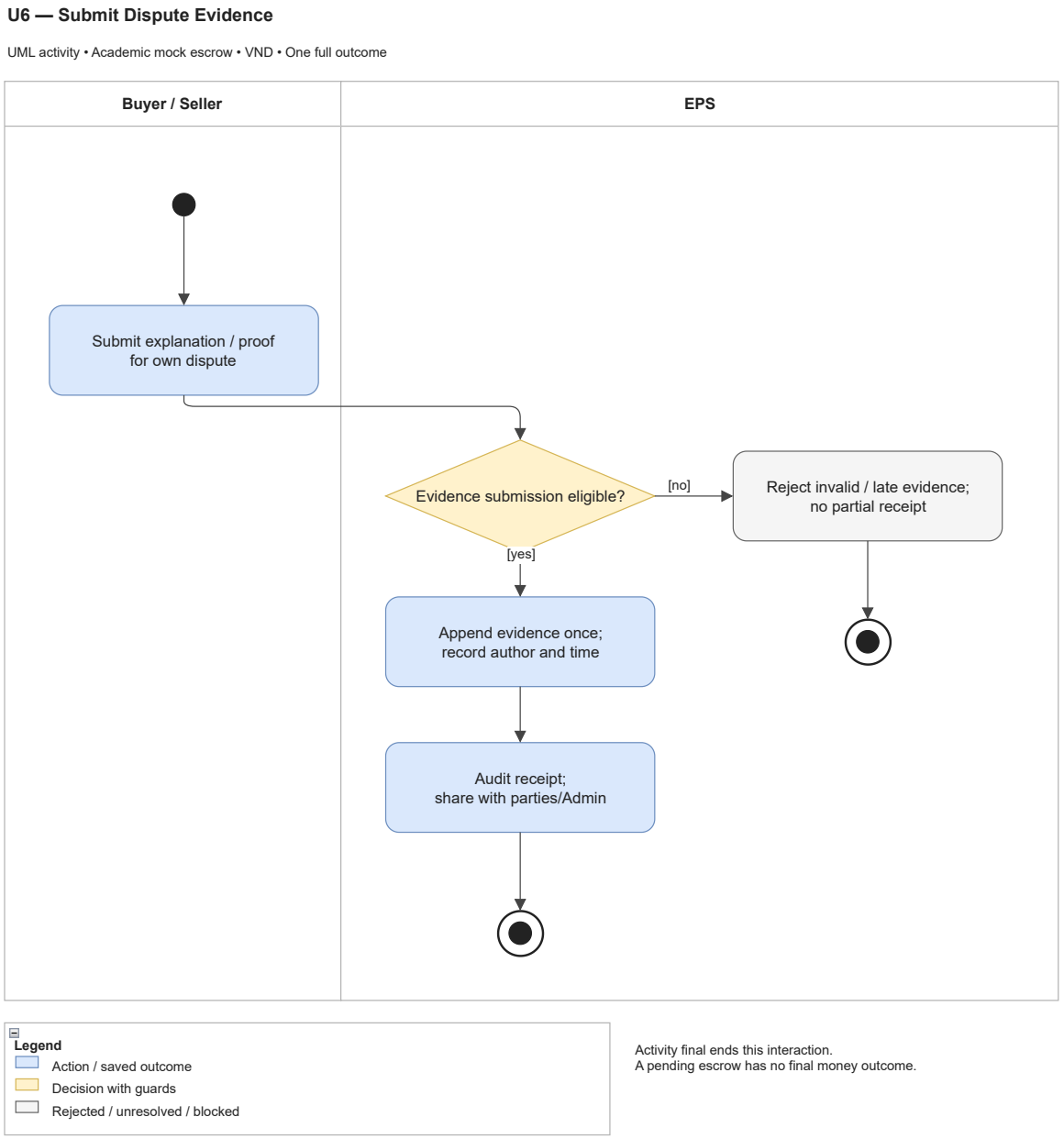


Figure 3.12: U6: Submit Dispute Evidence

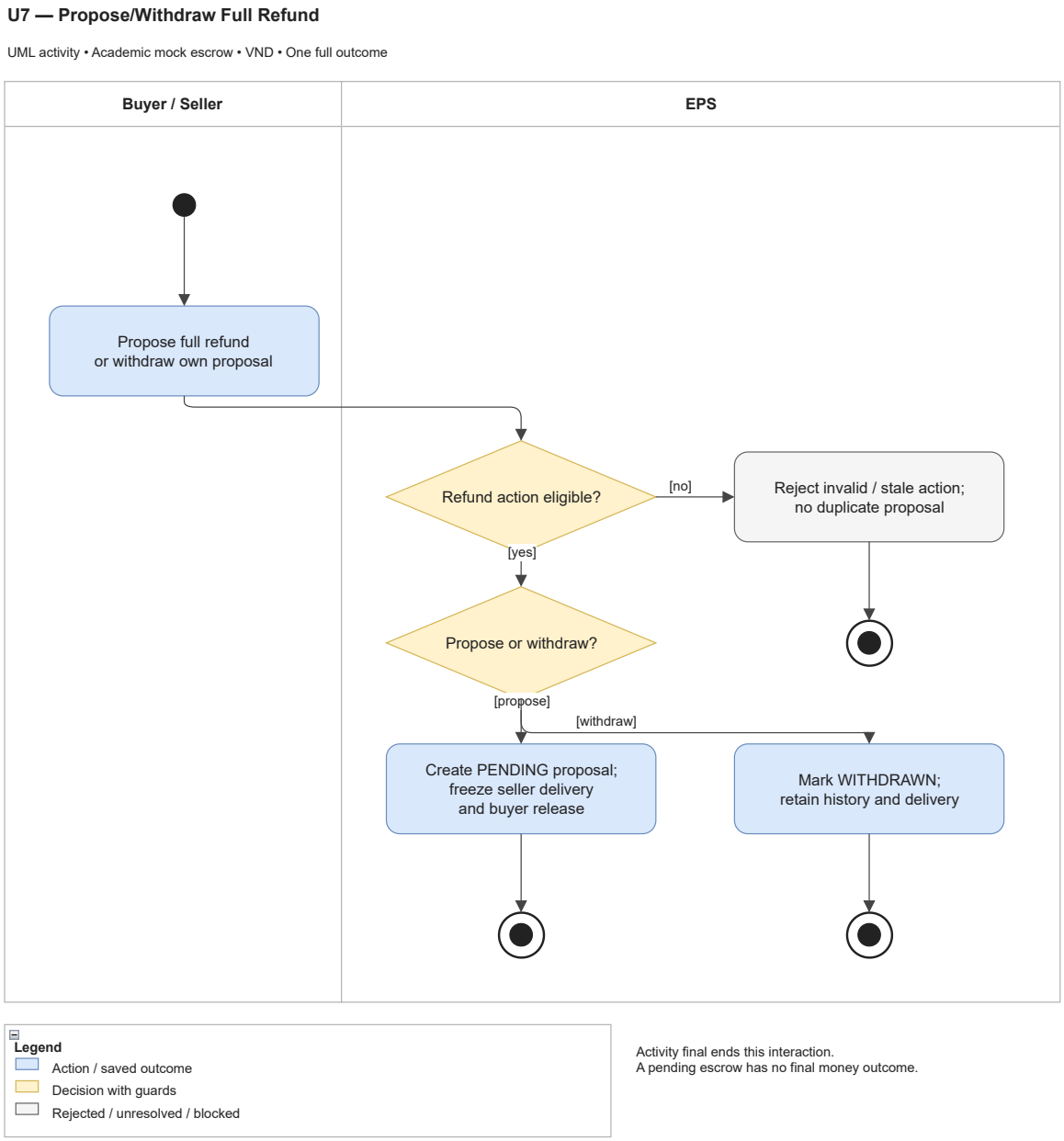


Figure 3.13: U7: Propose/Withdraw Full Refund

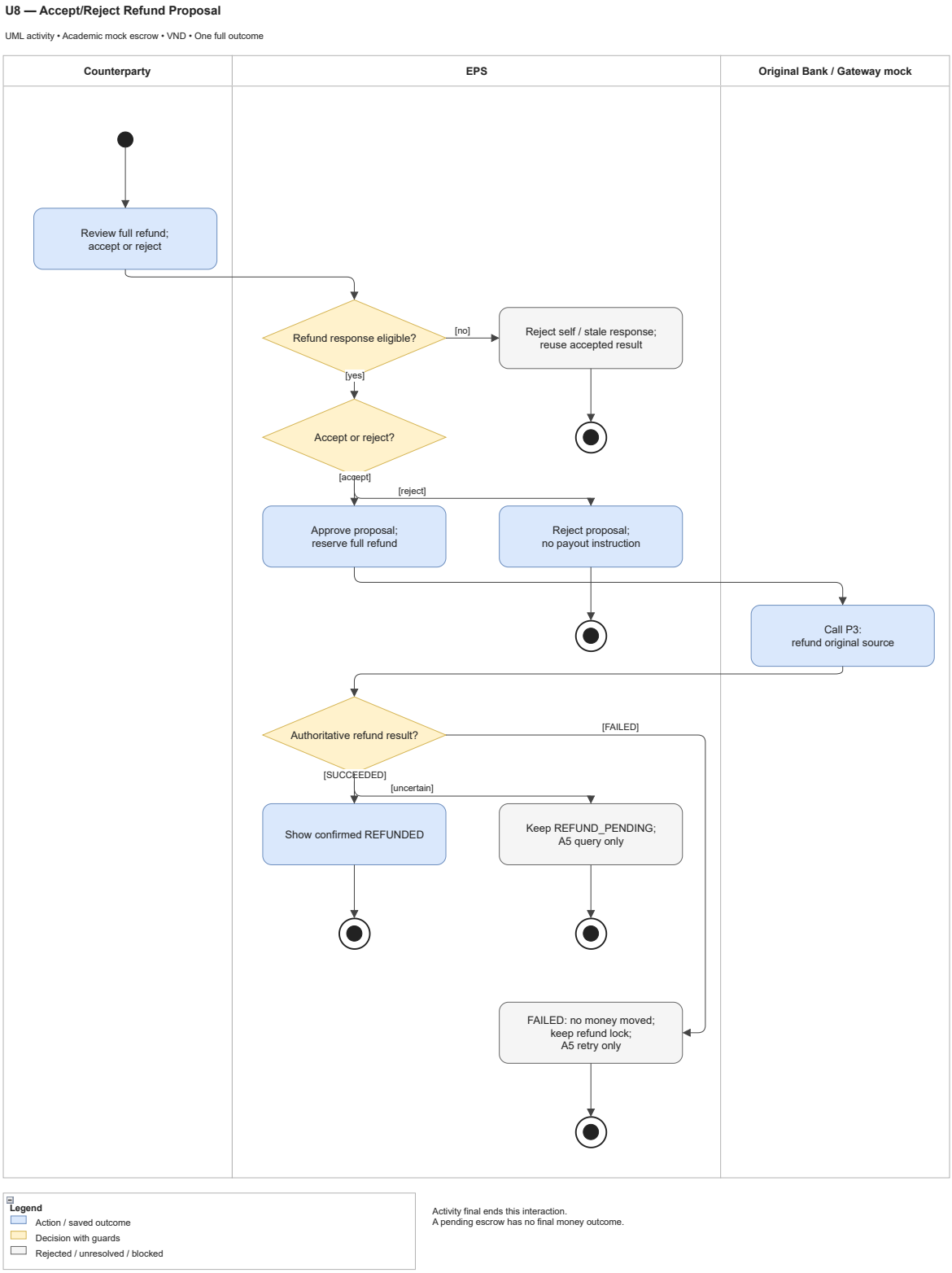


Figure 3.14: U8: Accept/Reject Refund Proposal

3.3.2 Buyer's Use cases



Figure 3.15: B1: Create Escrow

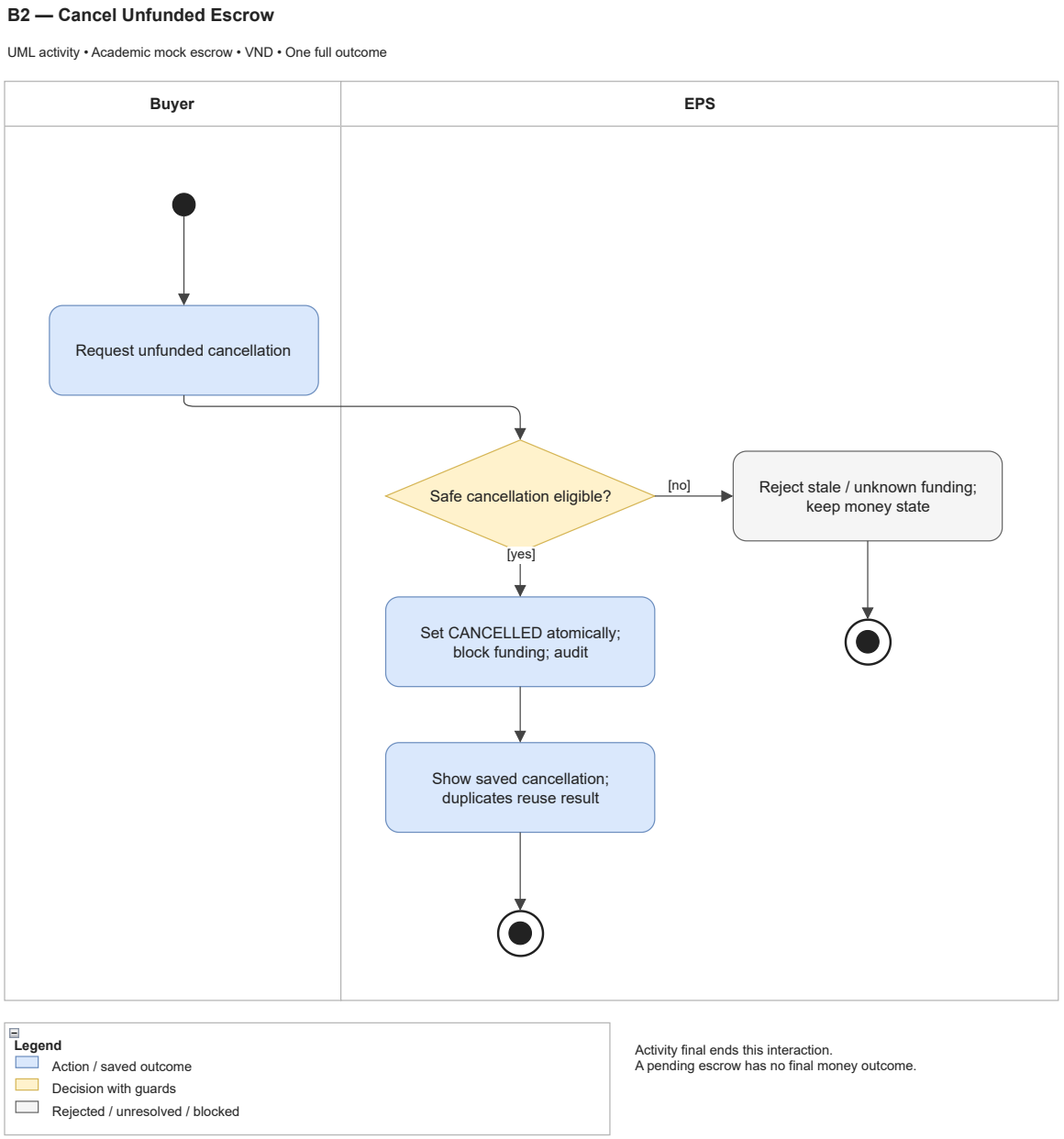


Figure 3.16: B2: Cancel Unfunded Escrow

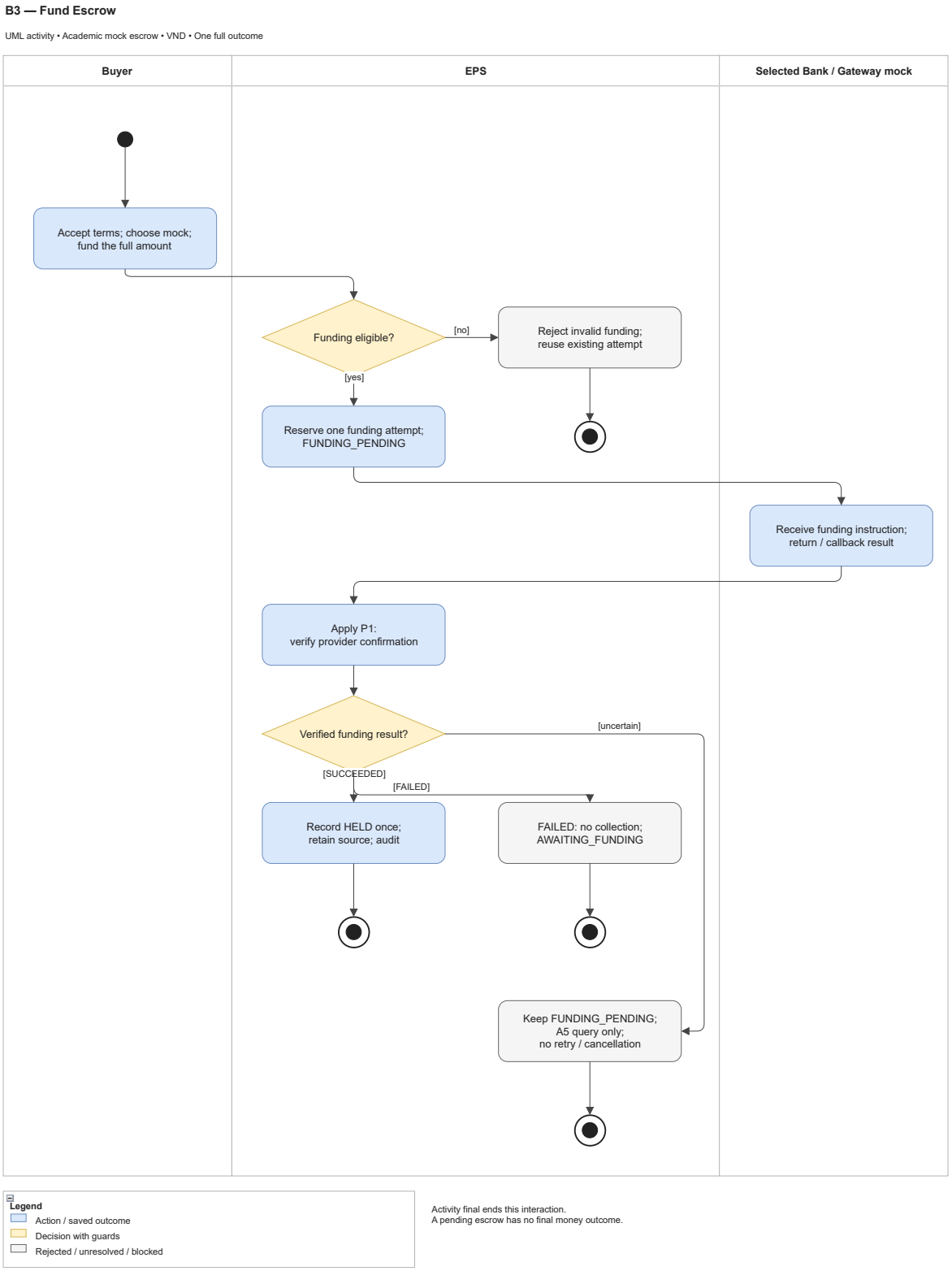


Figure 3.17: B3: Fund Escrow

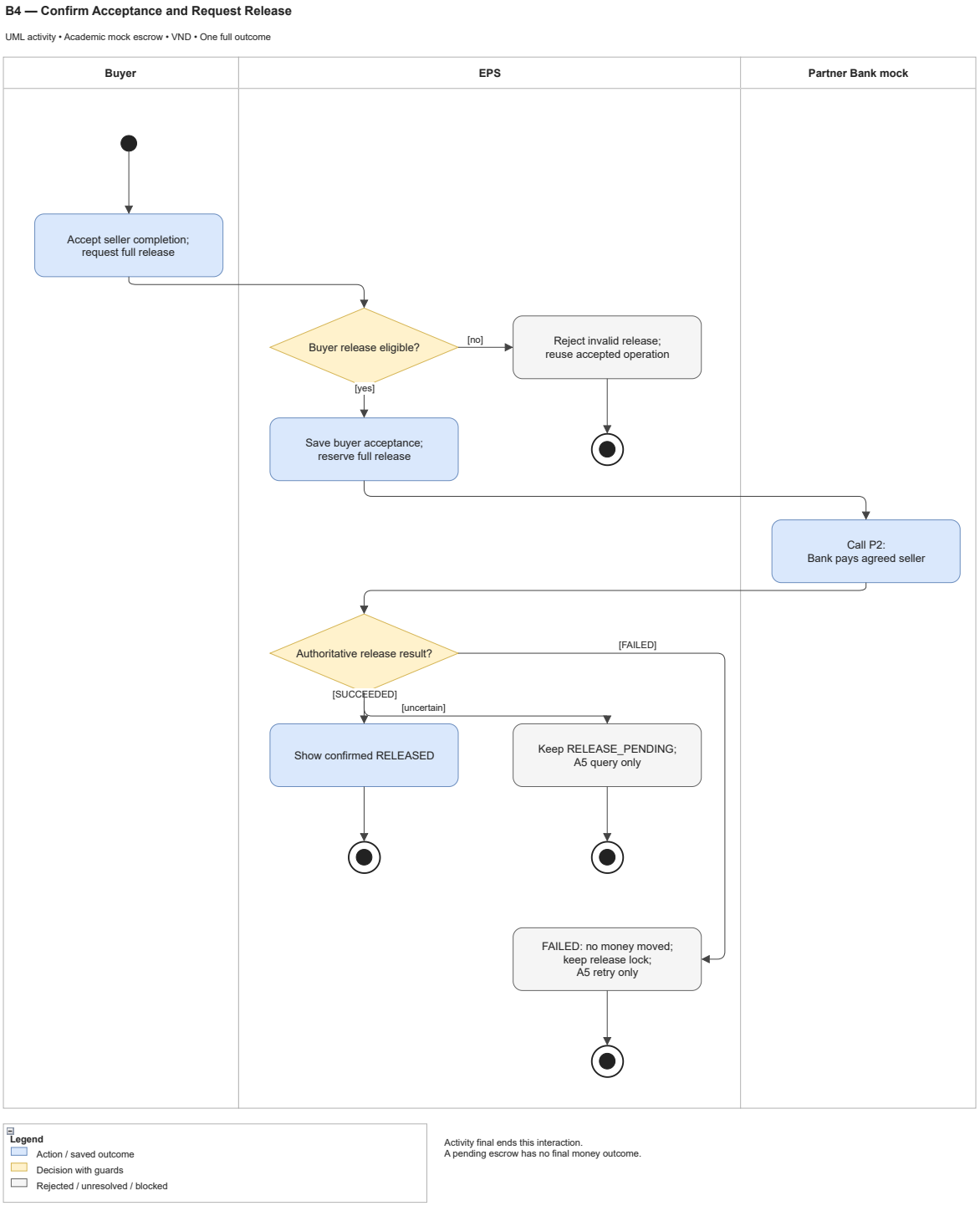


Figure 3.18: B4: Confirm Acceptance and Request Release

3.3.3 Seller's Use cases

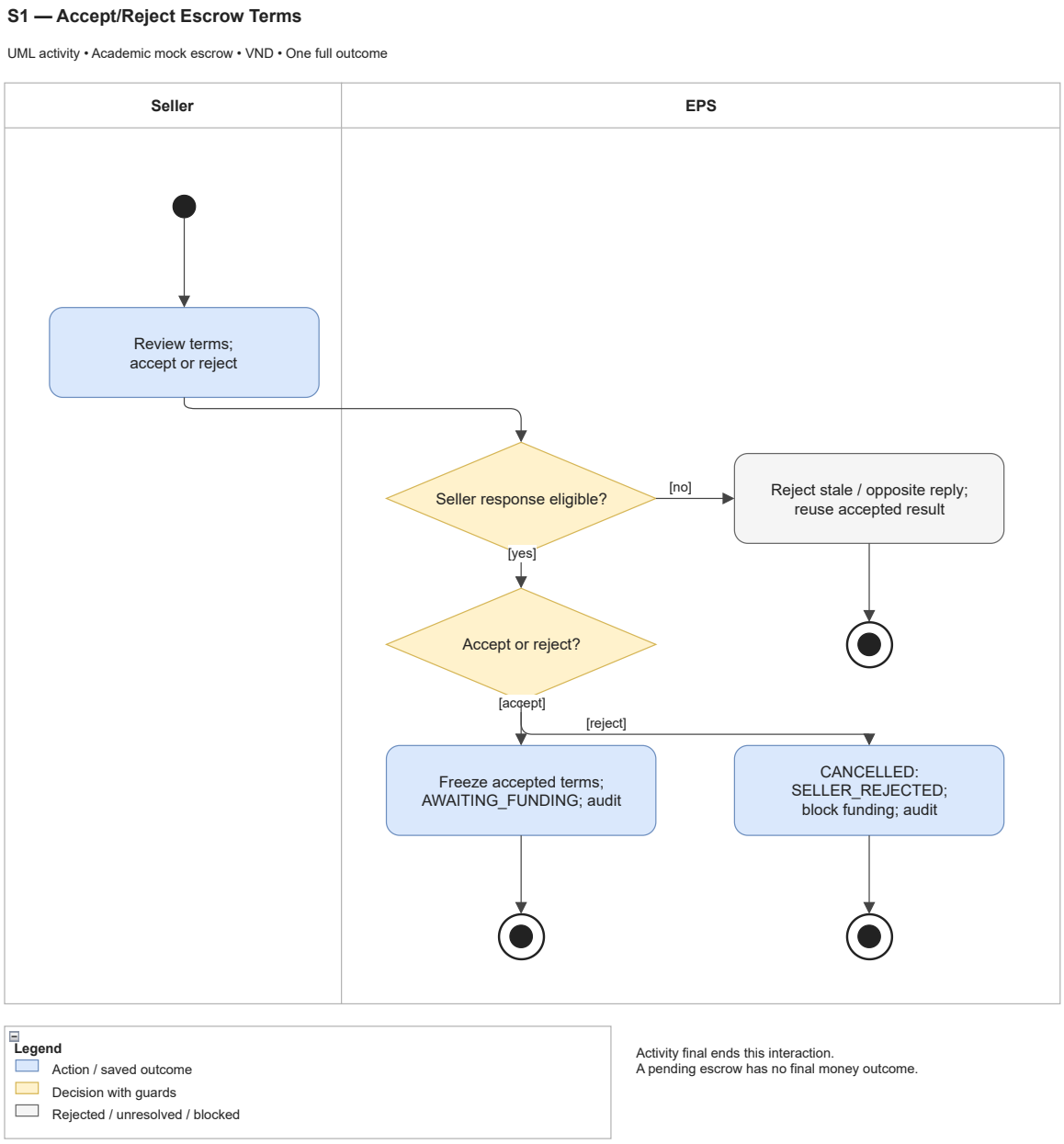


Figure 3.19: S1: Accept/Reject Escrow Terms

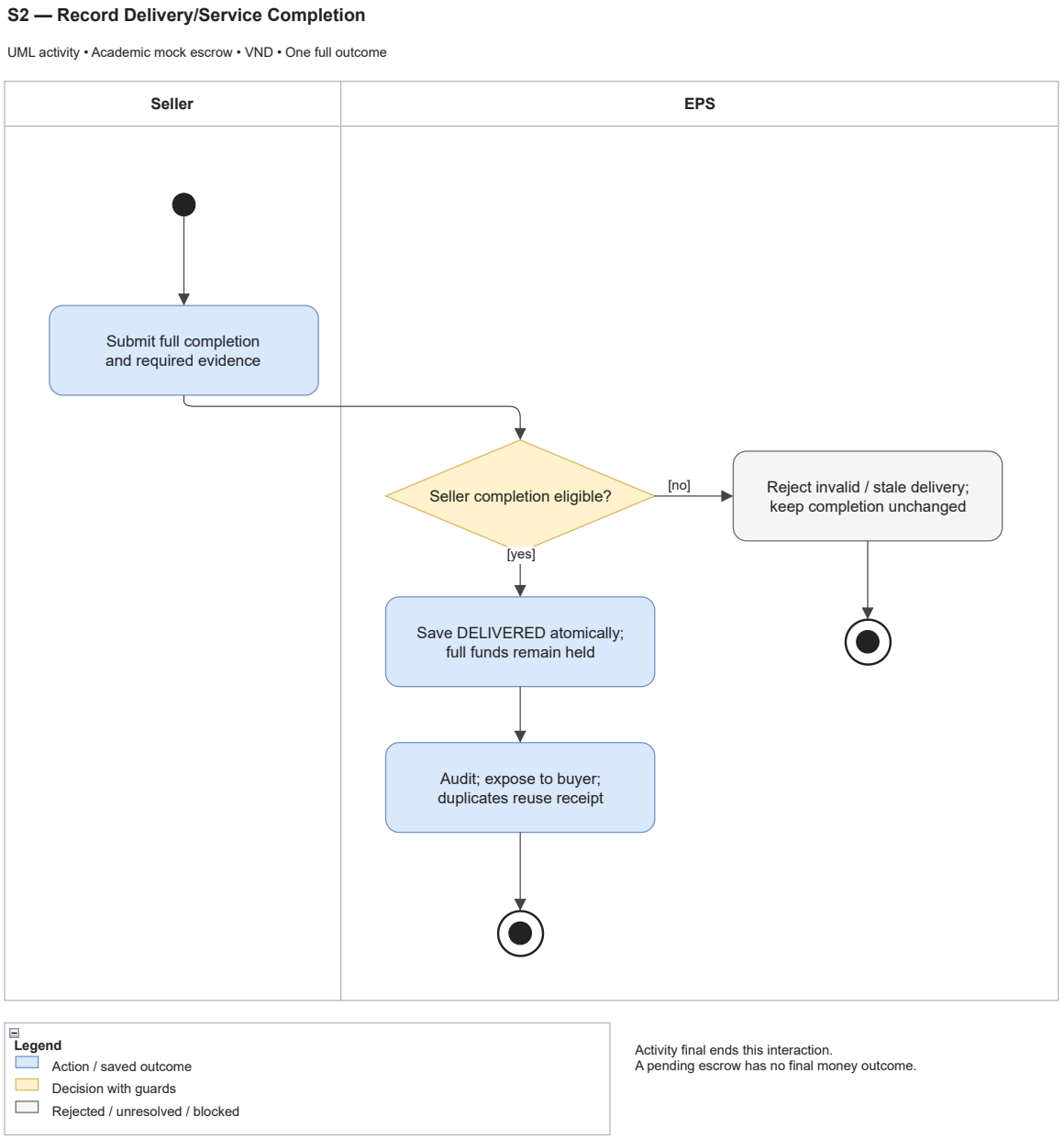


Figure 3.20: S2: Record Delivery/Service Completion

3.3.4 Admin's Use cases

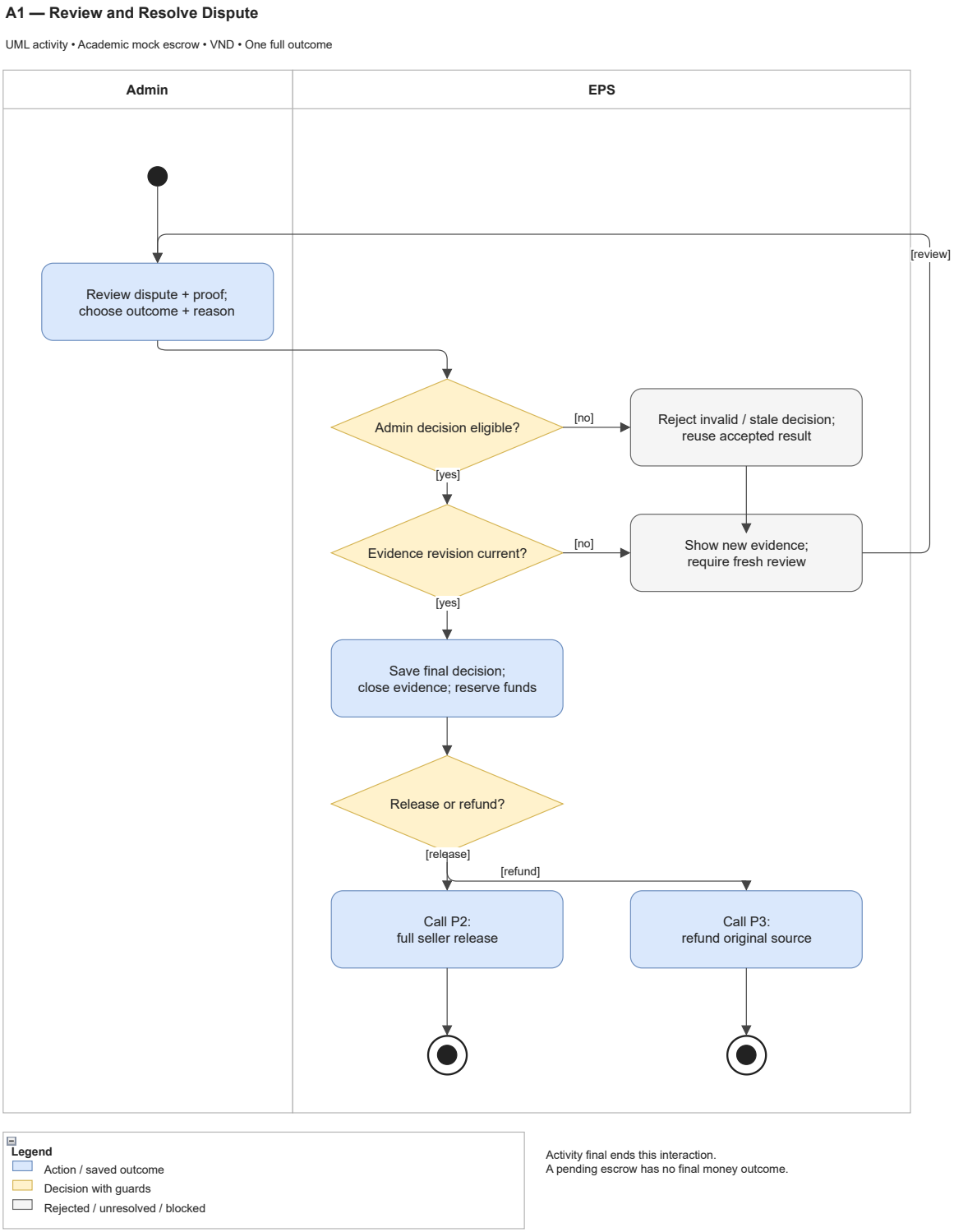


Figure 3.21: A1: Review and Resolve Dispute

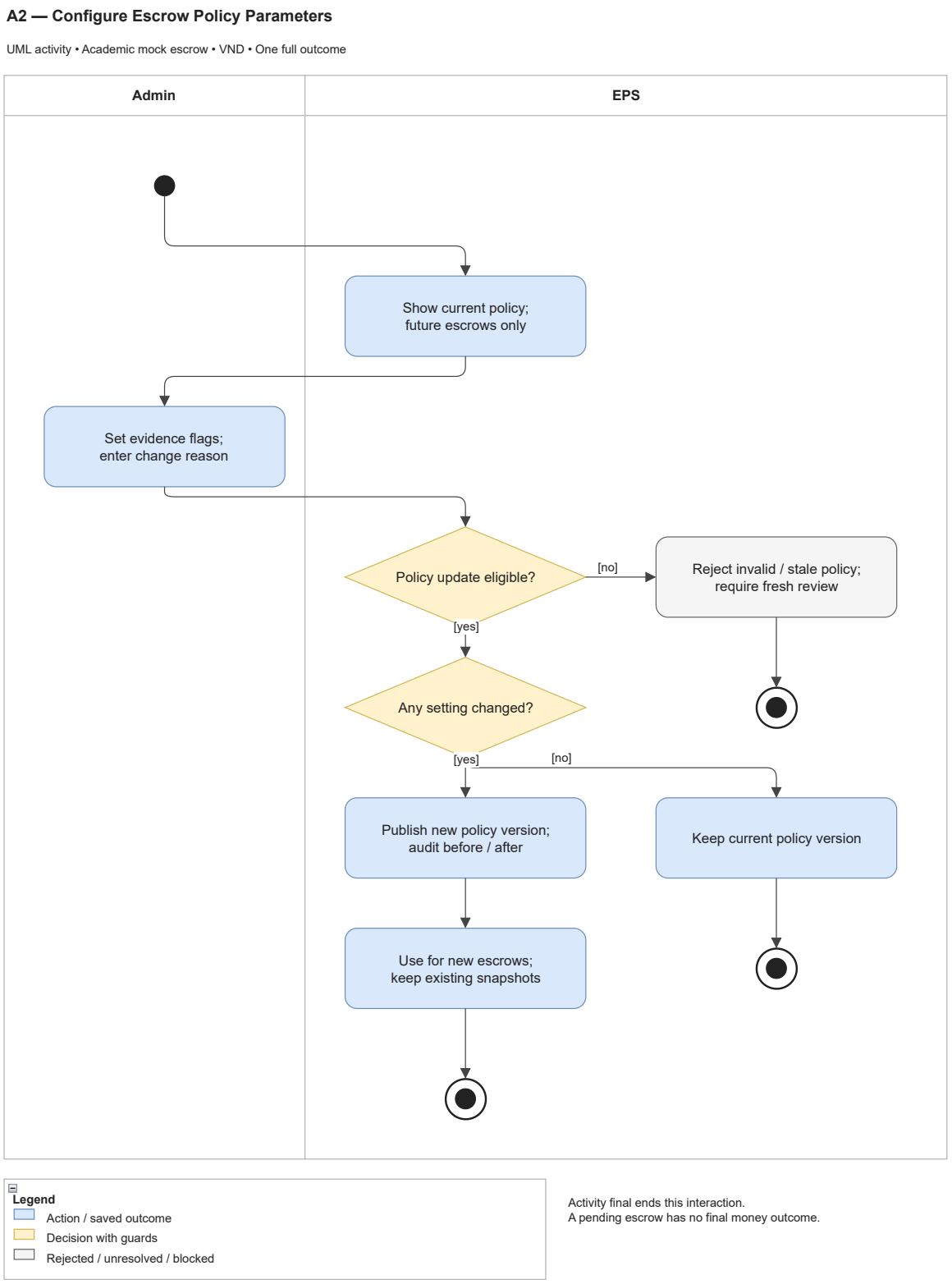


Figure 3.22: A2: Configure Escrow Policy Parameters

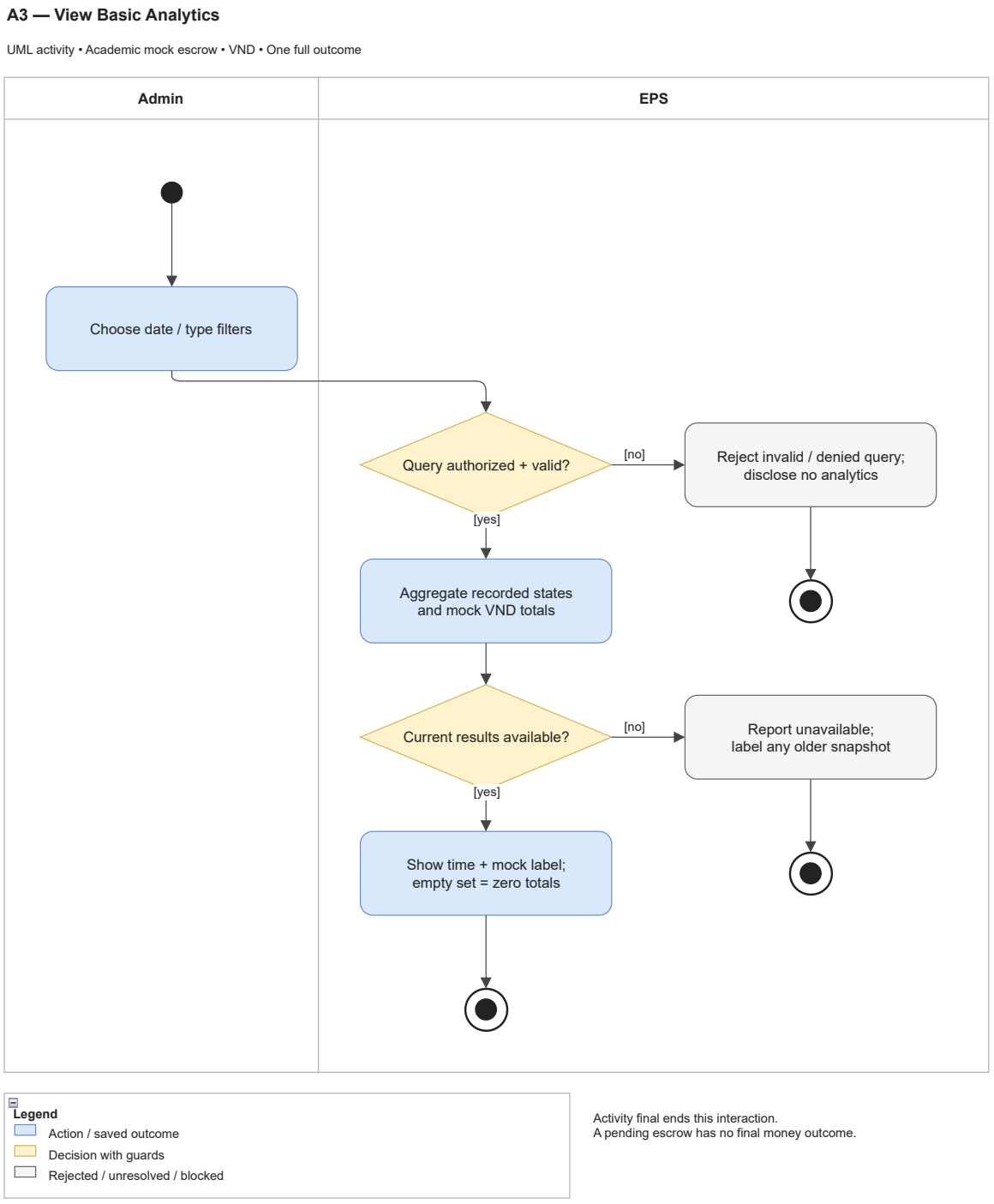


Figure 3.23: A3: View Basic Analytics

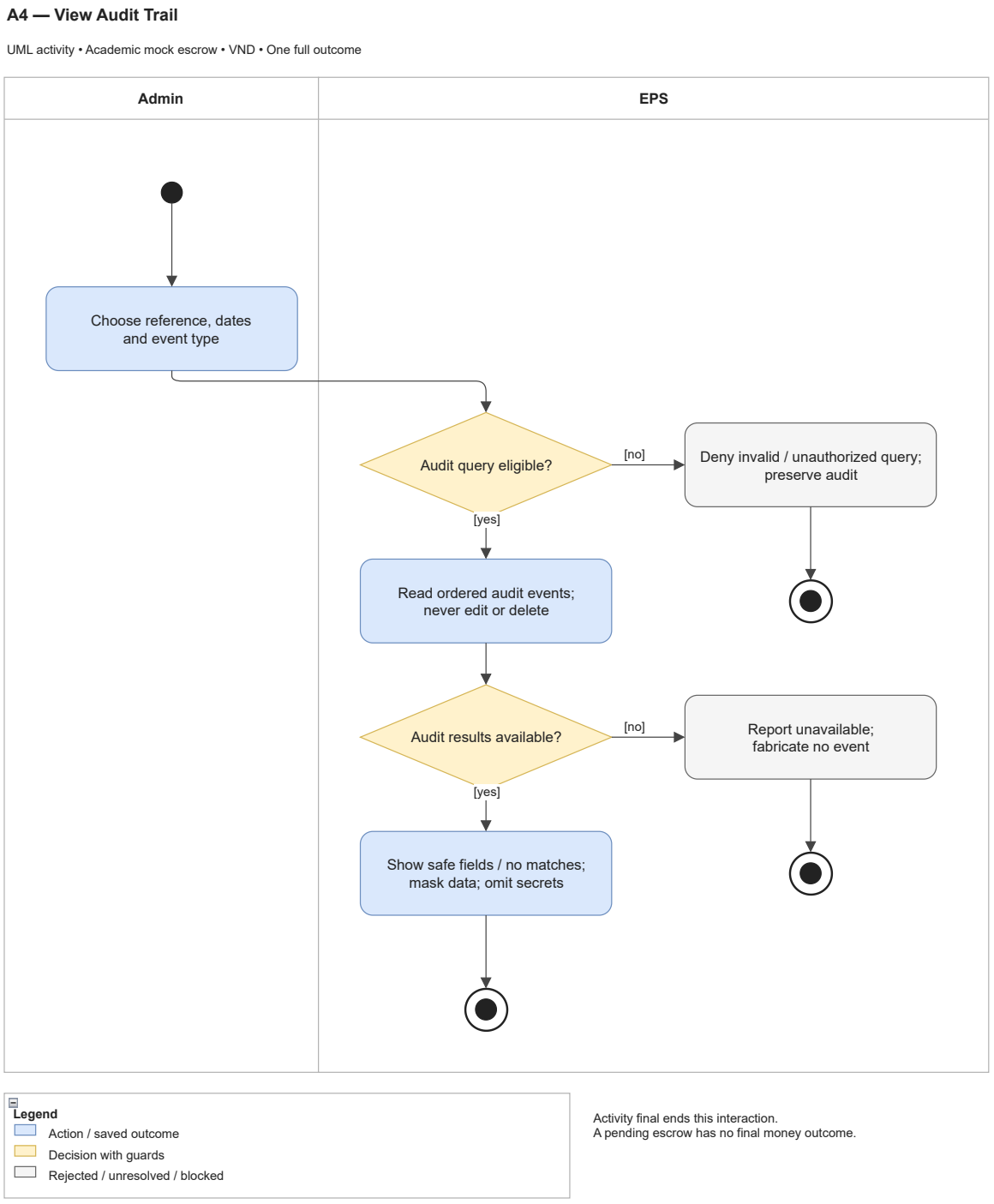


Figure 3.24: A4: View Audit Trail

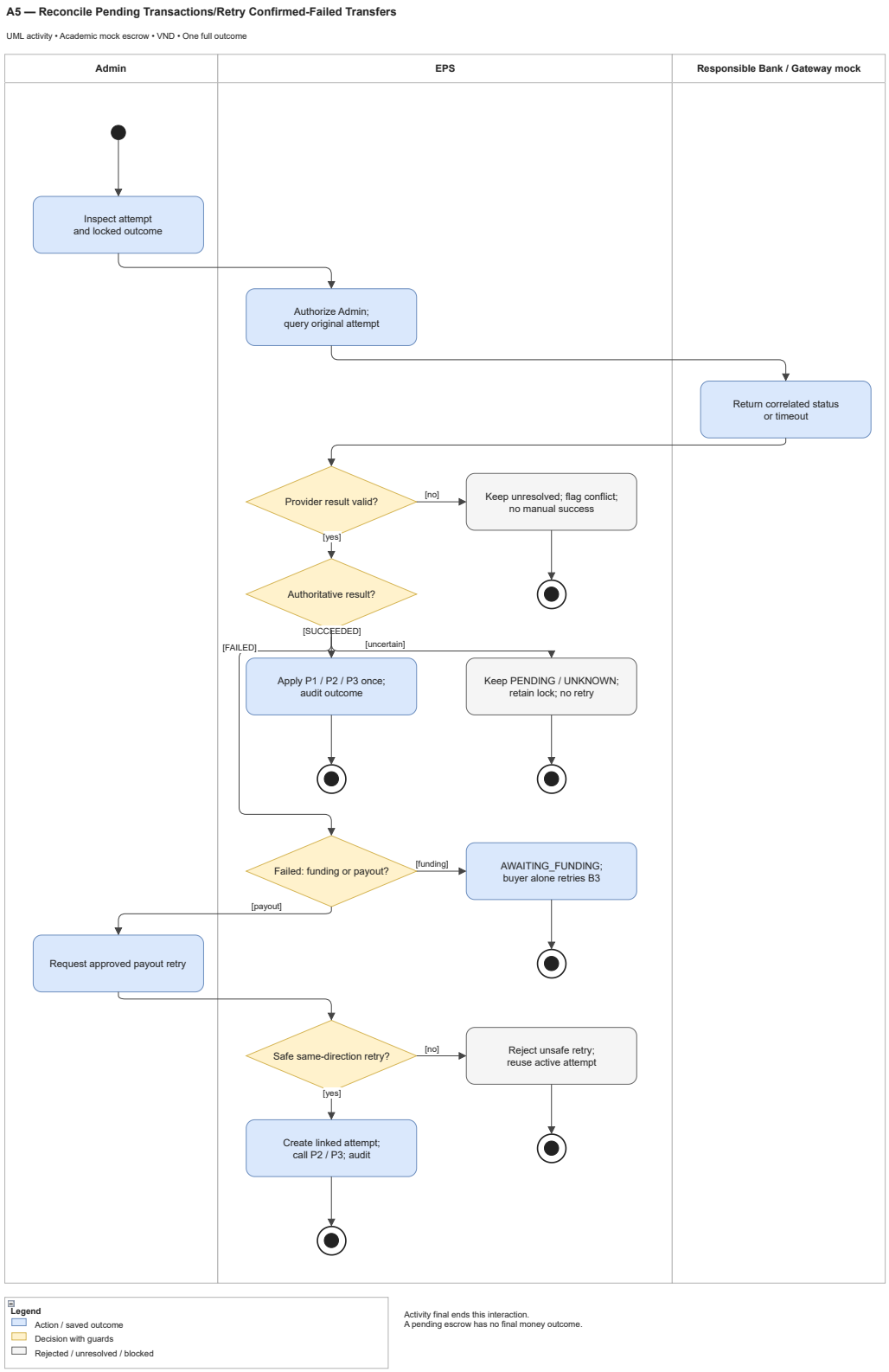


Figure 3.25: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers

3.3.5 Payment Integration's Use cases

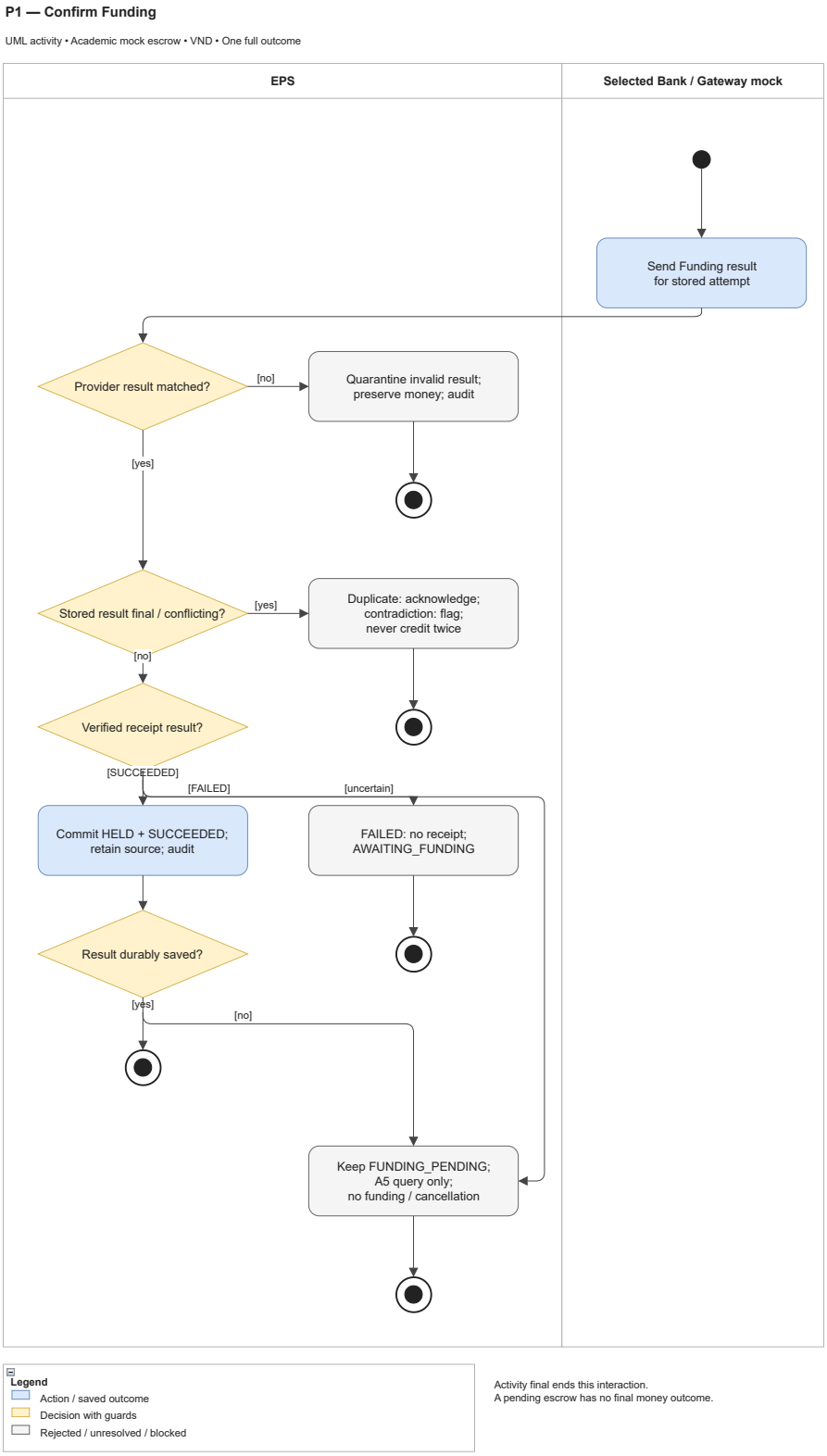


Figure 3.26: P1: Confirm Funding

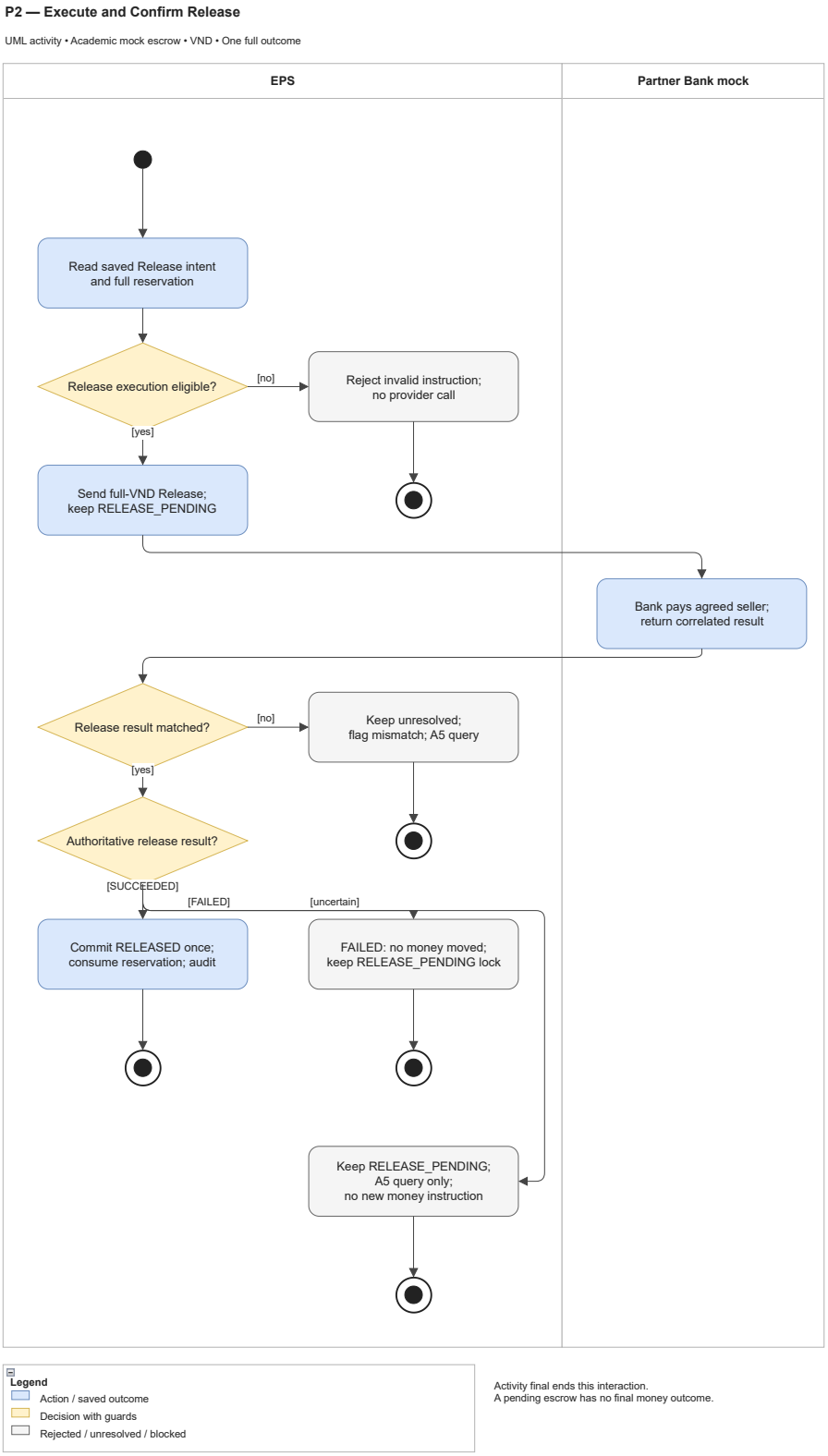


Figure 3.27: P2: Execute and Confirm Release

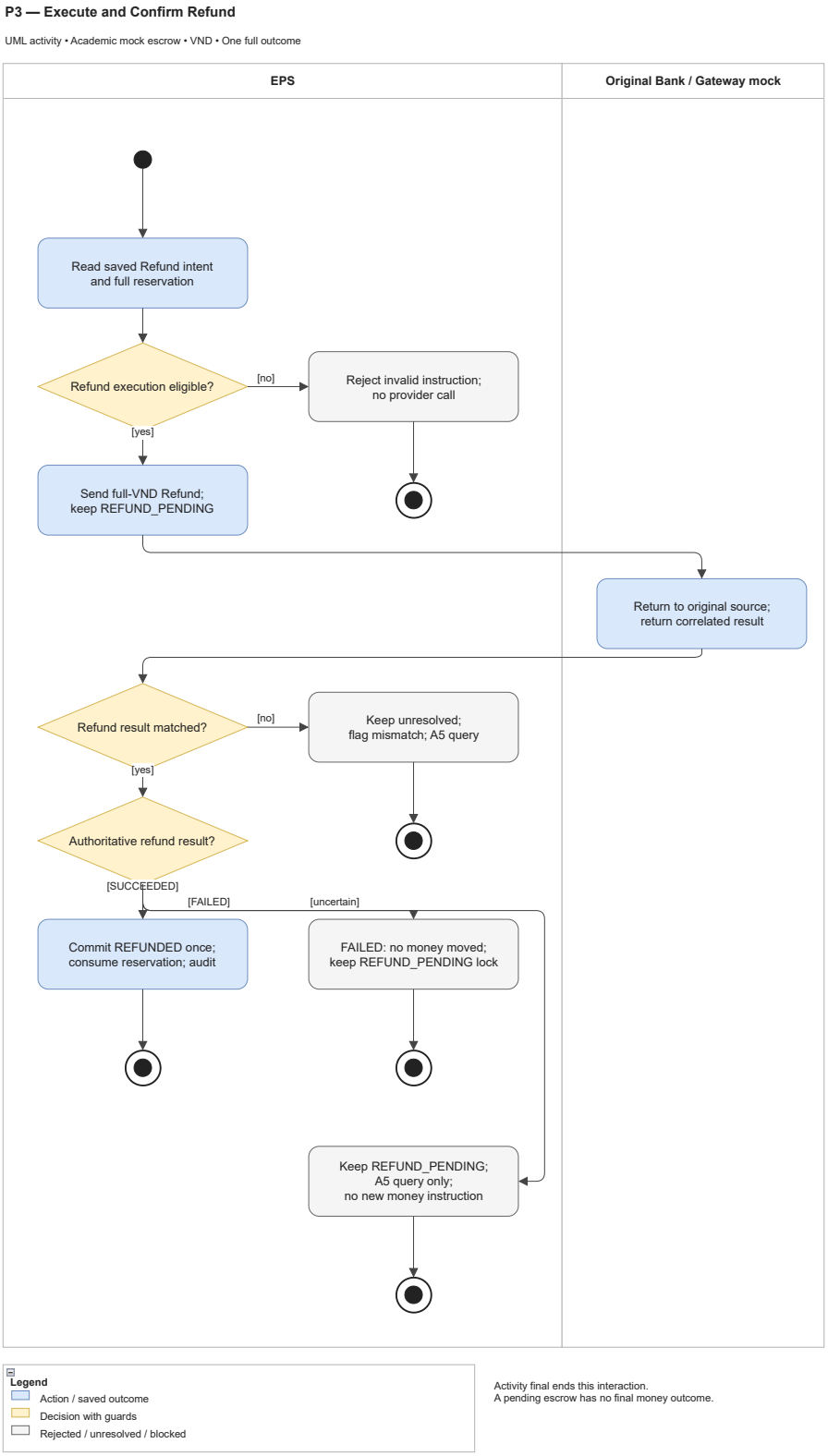


Figure 3.28: P3: Execute and Confirm Refund

3.4 Sequence Diagram

Sequence diagrams show actor, boundary, control and entity interactions. Large flows continue in the referenced modules; B1.API and U4.API use the same business rules as their user-interface variants.

3.4.1 User's Use cases

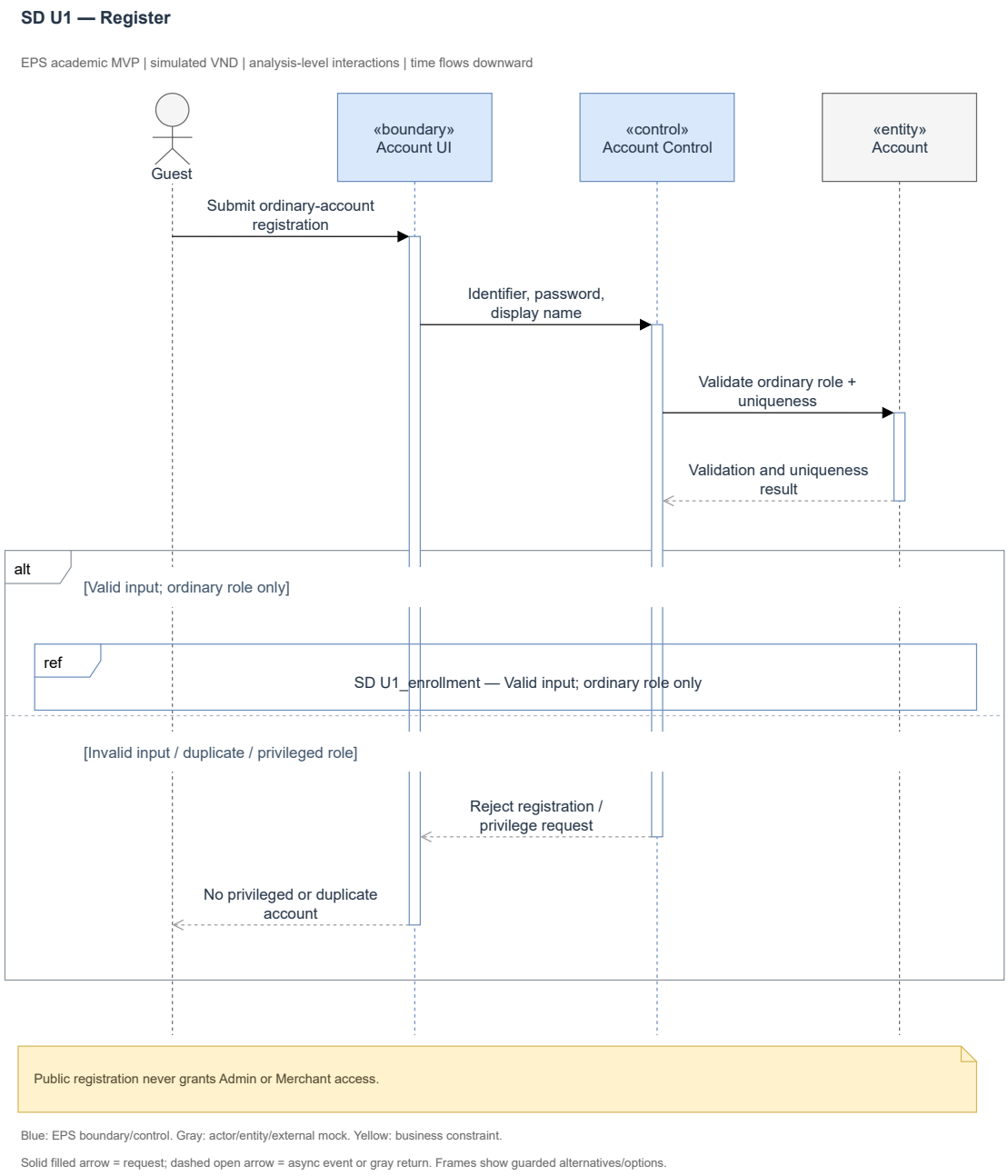


Figure 3.29: U1: Register

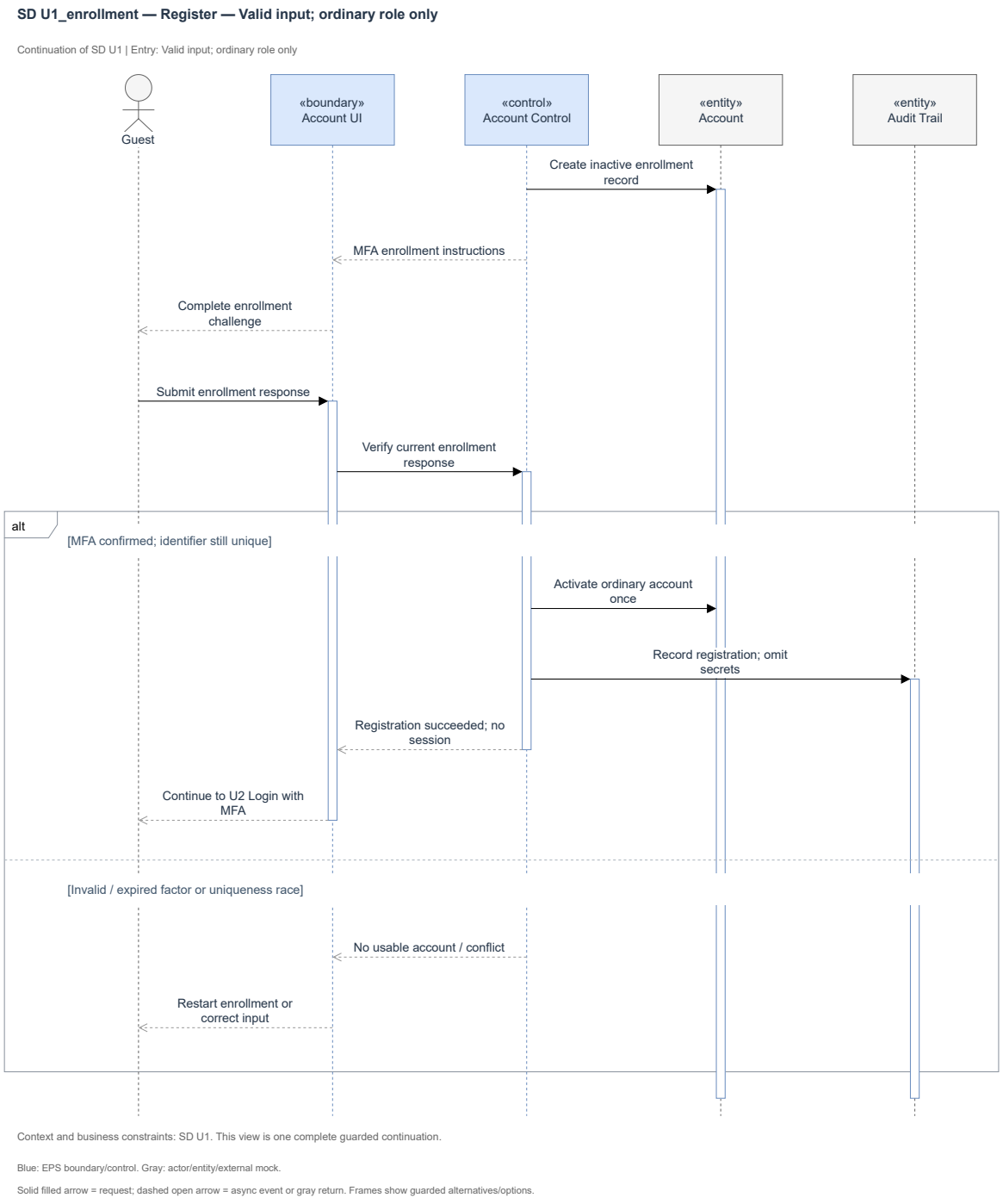


Figure 3.30: U1: Register (continued: U1_enrollment)

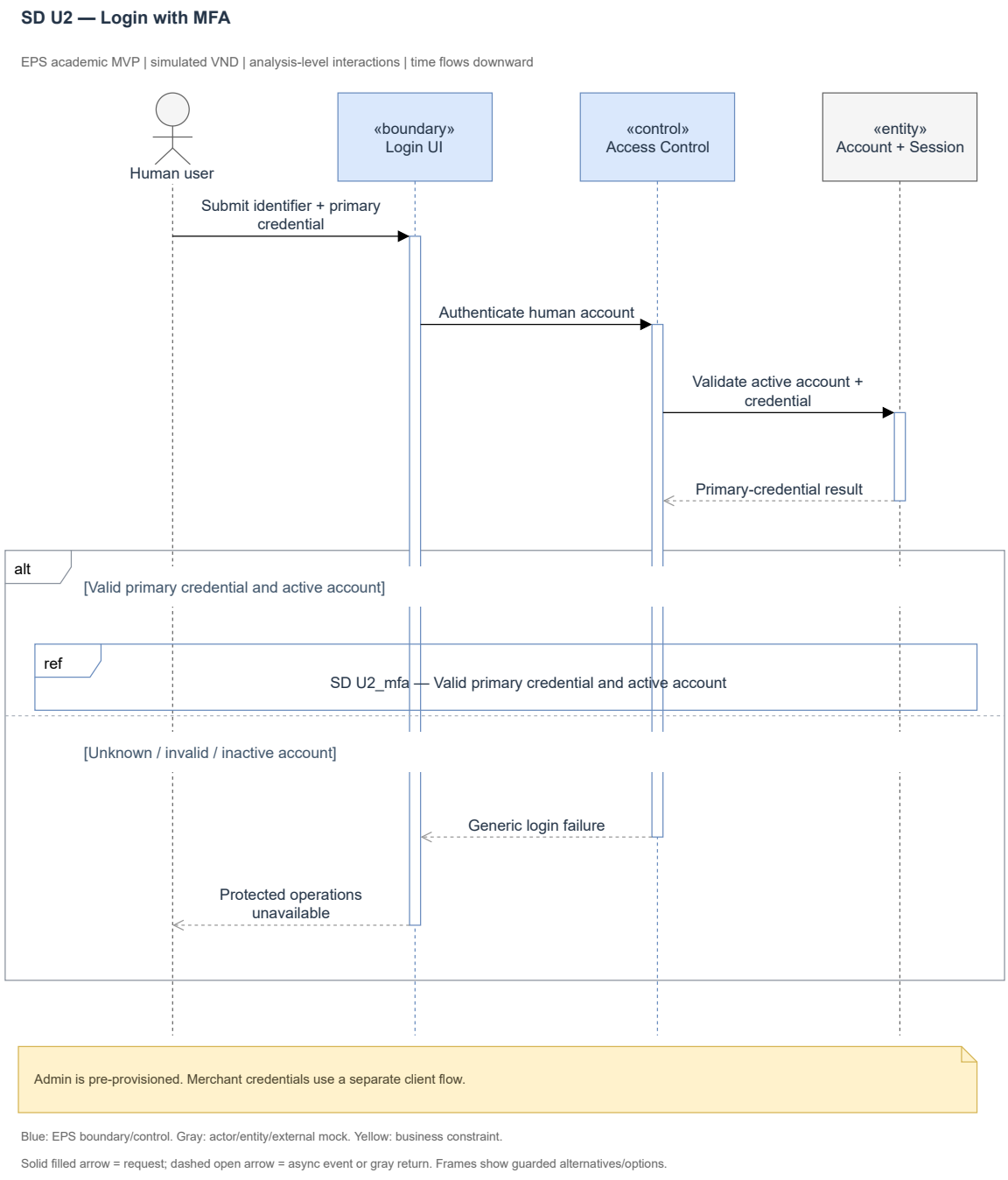


Figure 3.31: U2: Login with MFA

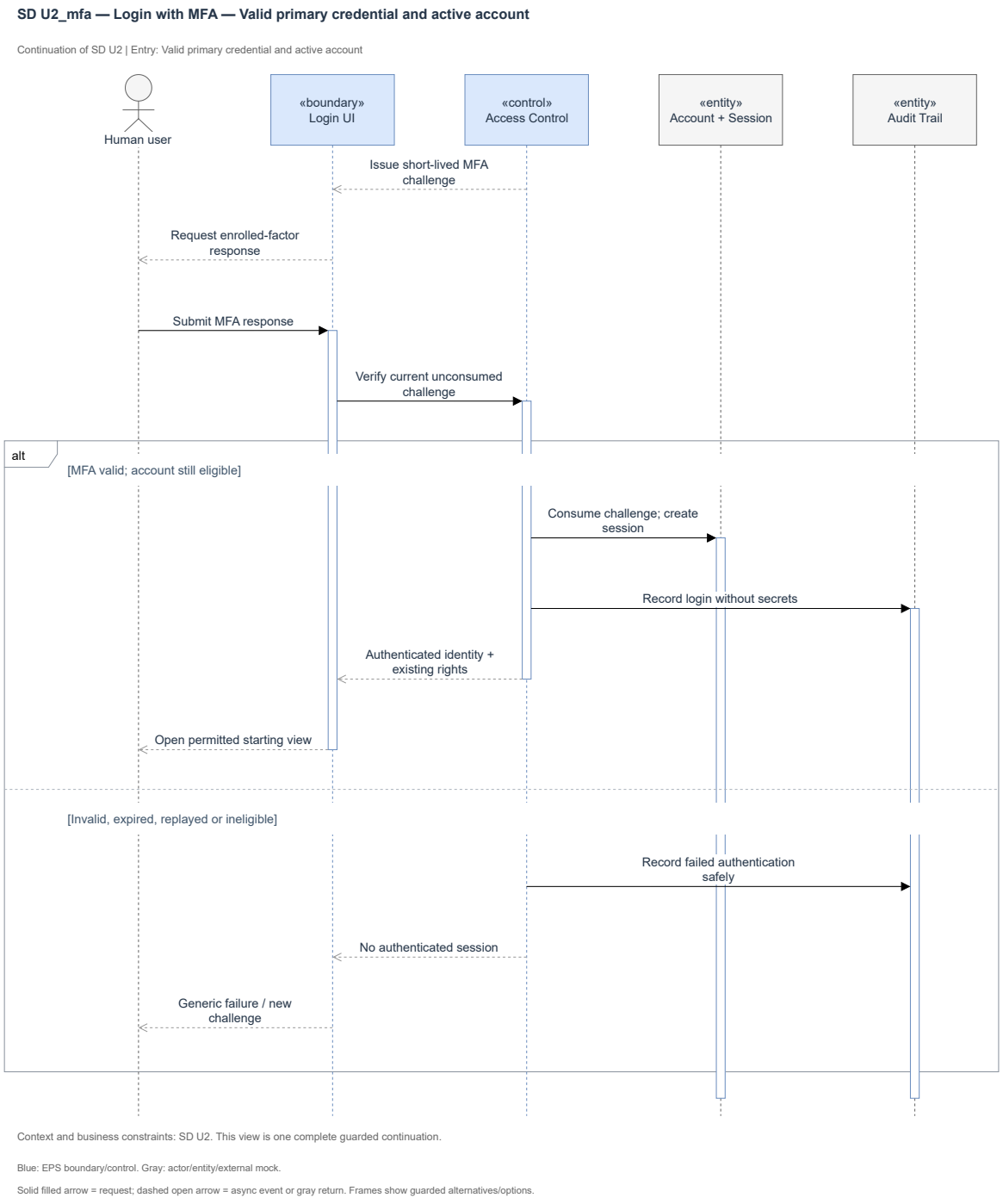


Figure 3.32: U2: Login with MFA (continued: U2_mfa)

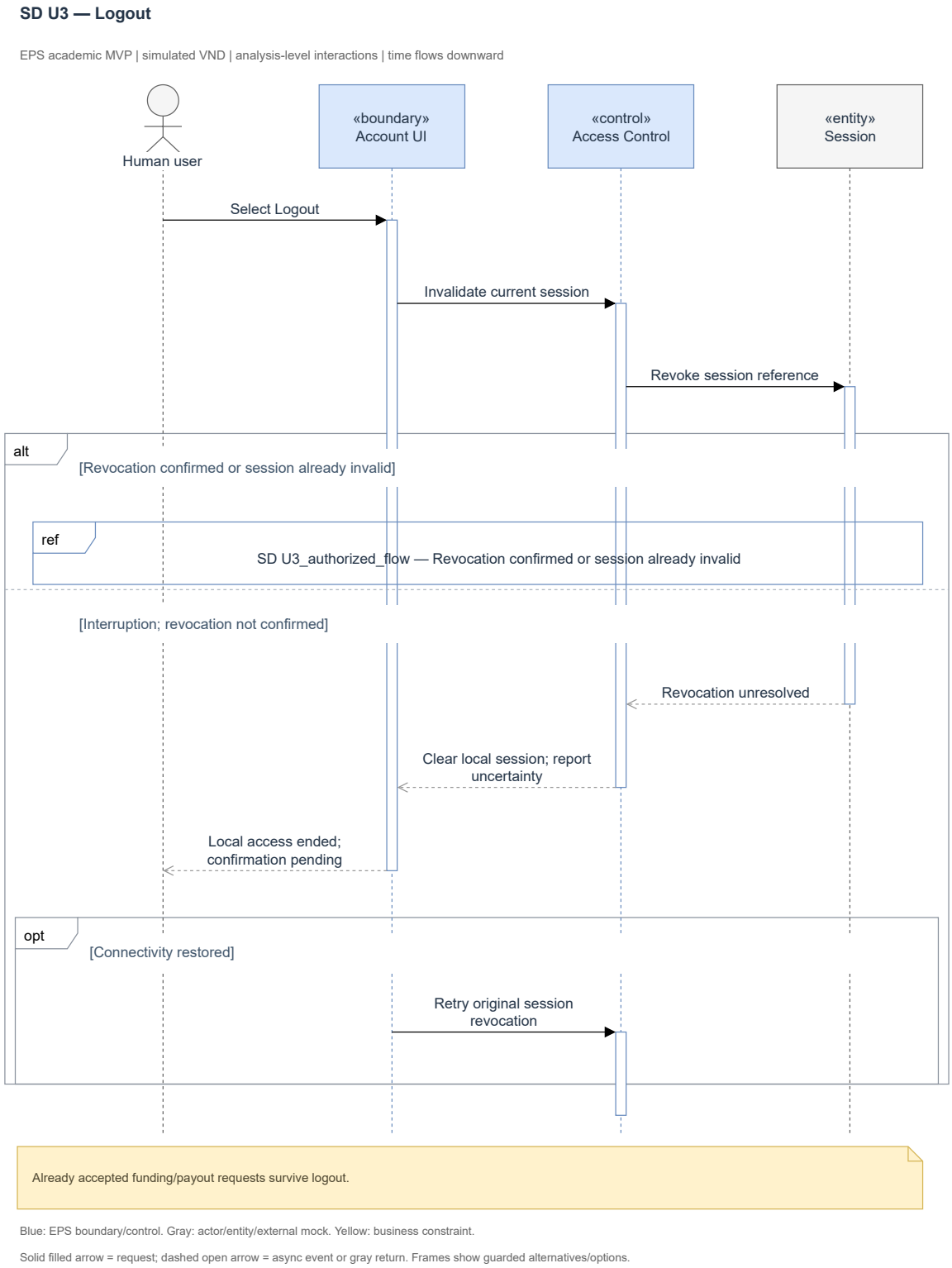


Figure 3.33: U3: Logout

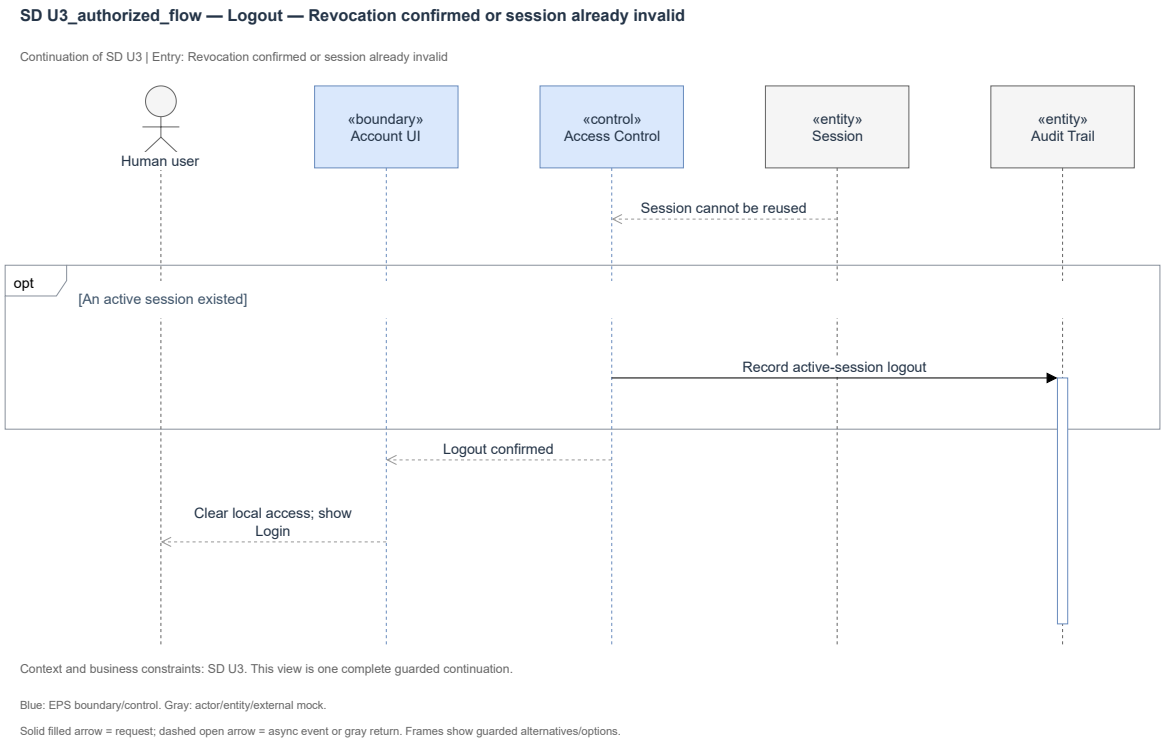


Figure 3.34: U3: Logout (continued: U3_authorized_flow)

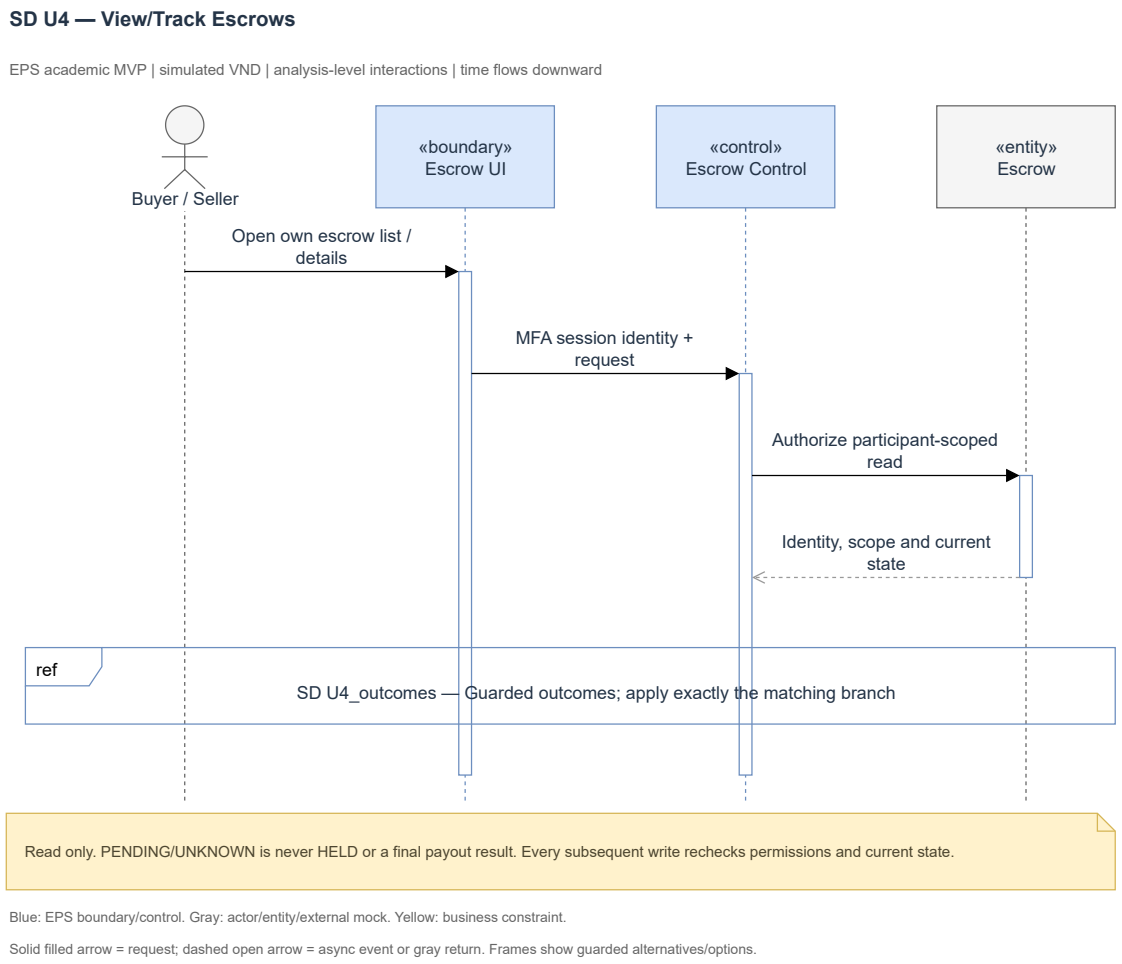


Figure 3.35: U4: View/Track Escrows

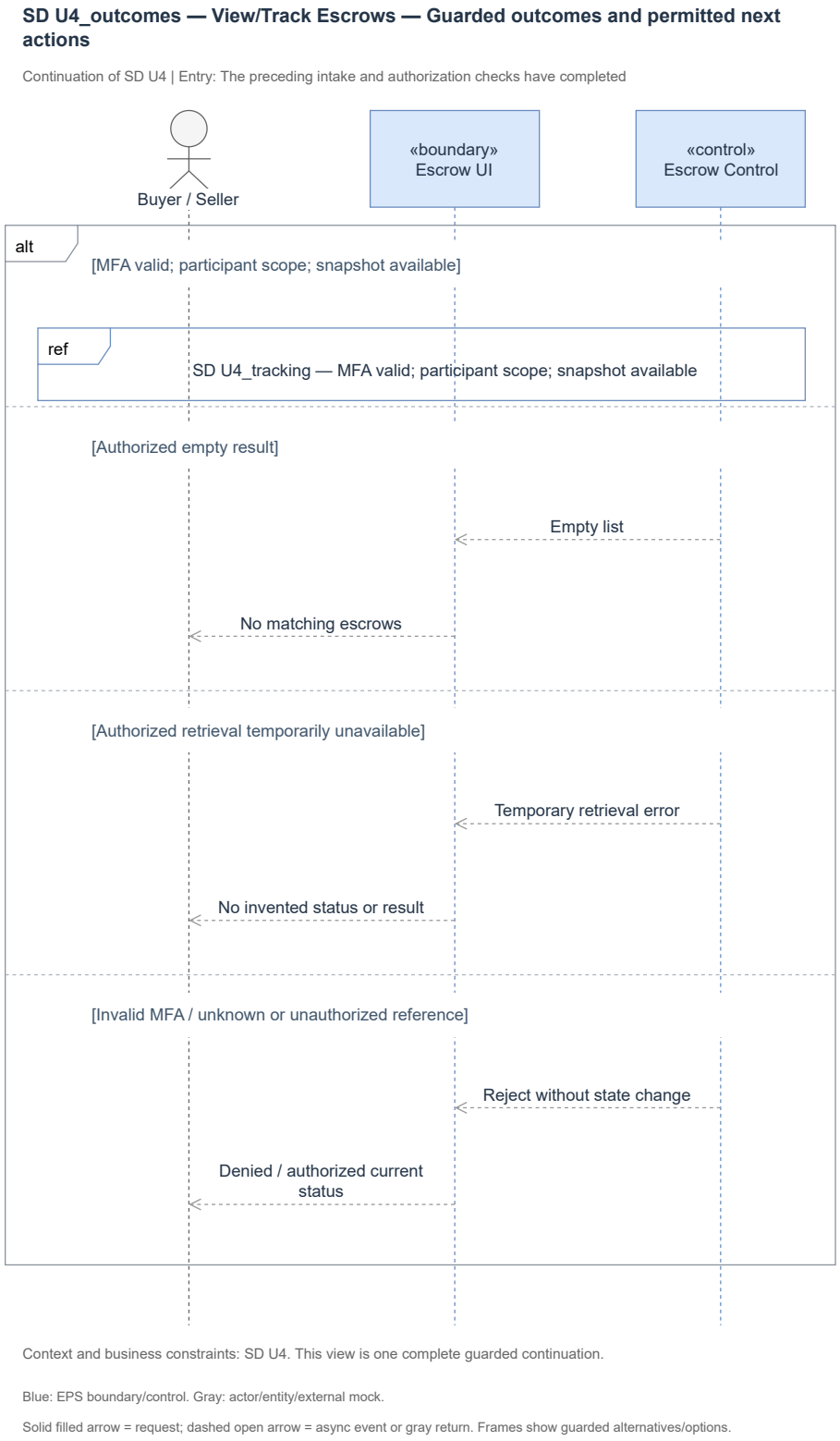


Figure 3.36: U4: View/Track Escrows (continued: U4_outcomes)

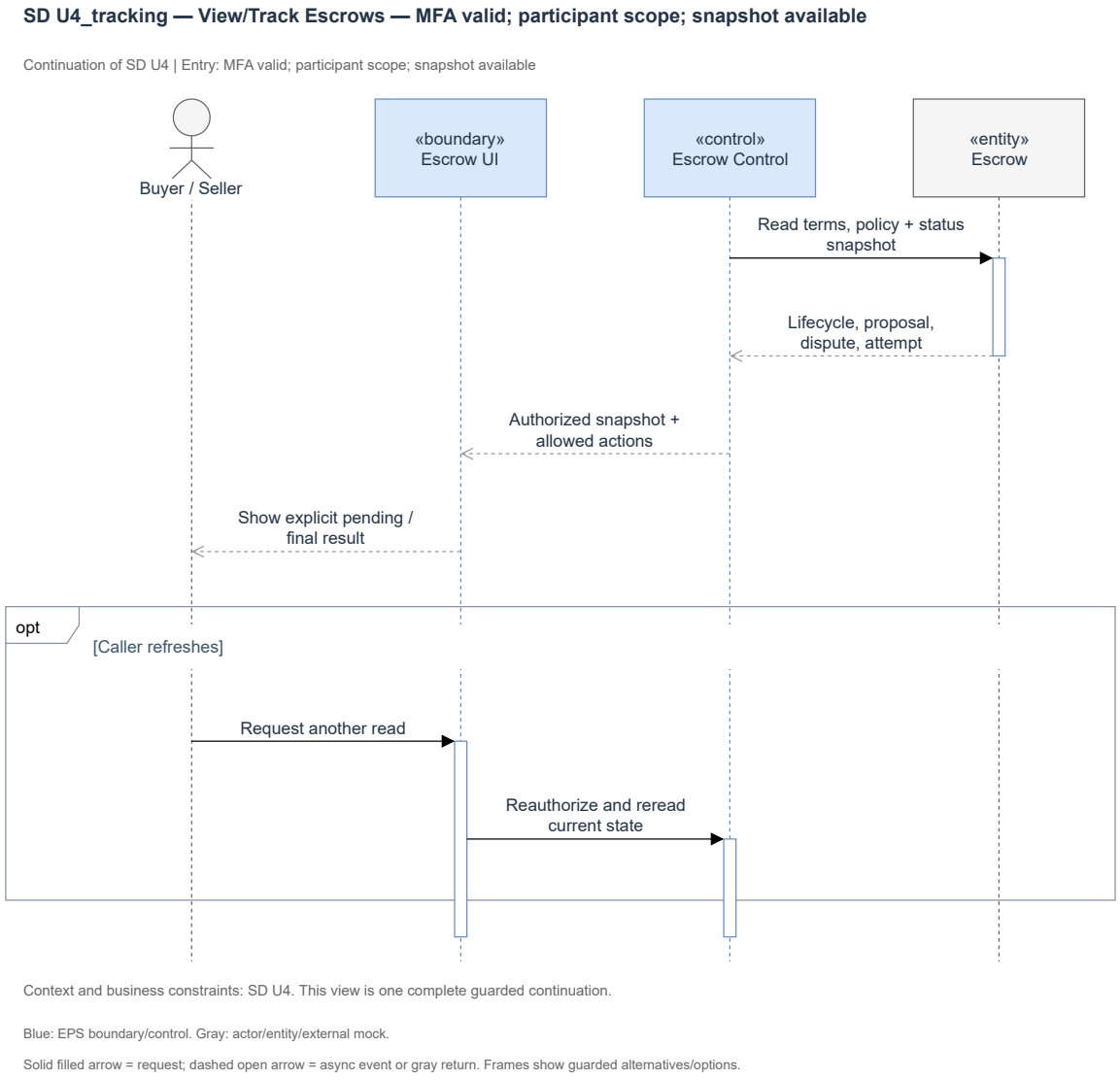


Figure 3.37: U4: View/Track Escrows (continued: U4_tracking)

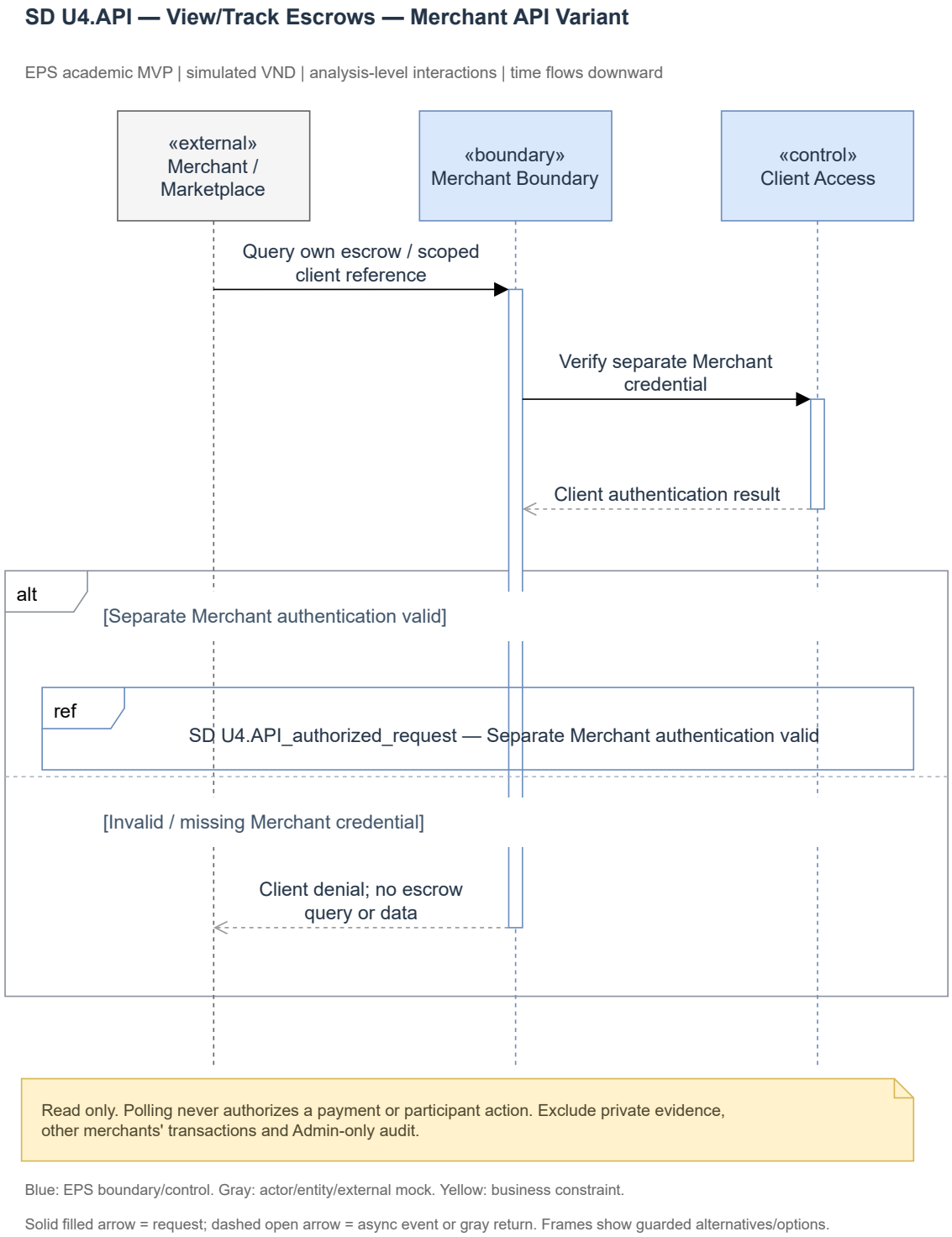


Figure 3.38: U4.API: View/Track Escrows

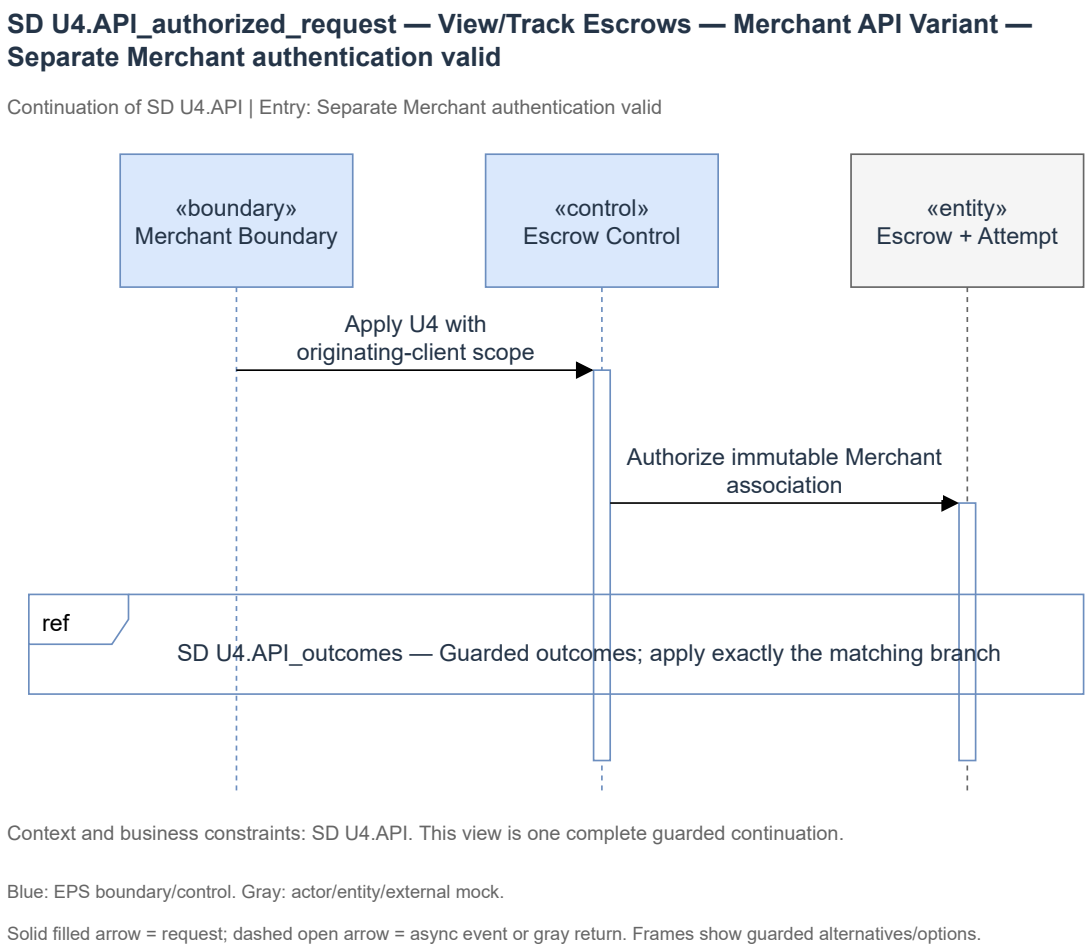


Figure 3.39: U4.API: View/Track Escrows (continued: U4.API_authorized_request)

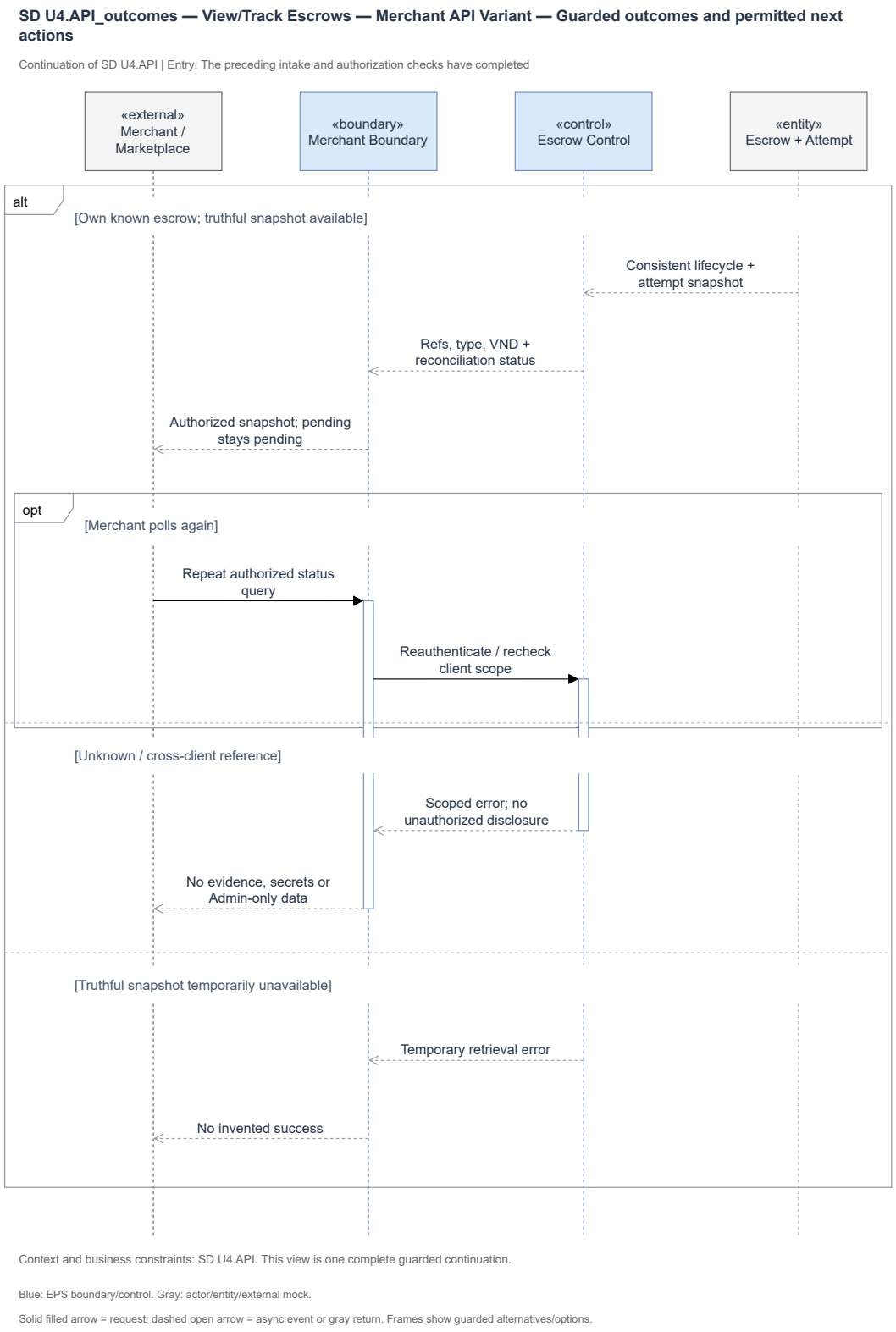


Figure 3.40: U4.API: View/Track Escrows (continued: U4.API_outcomes)

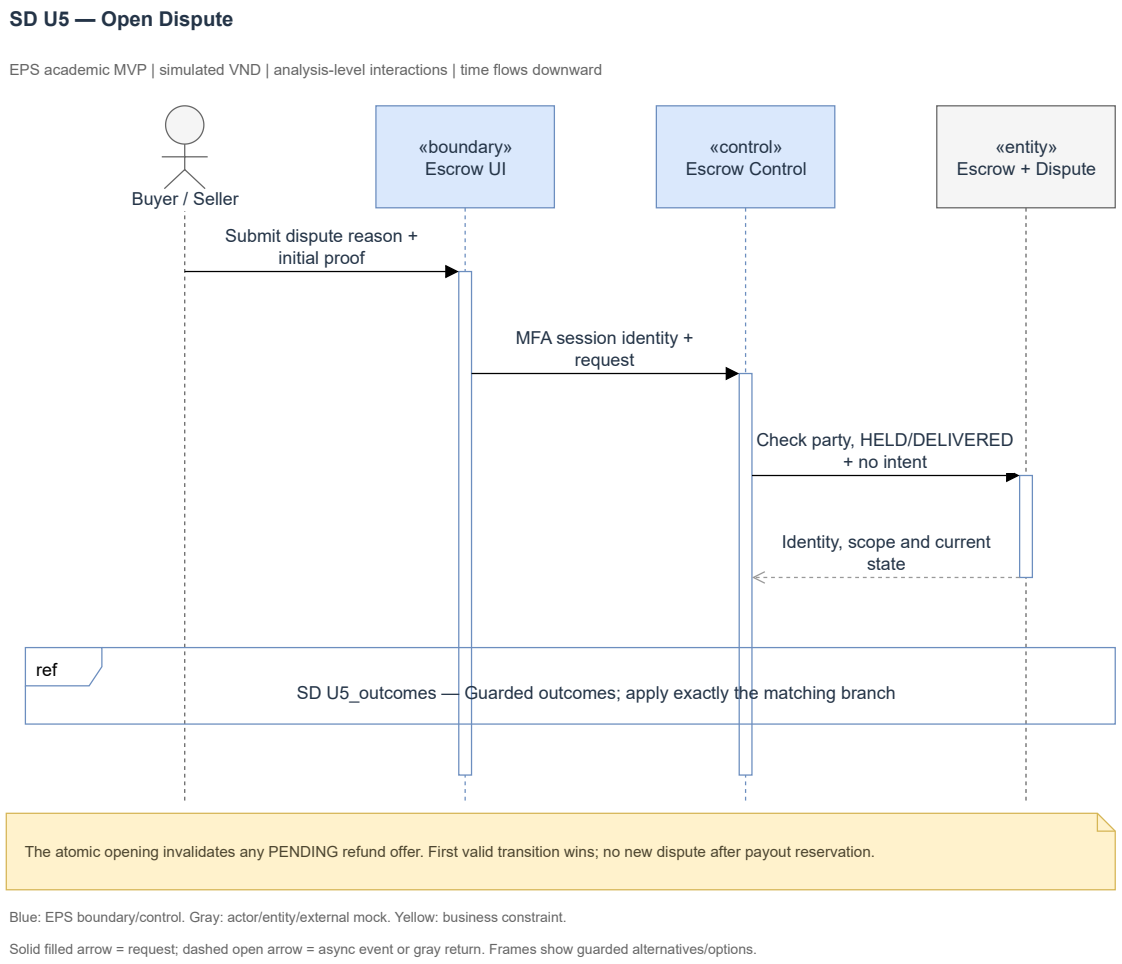


Figure 3.41: U5: Open Dispute

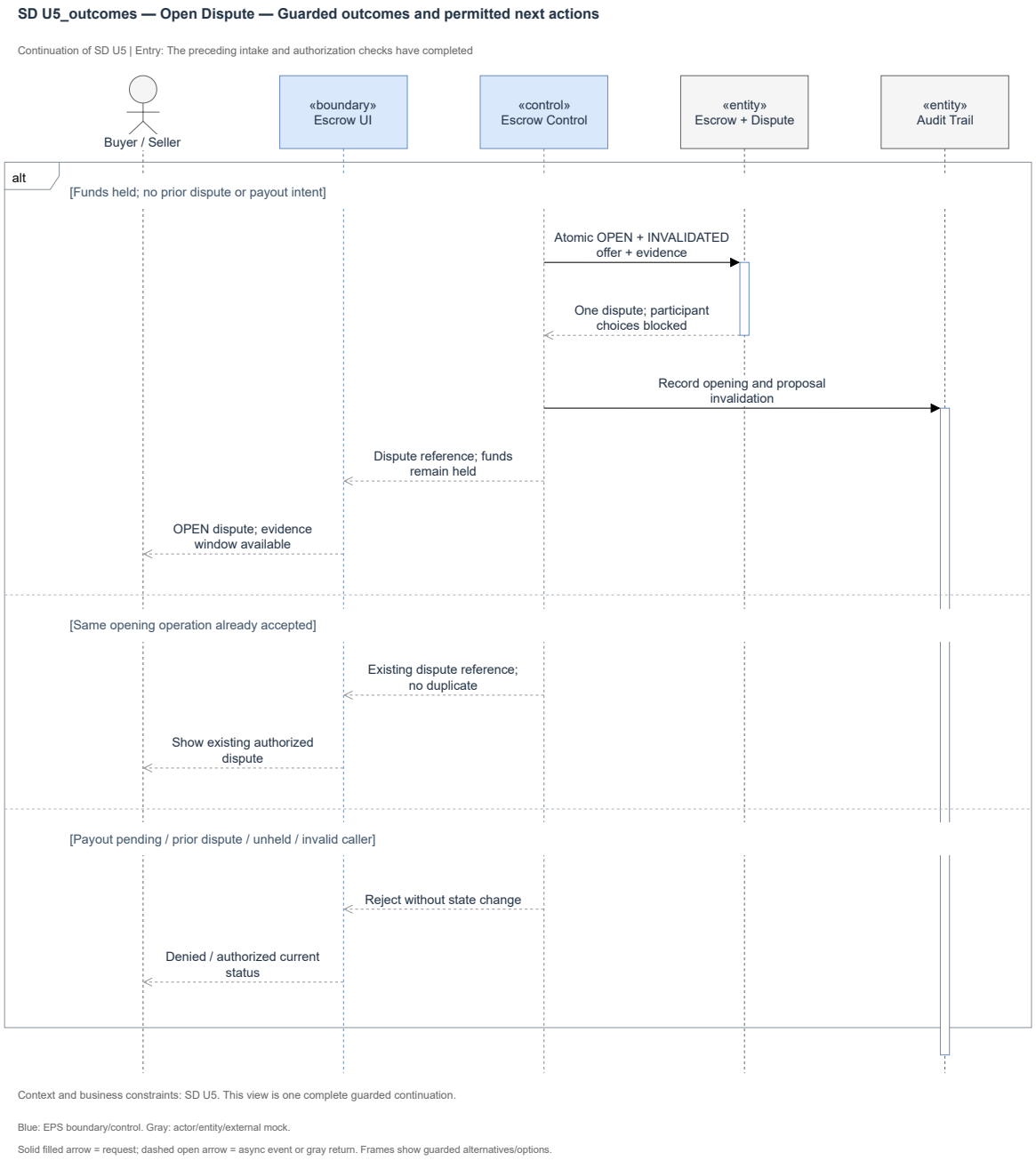


Figure 3.42: U5: Open Dispute (continued: U5_outcomes)

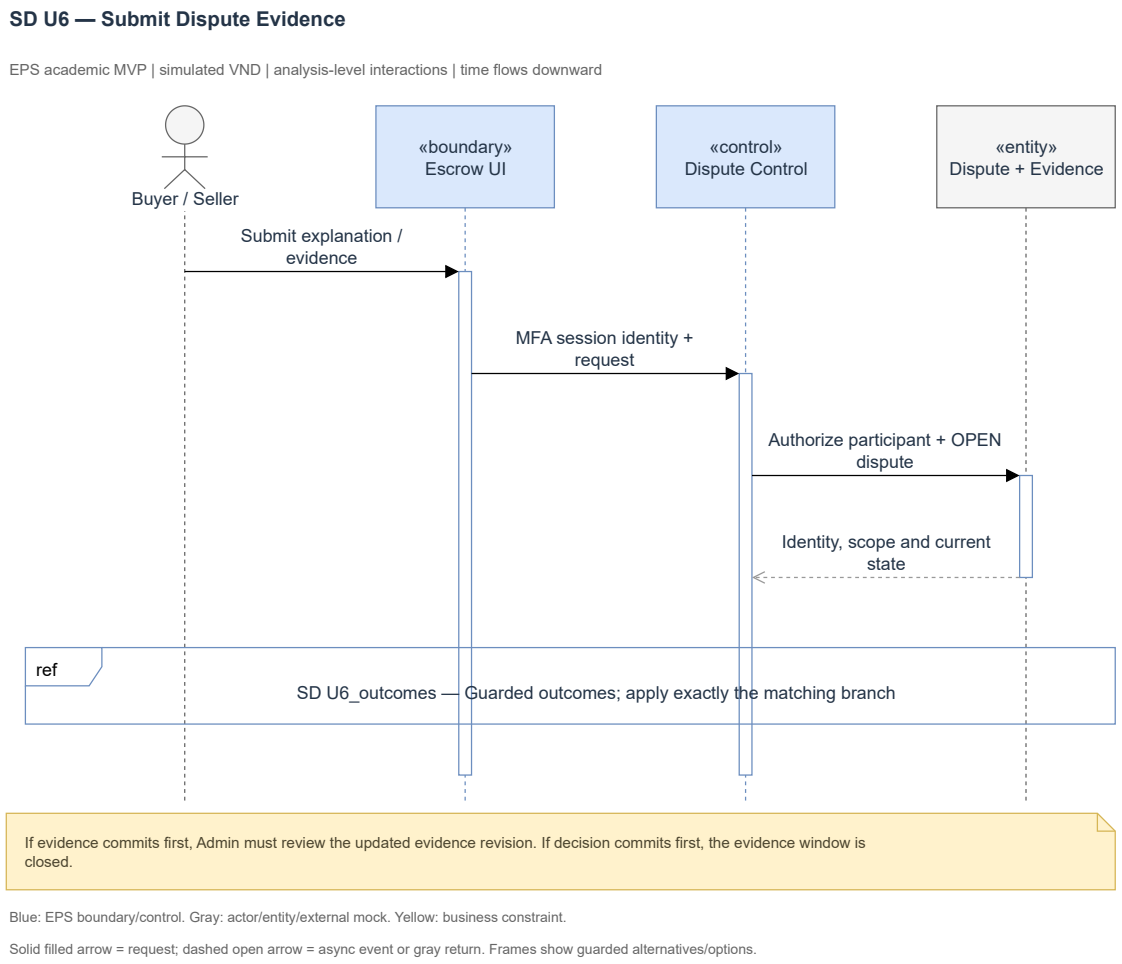


Figure 3.43: U6: Submit Dispute Evidence

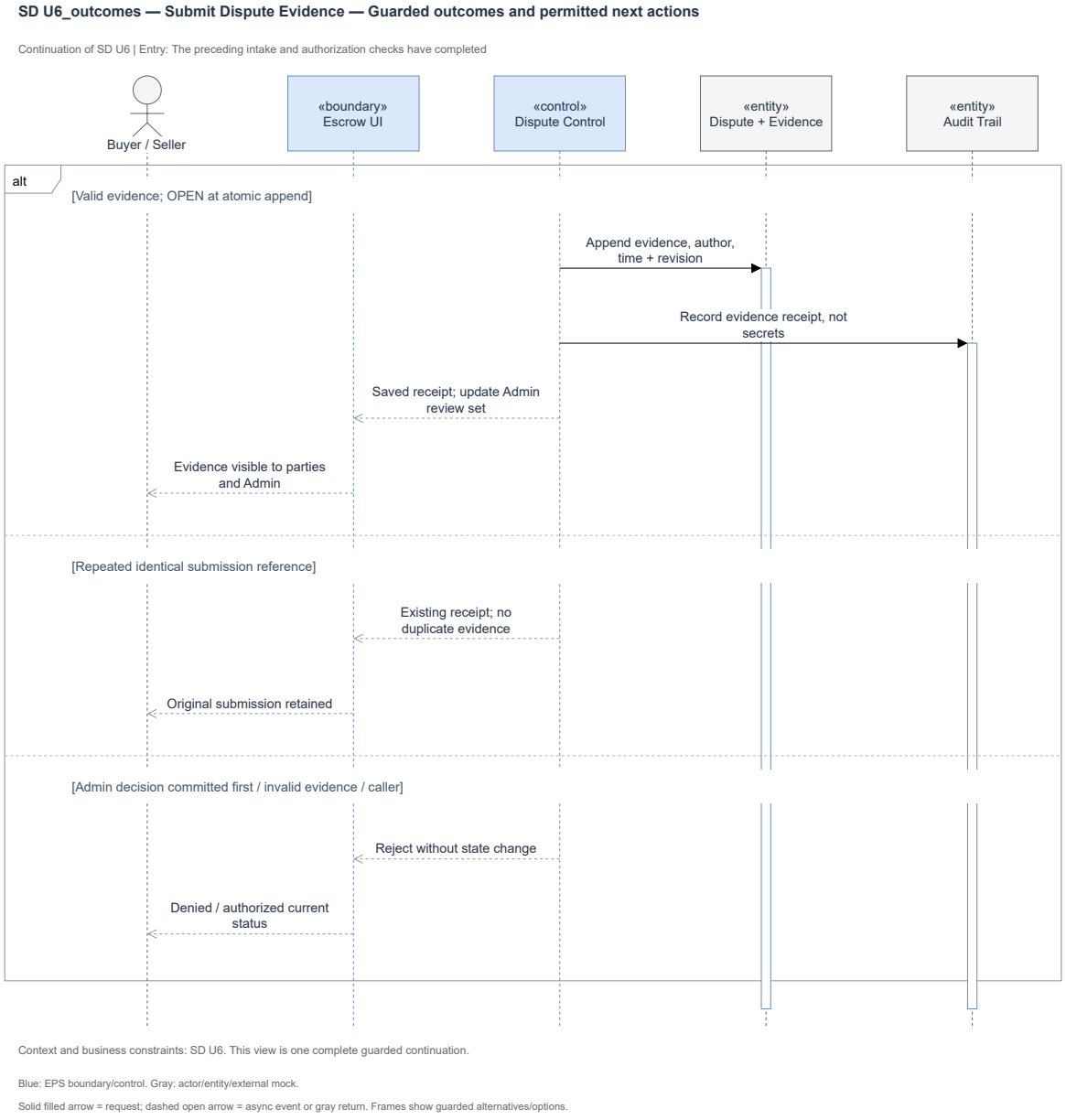


Figure 3.44: U6: Submit Dispute Evidence (continued: U6_outcomes)

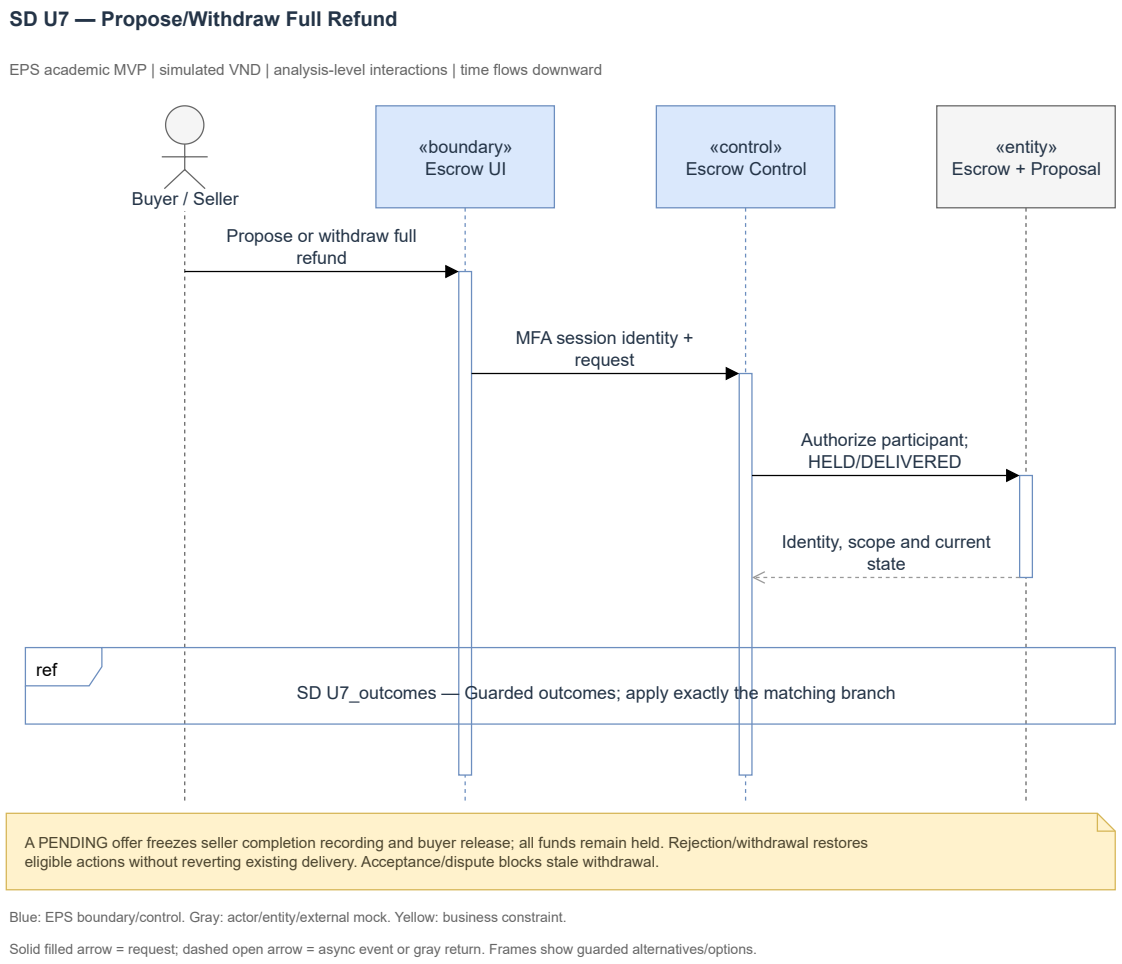


Figure 3.45: U7: Propose/Withdraw Full Refund

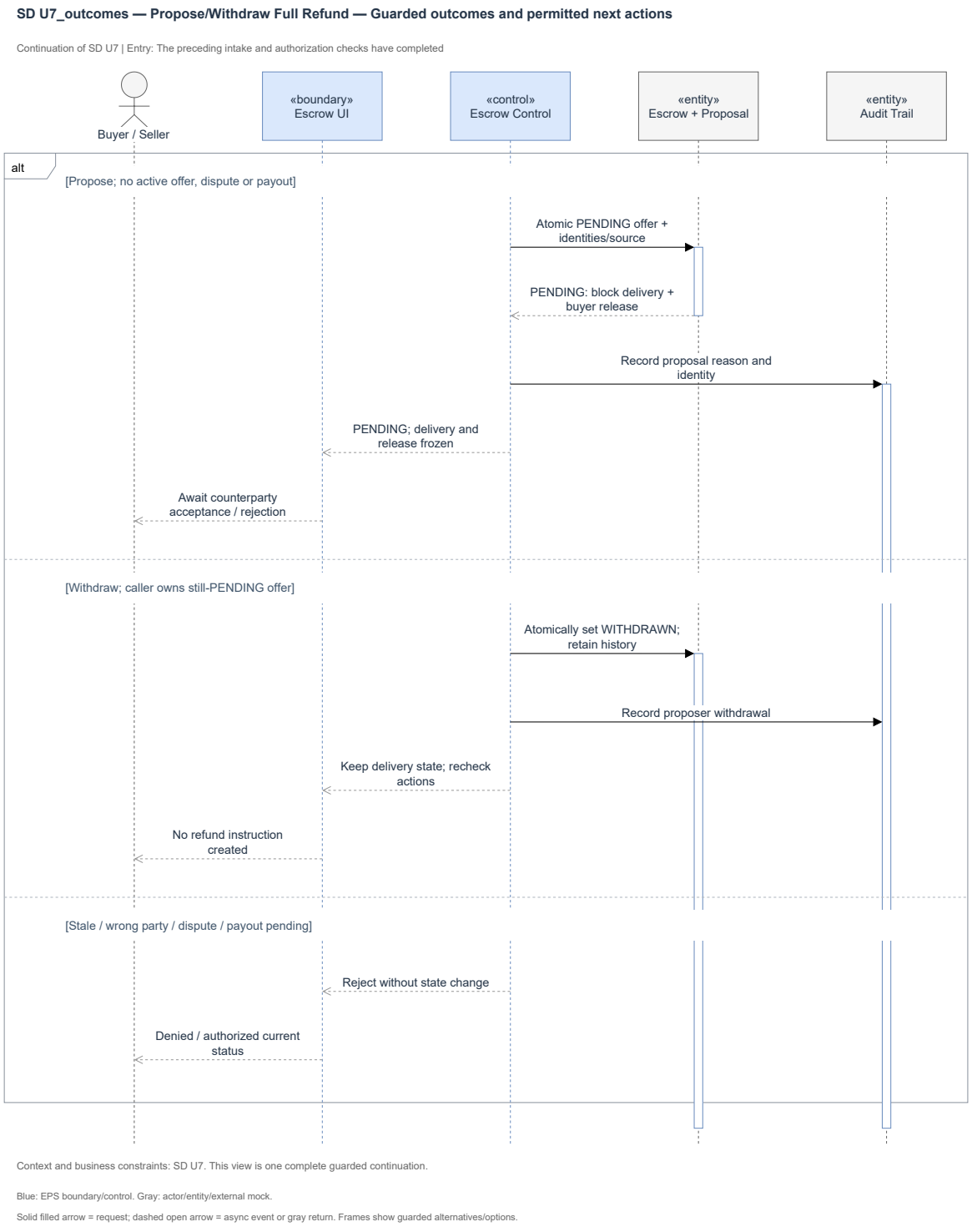


Figure 3.46: U7: Propose/Withdraw Full Refund (continued: U7_outcomes)

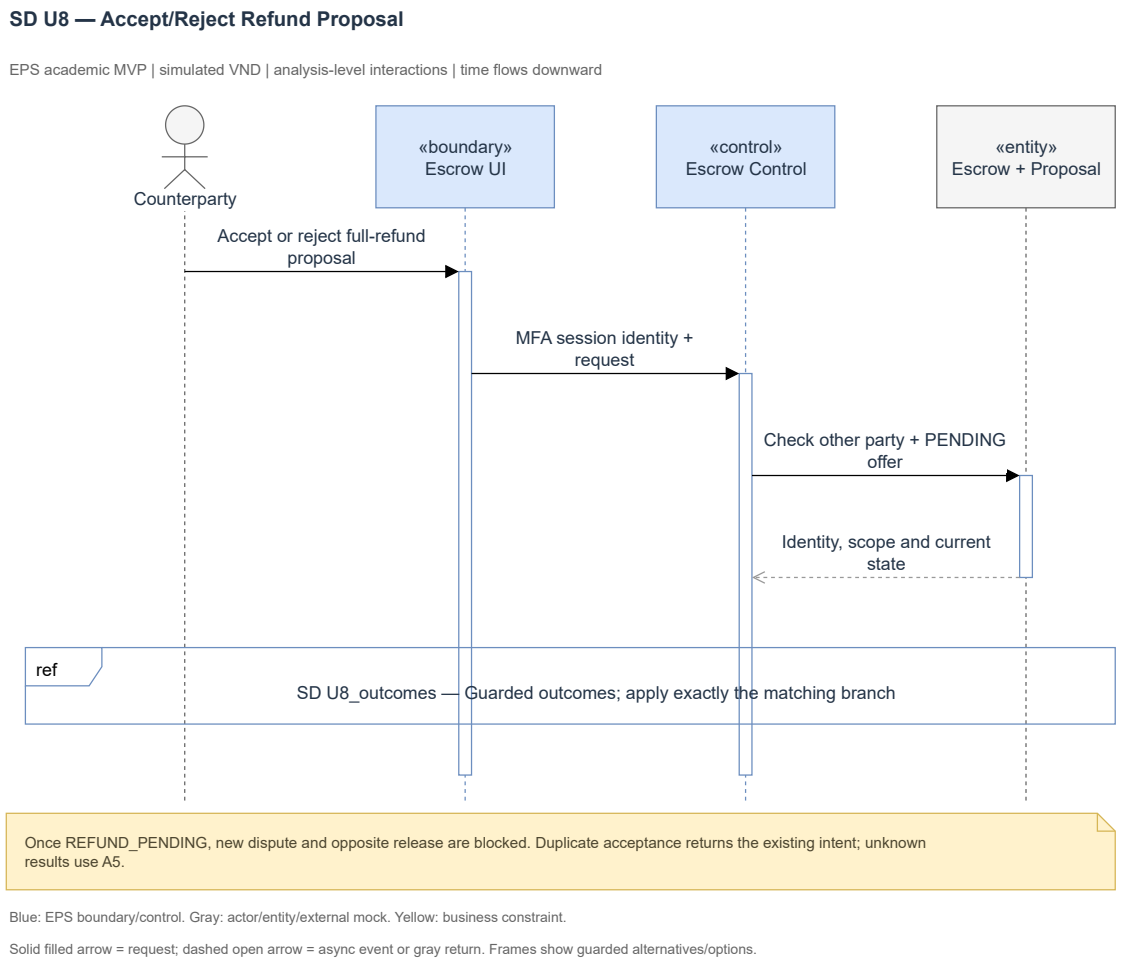


Figure 3.47: U8: Accept/Reject Refund Proposal

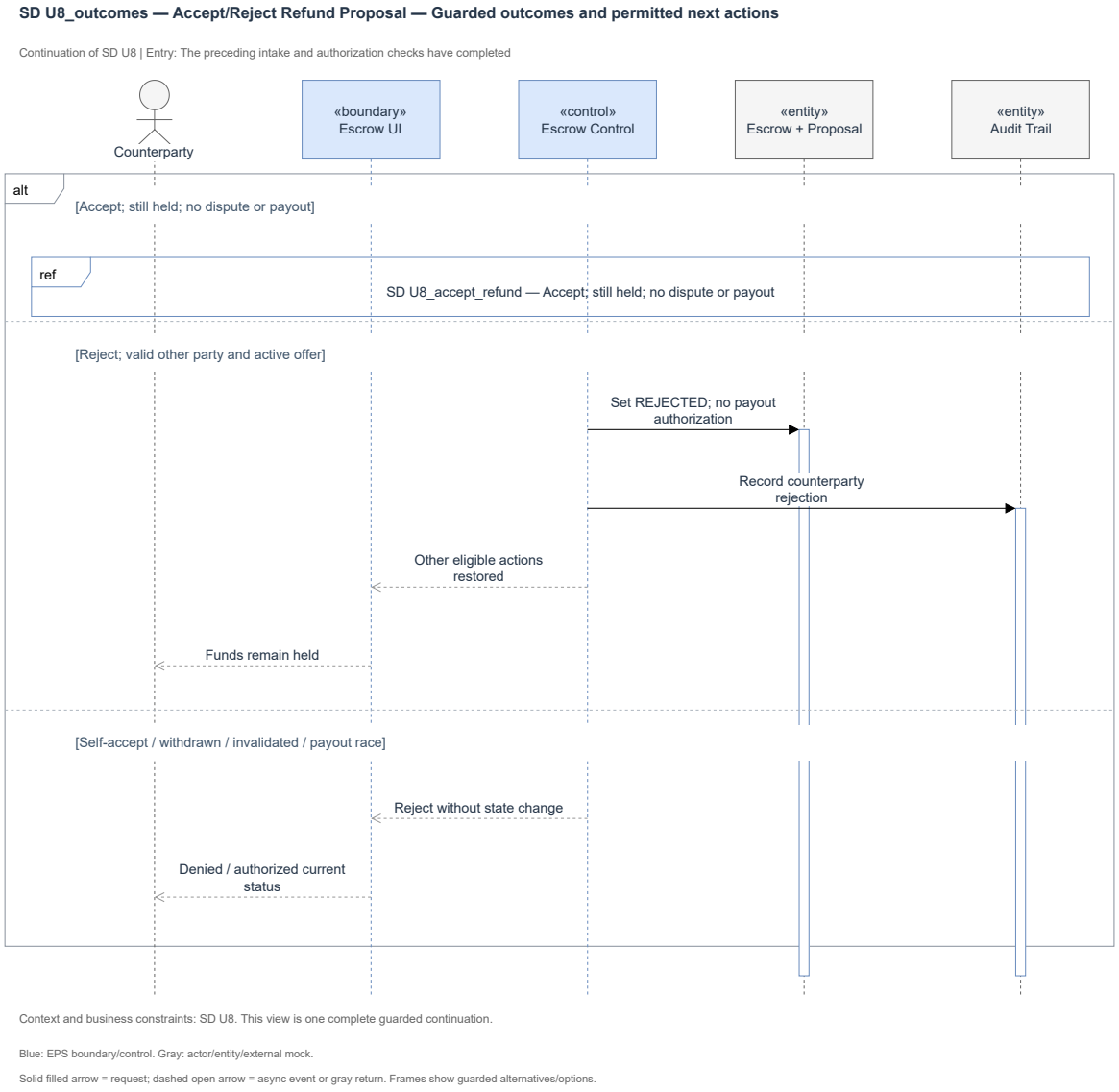


Figure 3.48: U8: Accept/Reject Refund Proposal (continued: U8_outcomes)

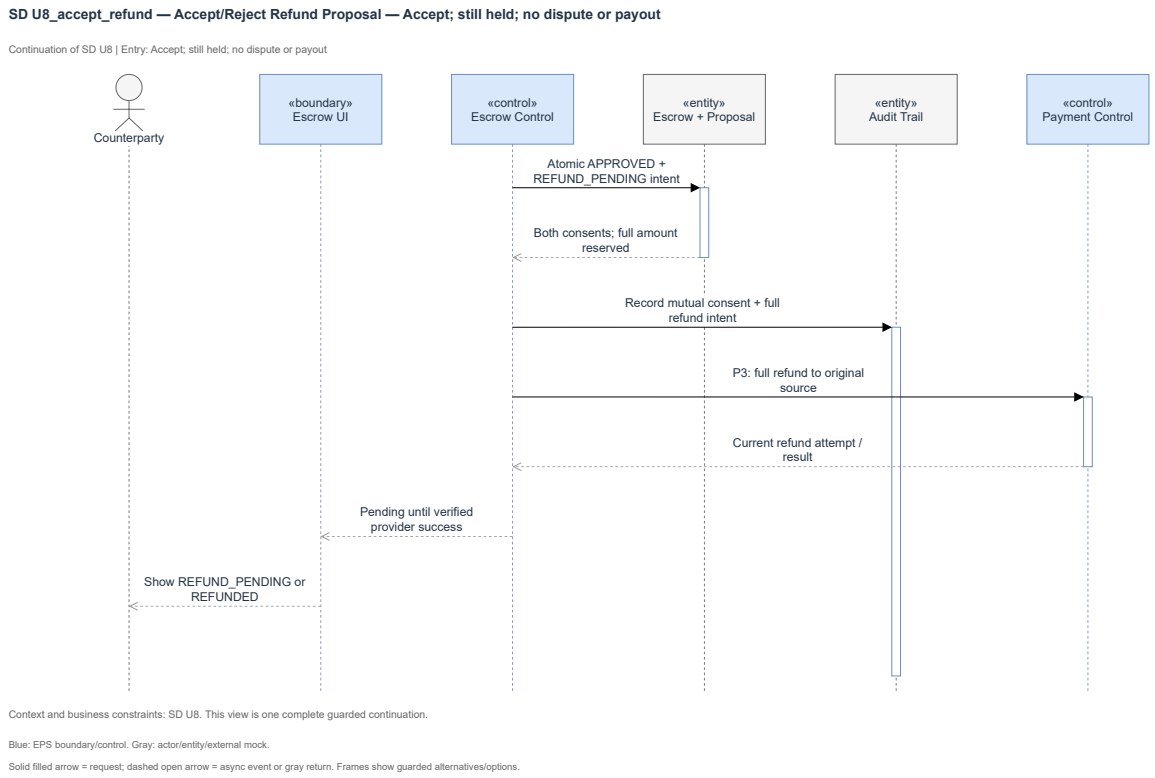


Figure 3.49: U8: Accept/Reject Refund Proposal (continued: U8_accept_refund)

3.4.2 Buyer's Use cases

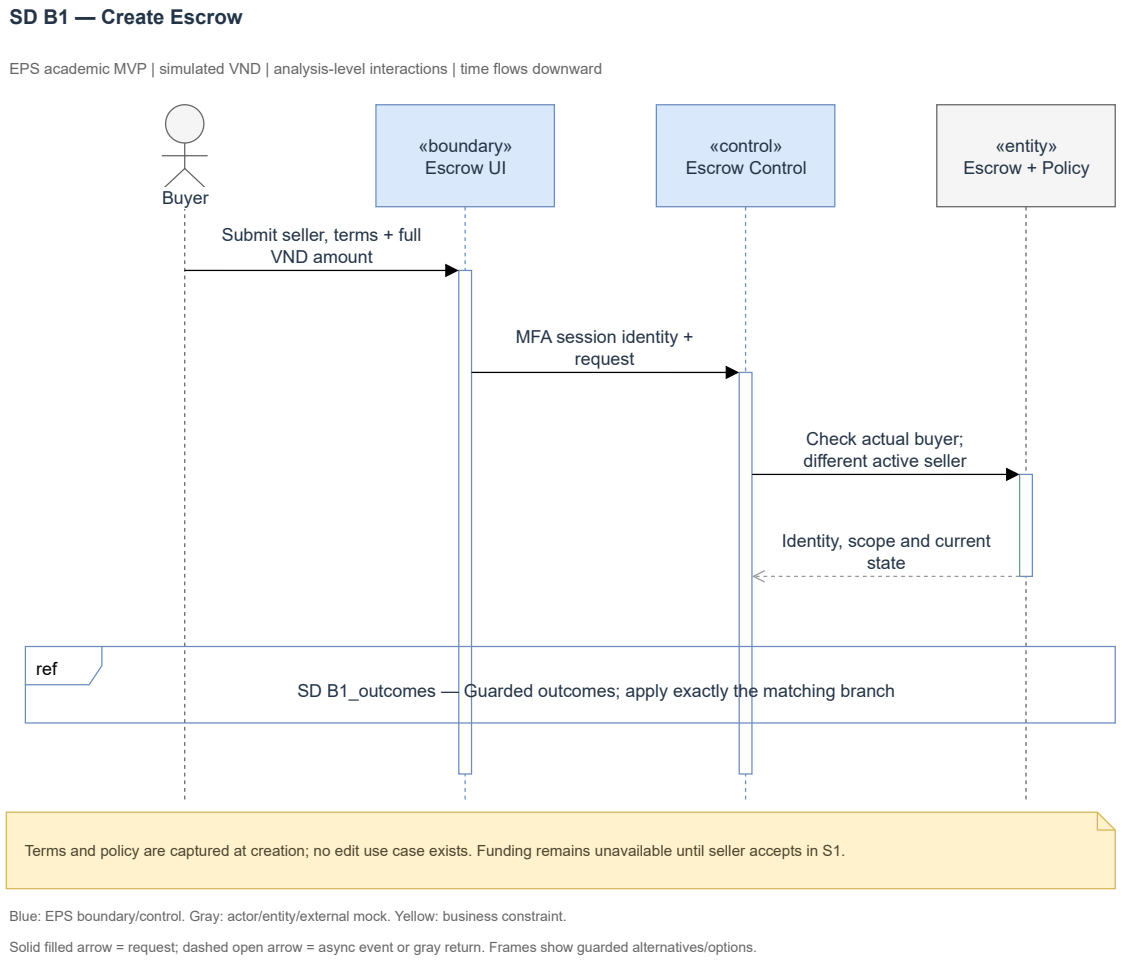


Figure 3.50: B1: Create Escrow

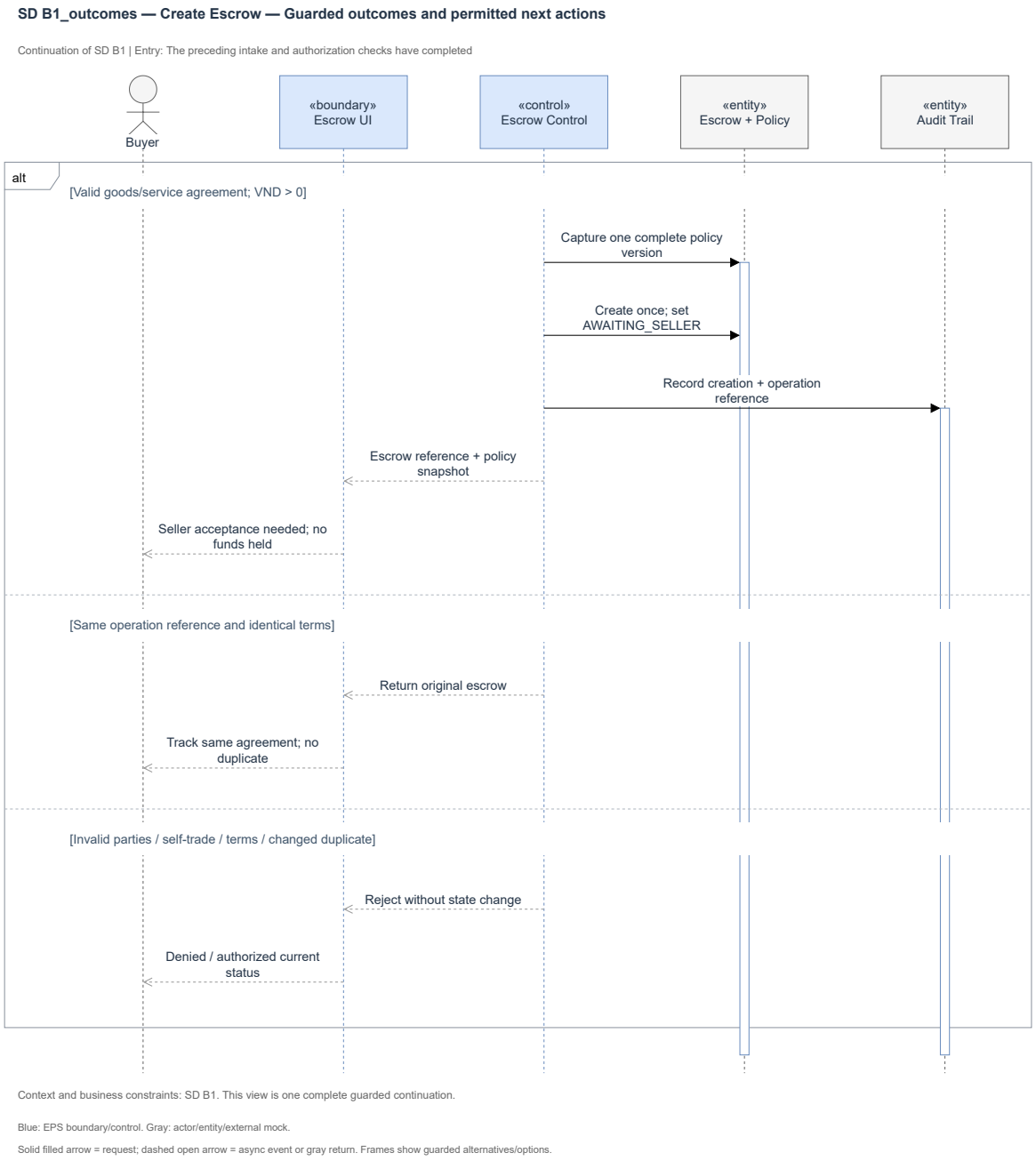


Figure 3.51: B1: Create Escrow (continued: B1_outcomes)

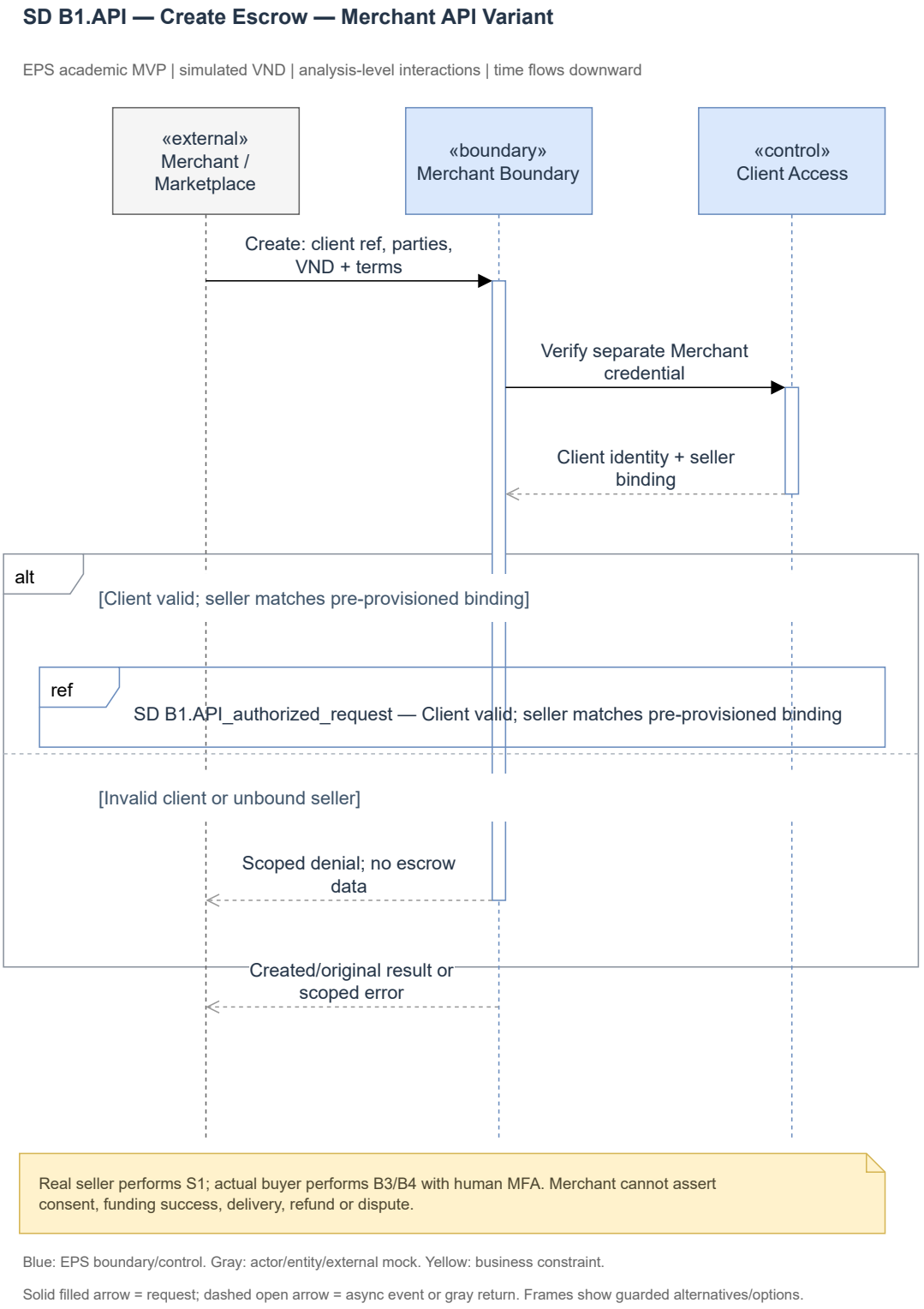


Figure 3.52: B1.API: Create Escrow

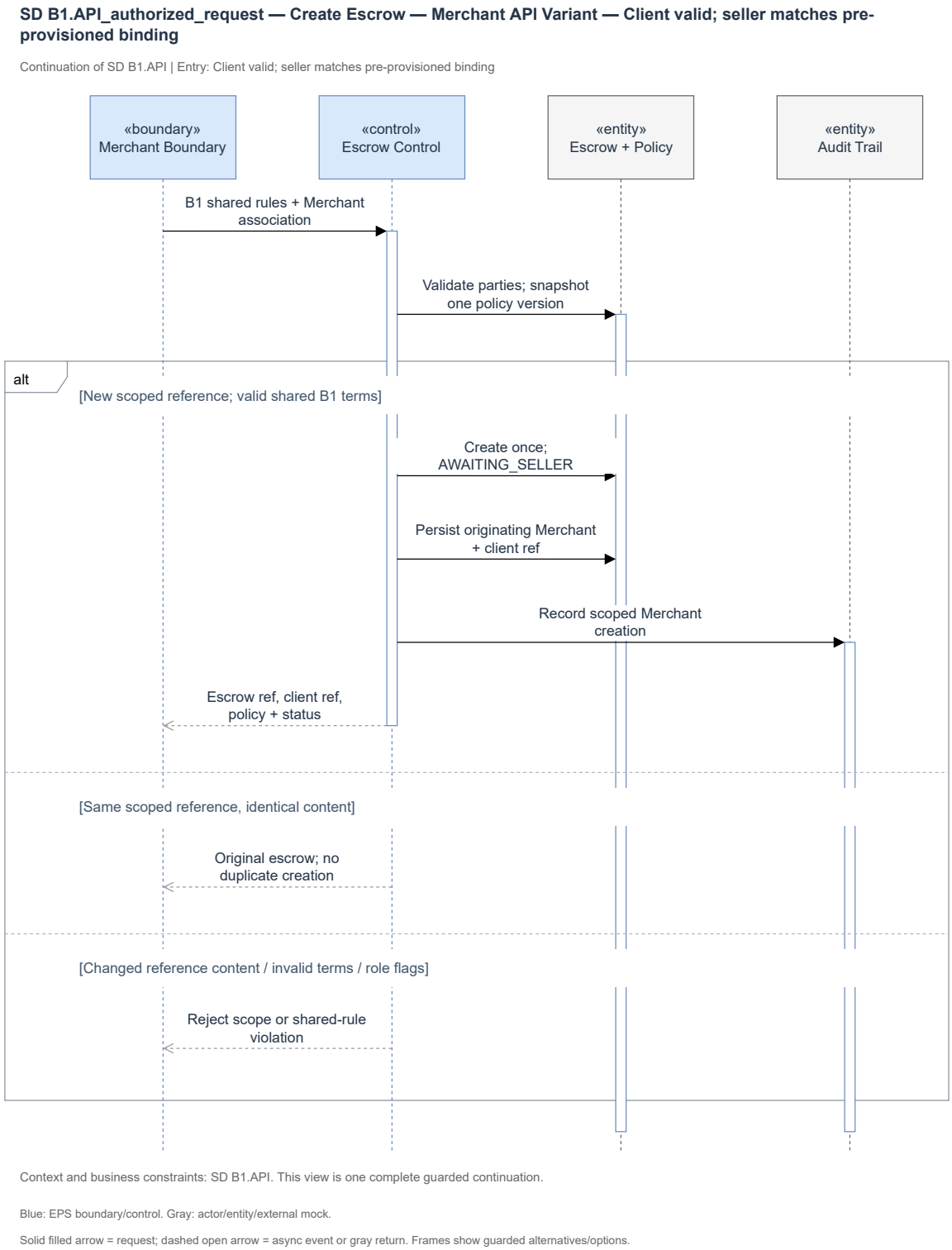


Figure 3.53: B1.API: Create Escrow (continued: B1.API_authorized_request)

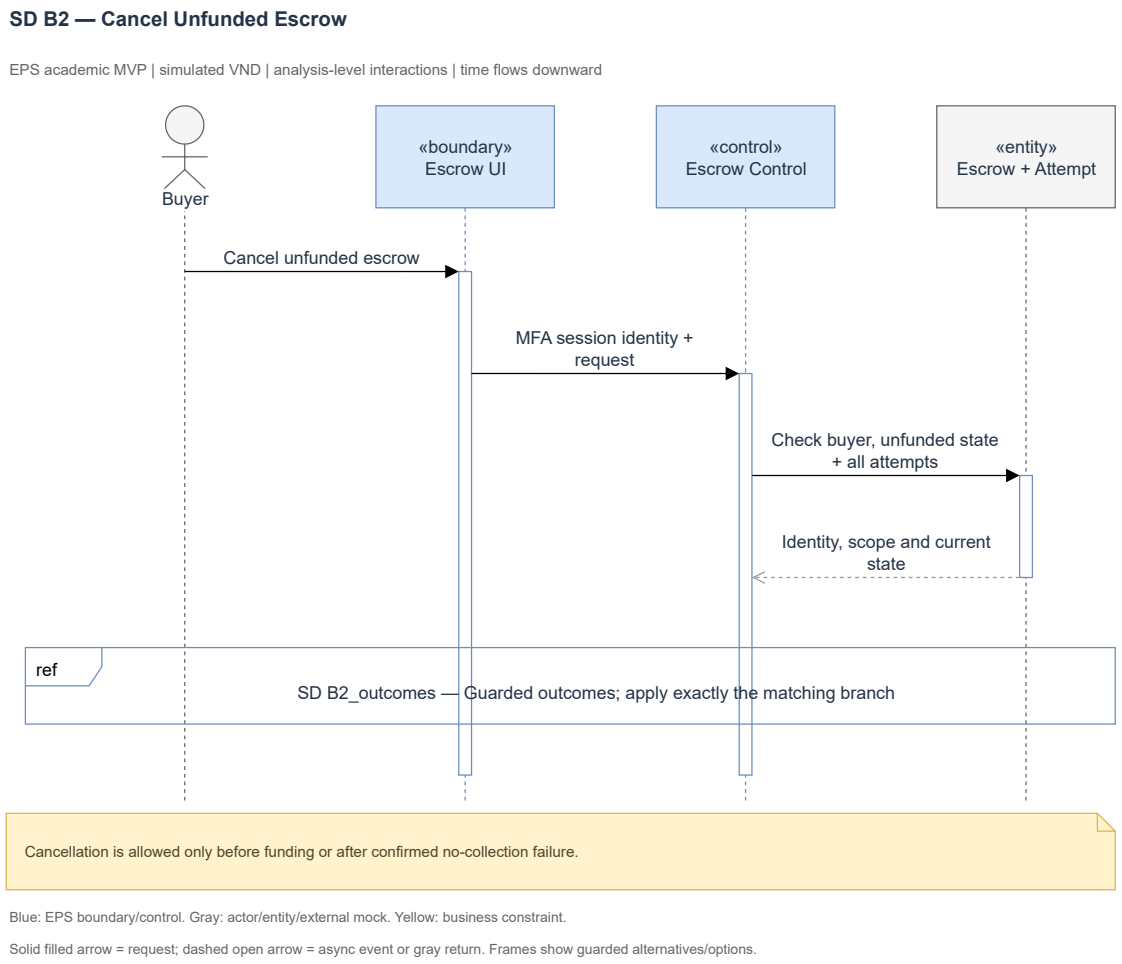


Figure 3.54: B2: Cancel Unfunded Escrow

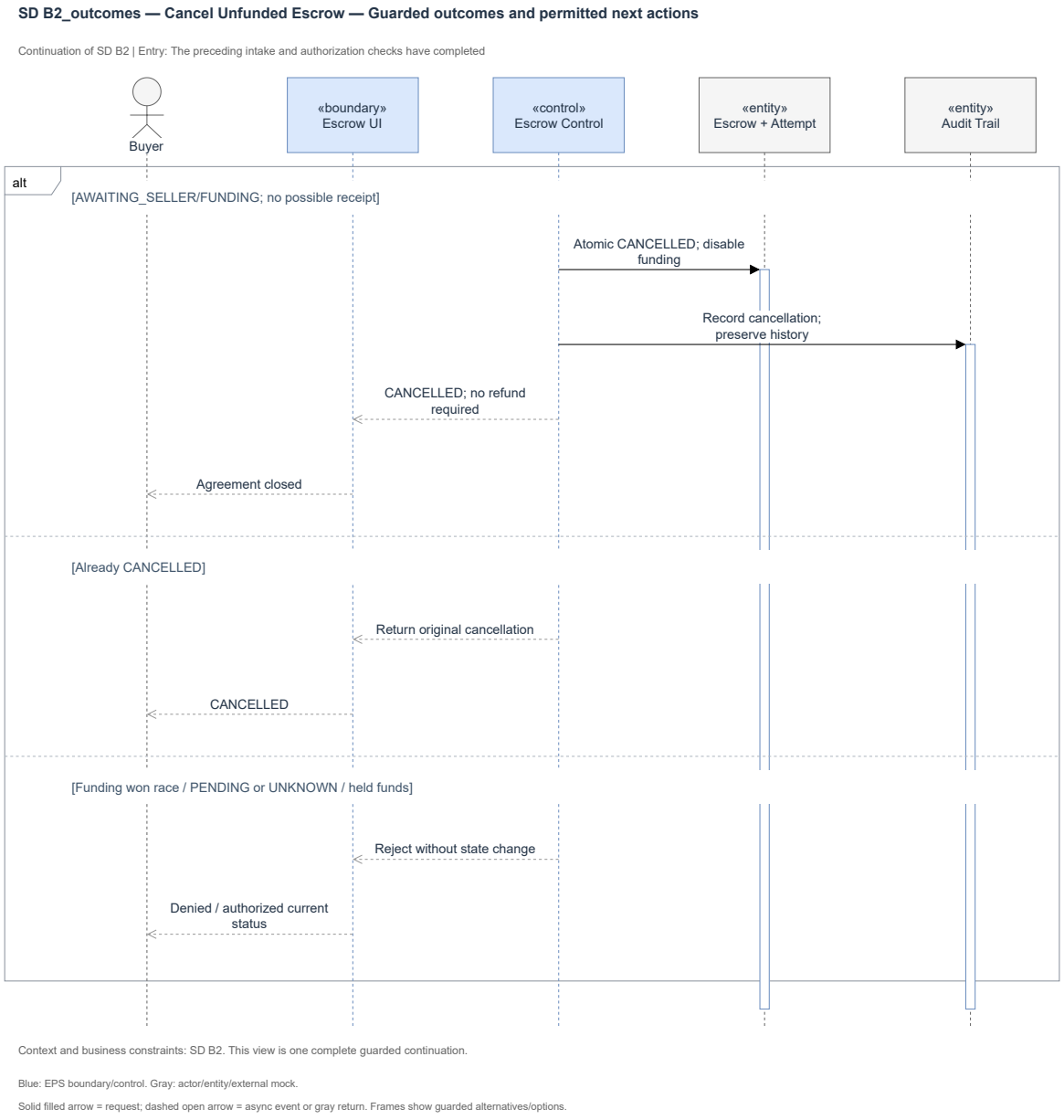


Figure 3.55: B2: Cancel Unfunded Escrow (continued: B2_outcomes)

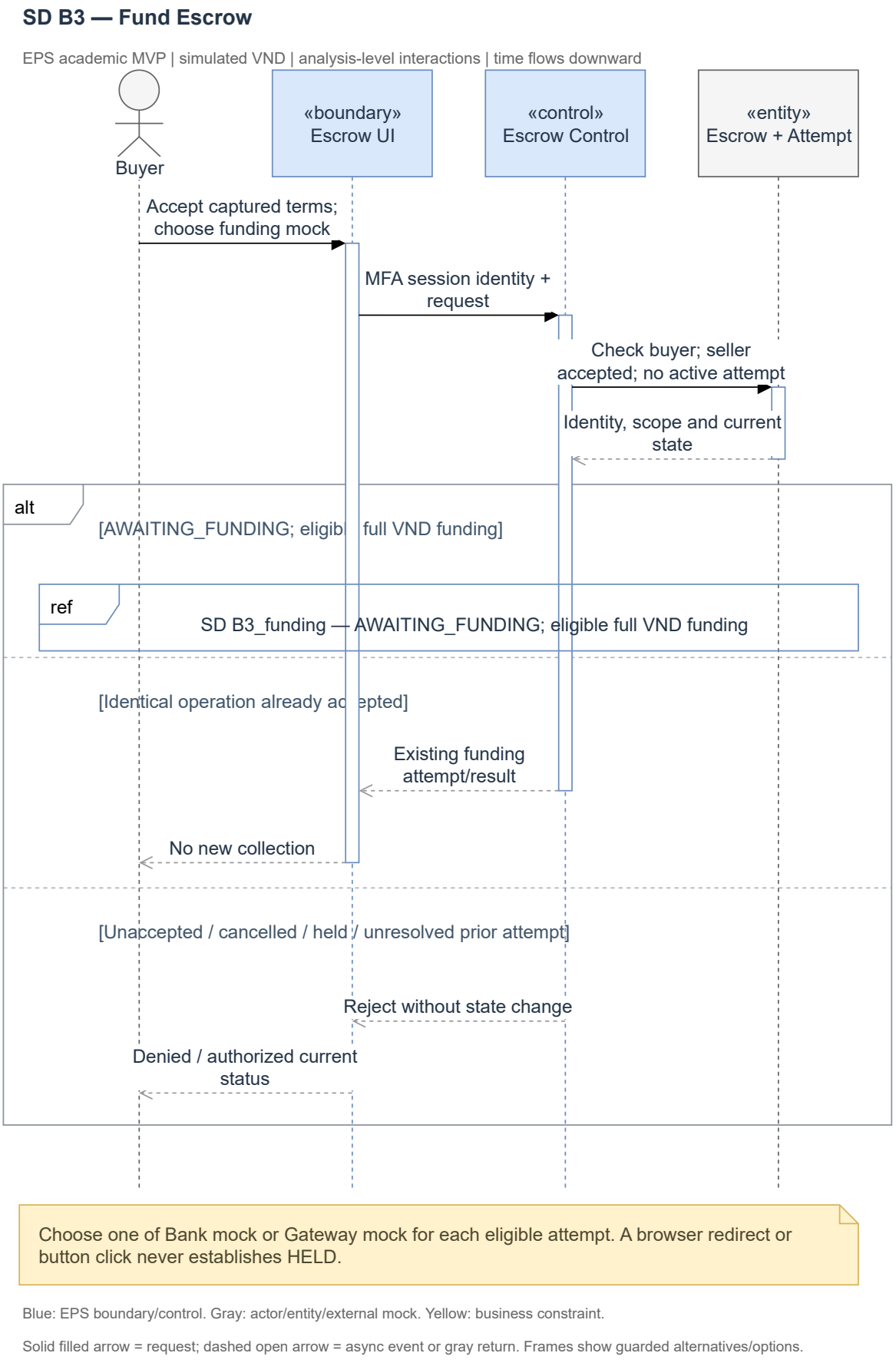


Figure 3.56: B3: Fund Escrow

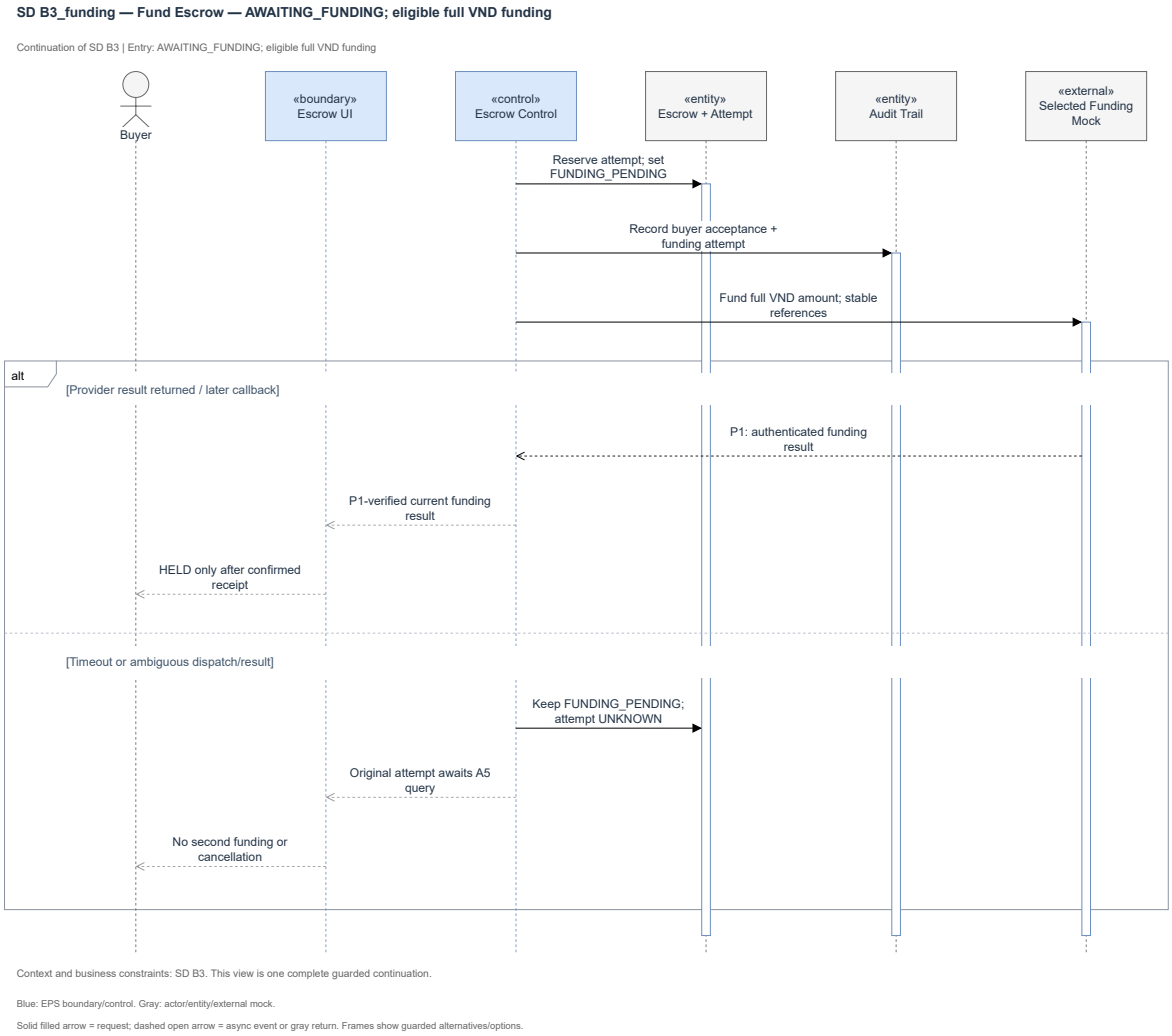


Figure 3.57: B3: Fund Escrow (continued: B3_funding)

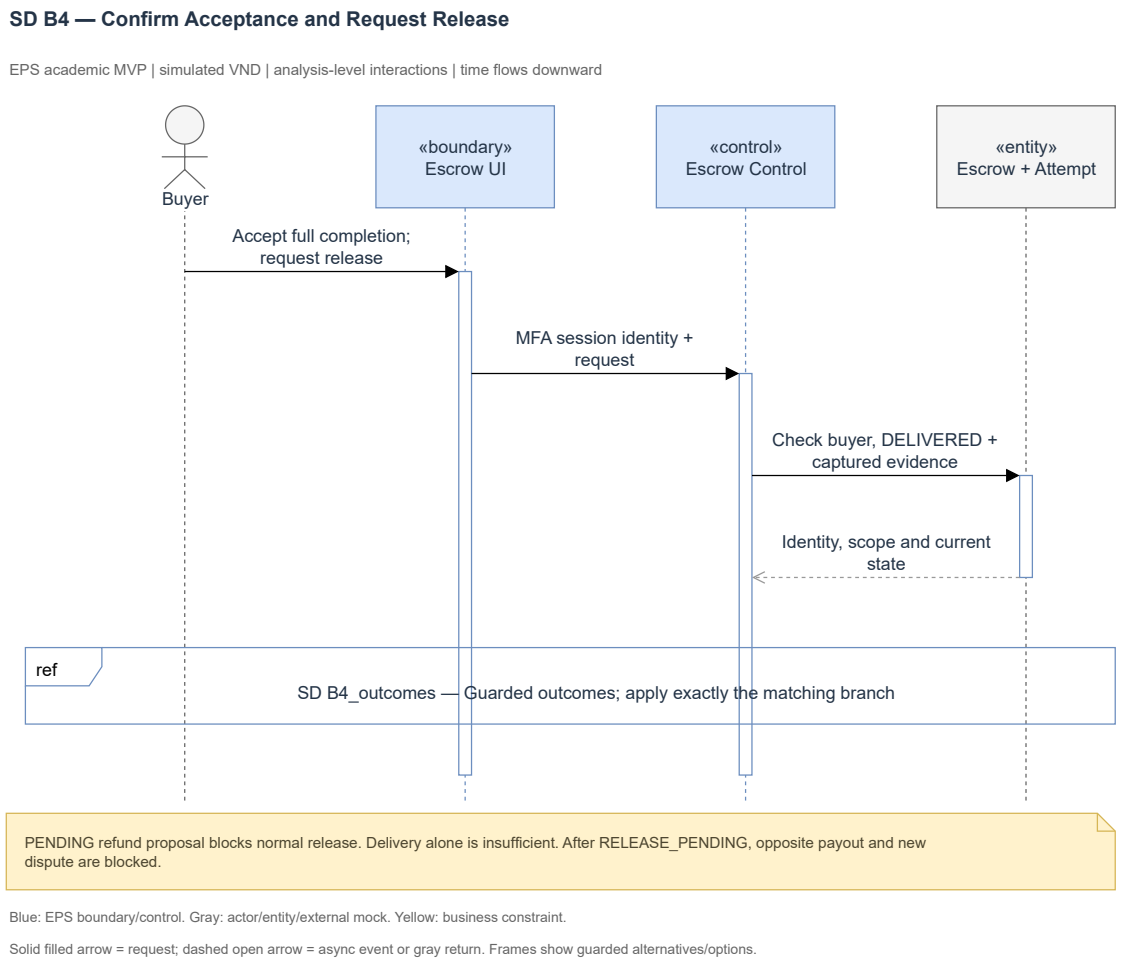


Figure 3.58: B4: Confirm Acceptance and Request Release

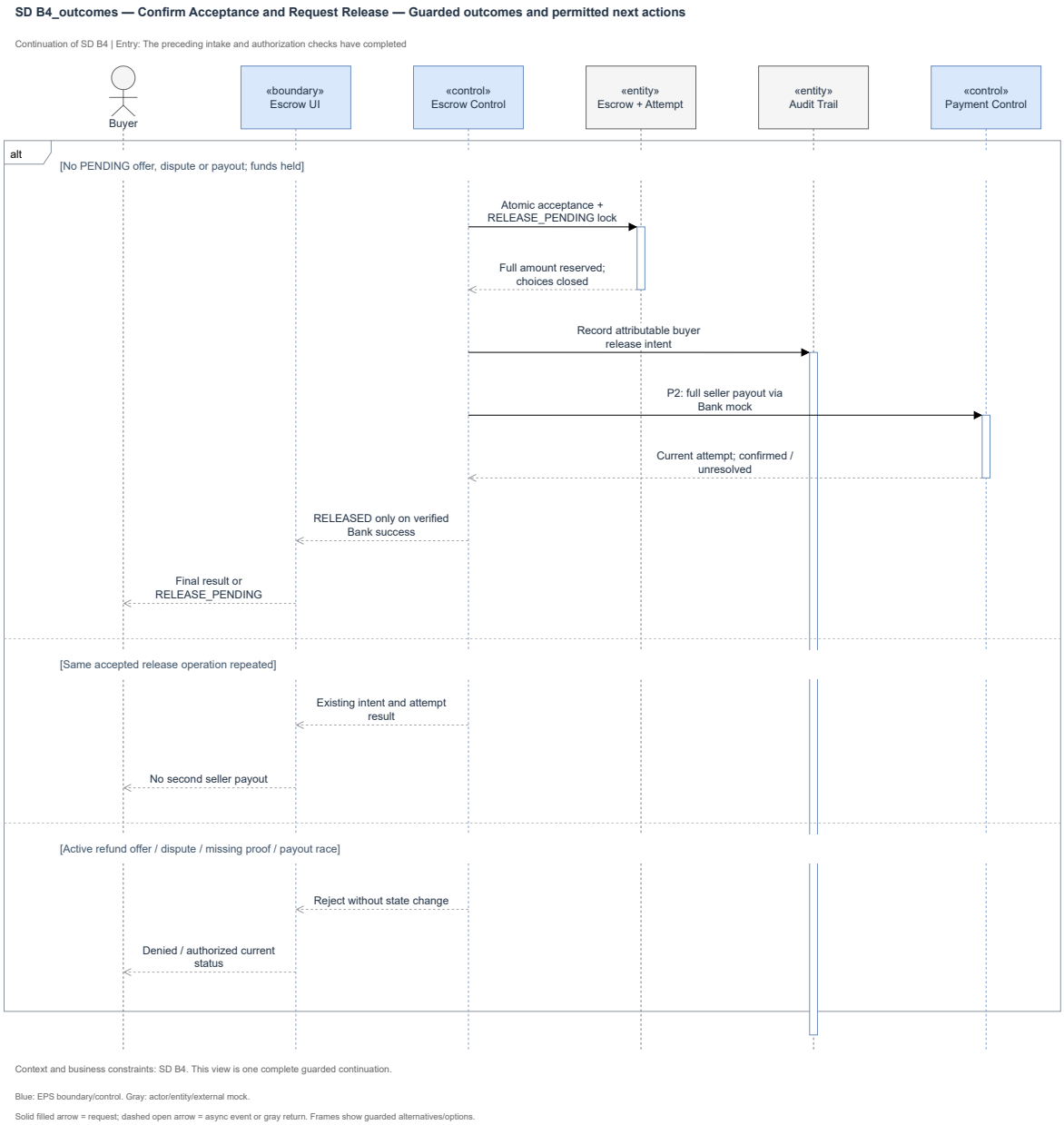


Figure 3.59: B4: Confirm Acceptance and Request Release (continued: B4_outcomes)

3.4.3 Seller's Use cases

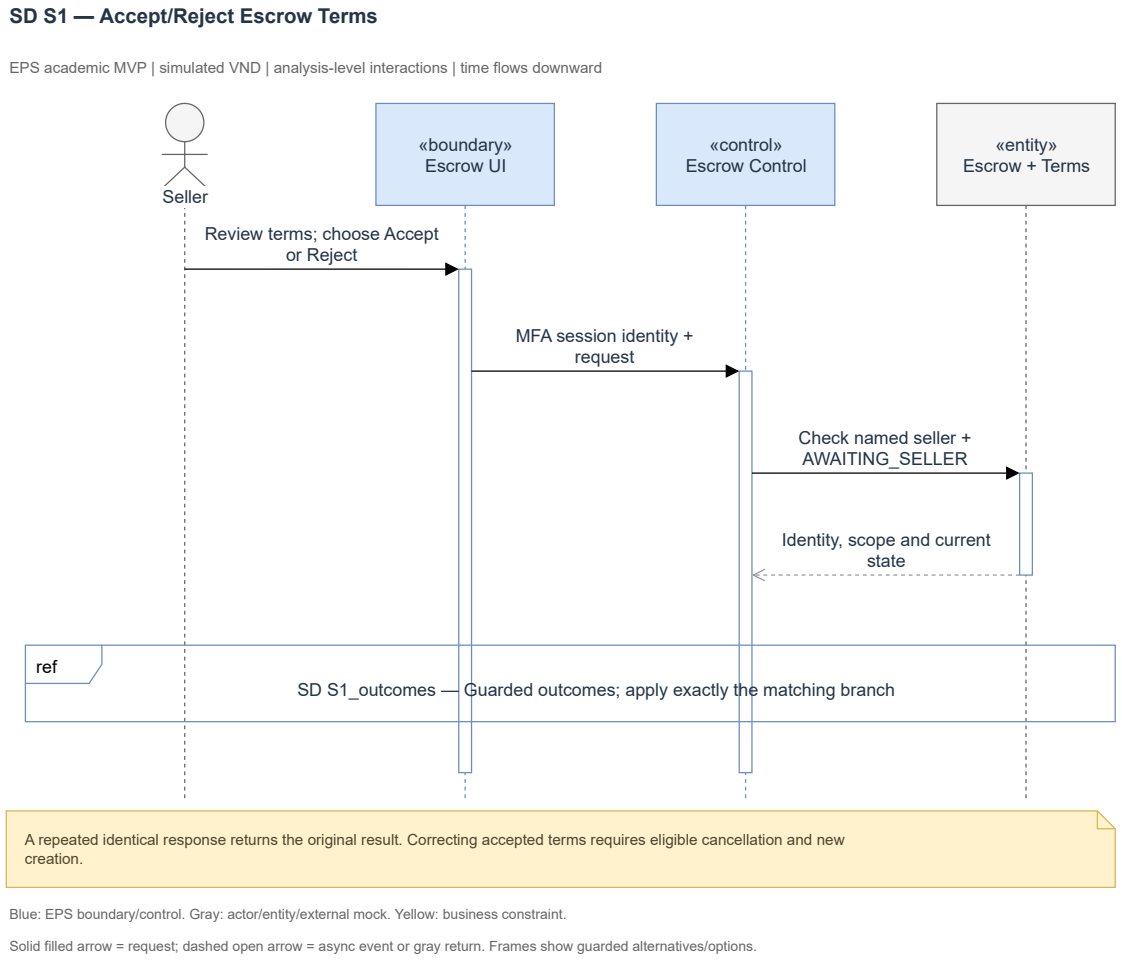


Figure 3.60: S1: Accept/Reject Escrow Terms

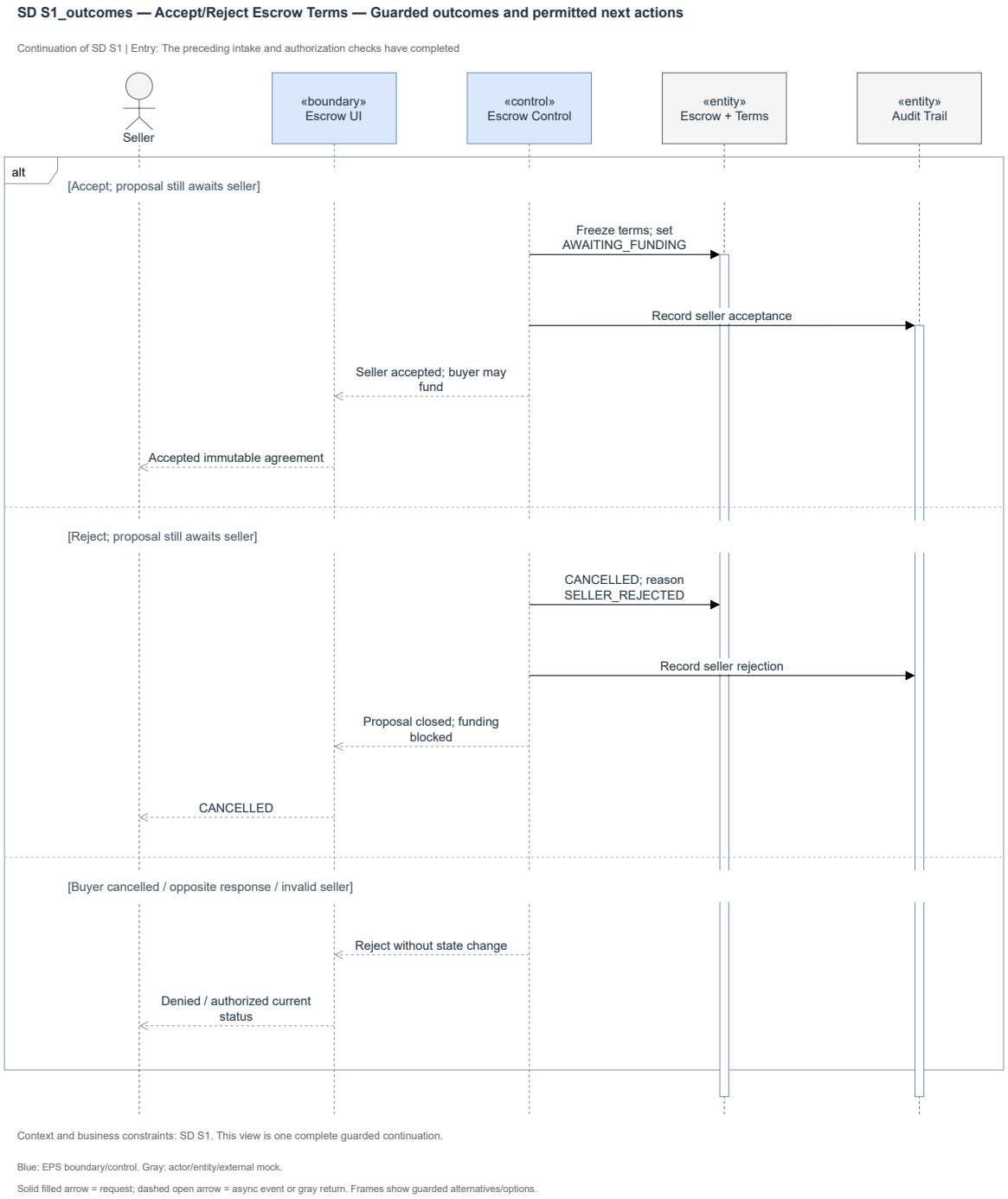


Figure 3.61: S1: Accept/Reject Escrow Terms (continued: S1_outcomes)

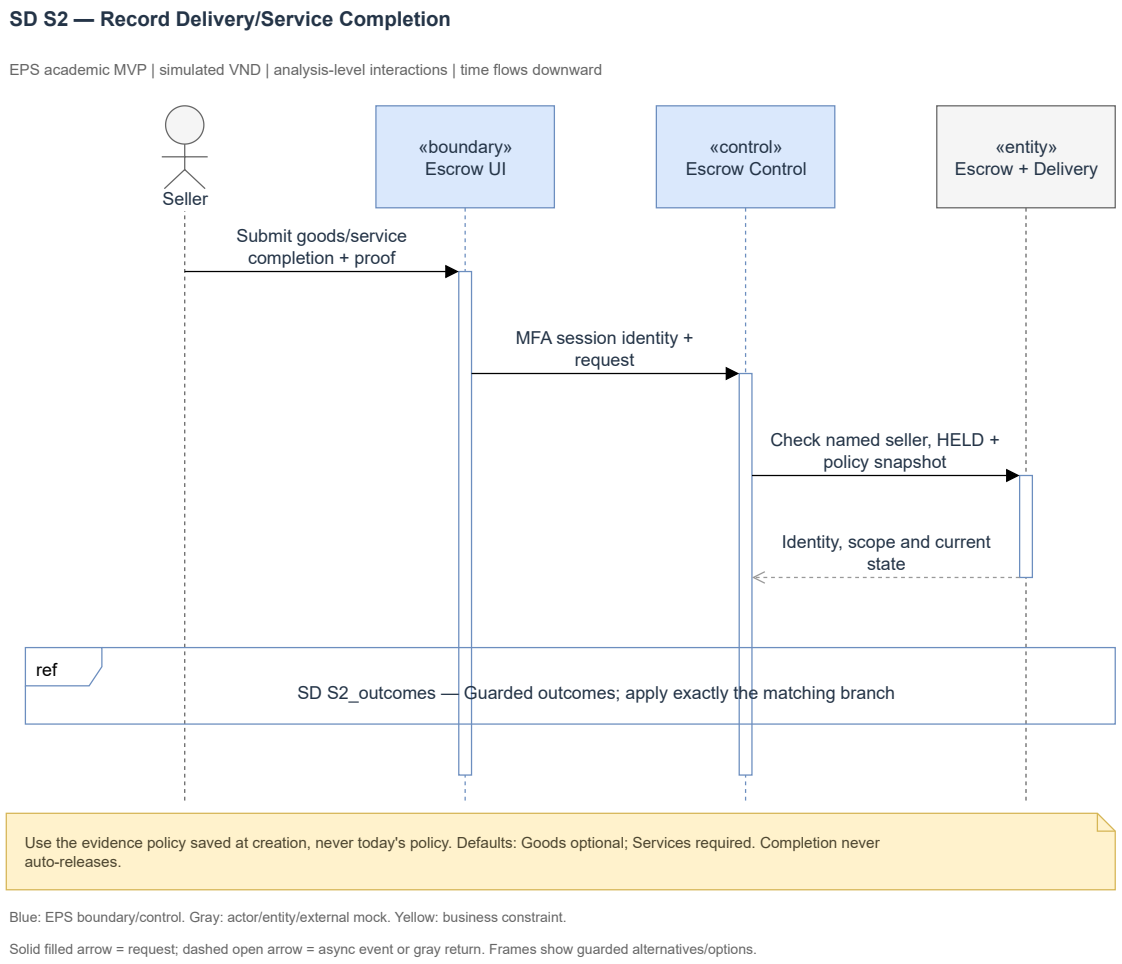


Figure 3.62: S2: Record Delivery/Service Completion

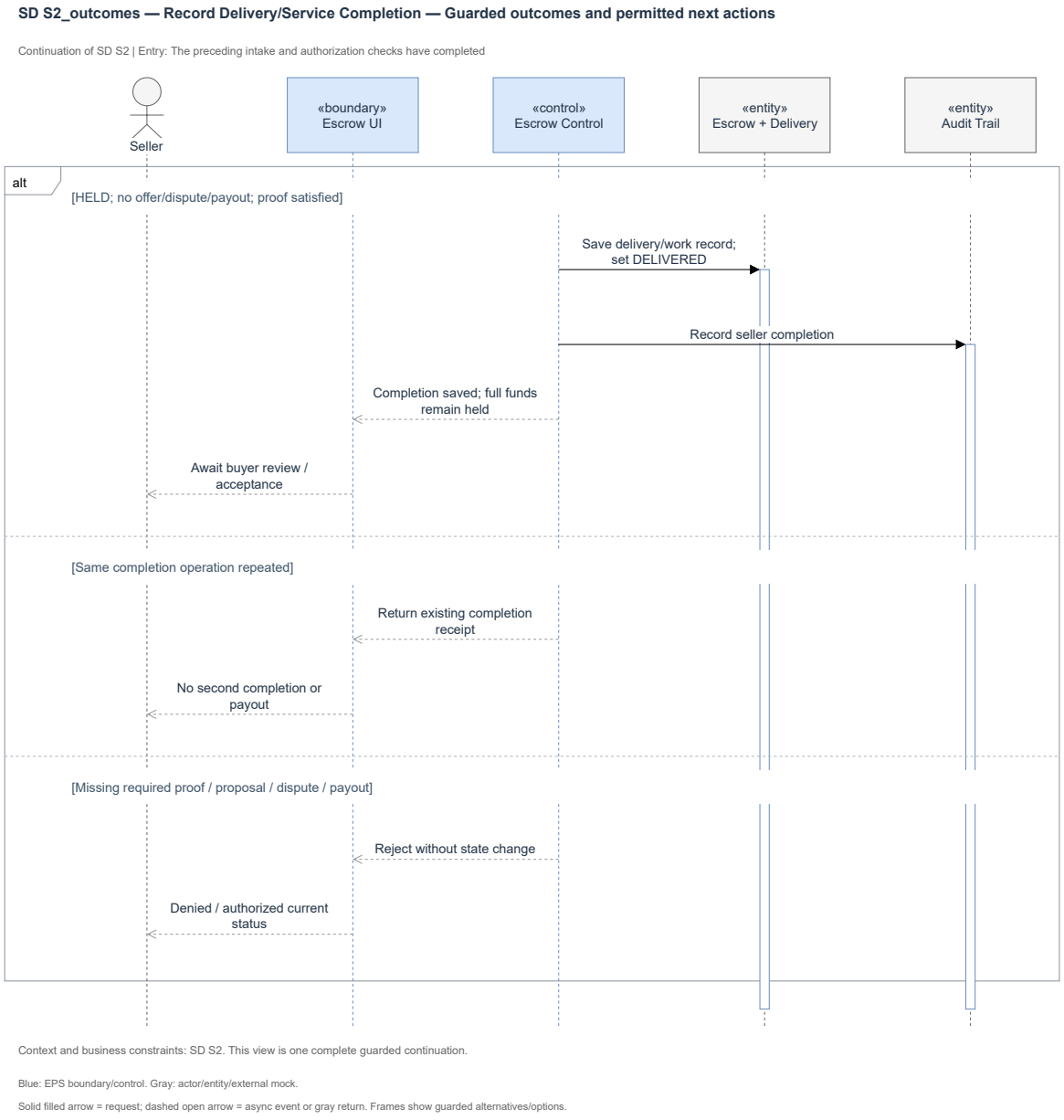


Figure 3.63: S2: Record Delivery/Service Completion (continued: S2_outcomes)

3.4.4 Admin's Use cases

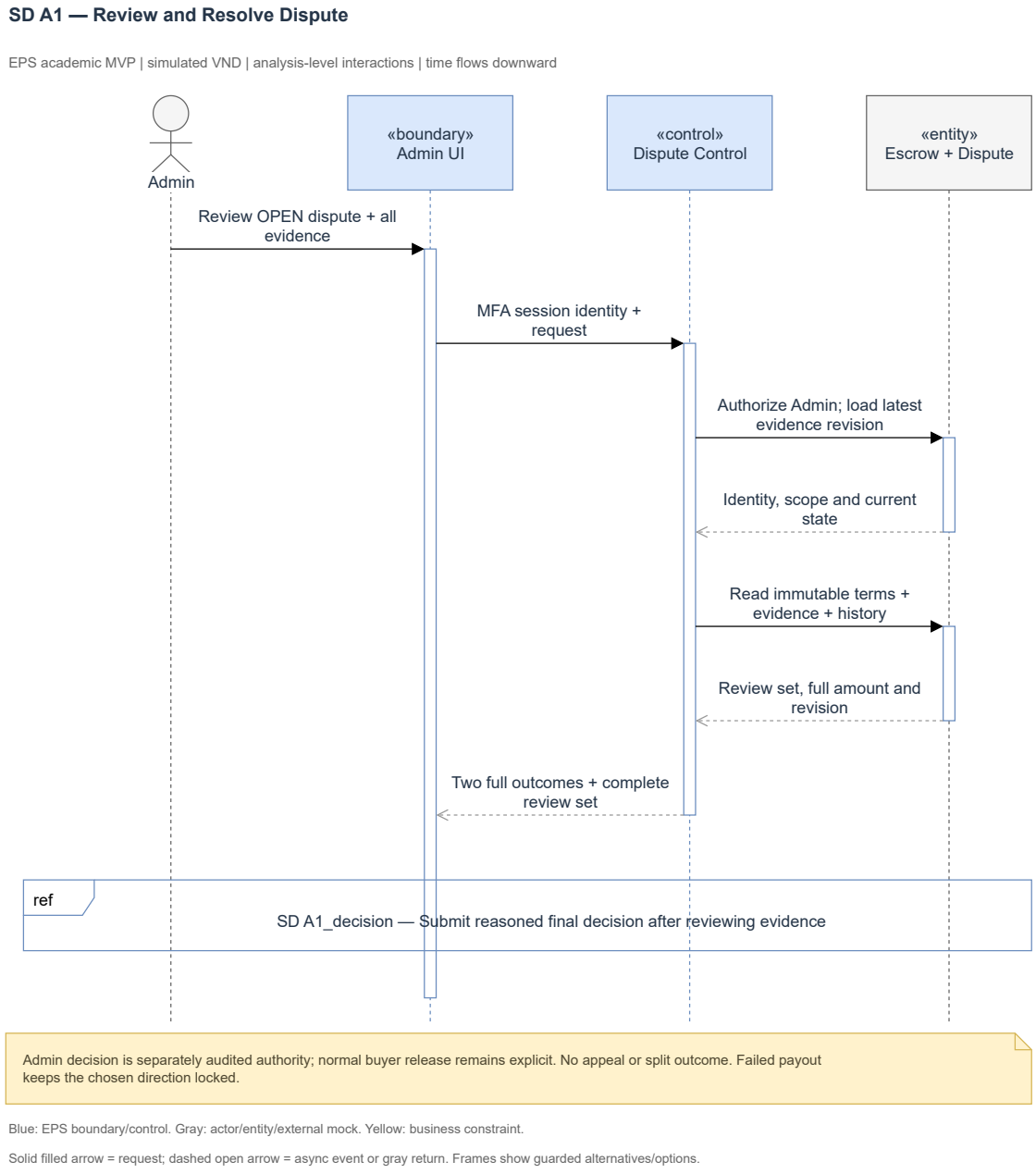


Figure 3.64: A1: Review and Resolve Dispute

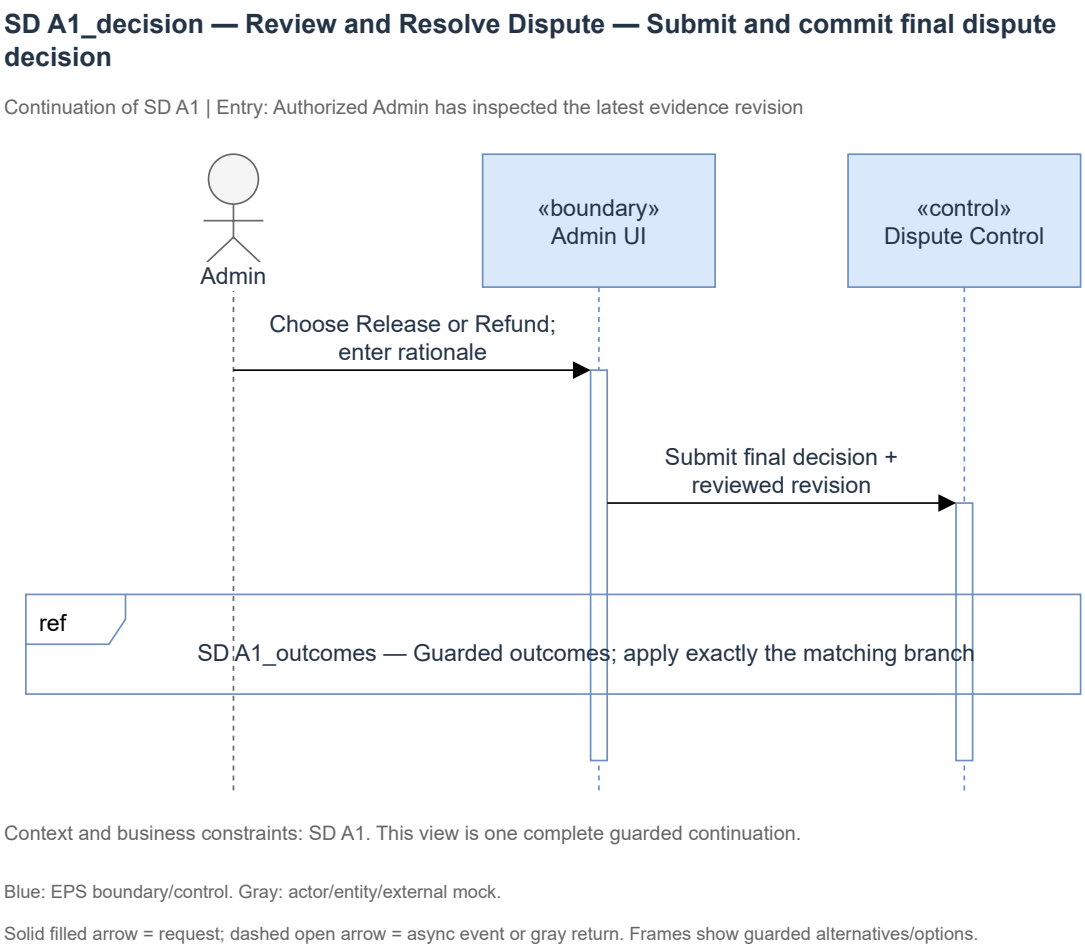


Figure 3.65: A1: Review and Resolve Dispute (continued: A1_decision)

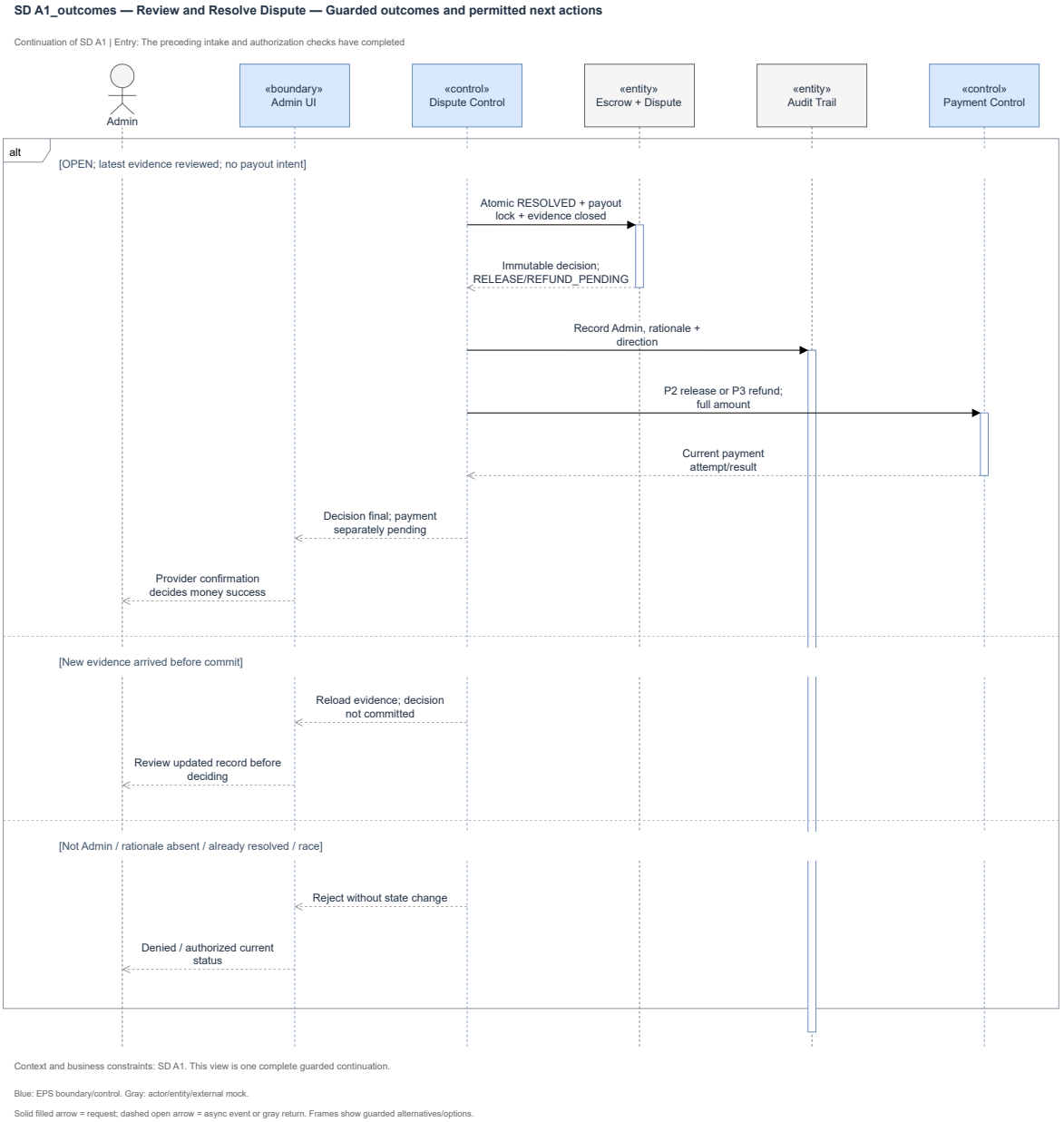


Figure 3.66: A1: Review and Resolve Dispute (continued: A1_outcomes)

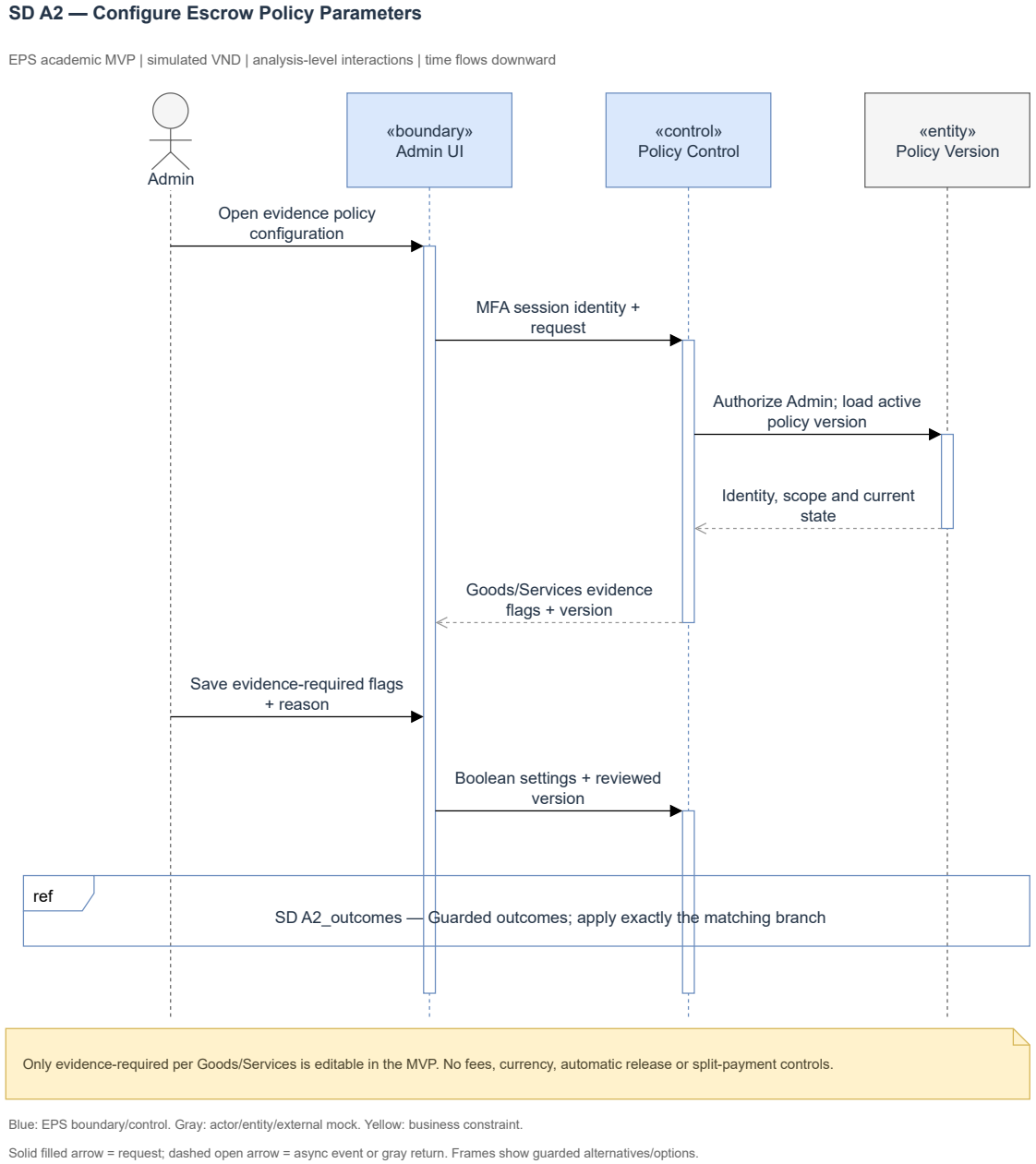


Figure 3.67: A2: Configure Escrow Policy Parameters

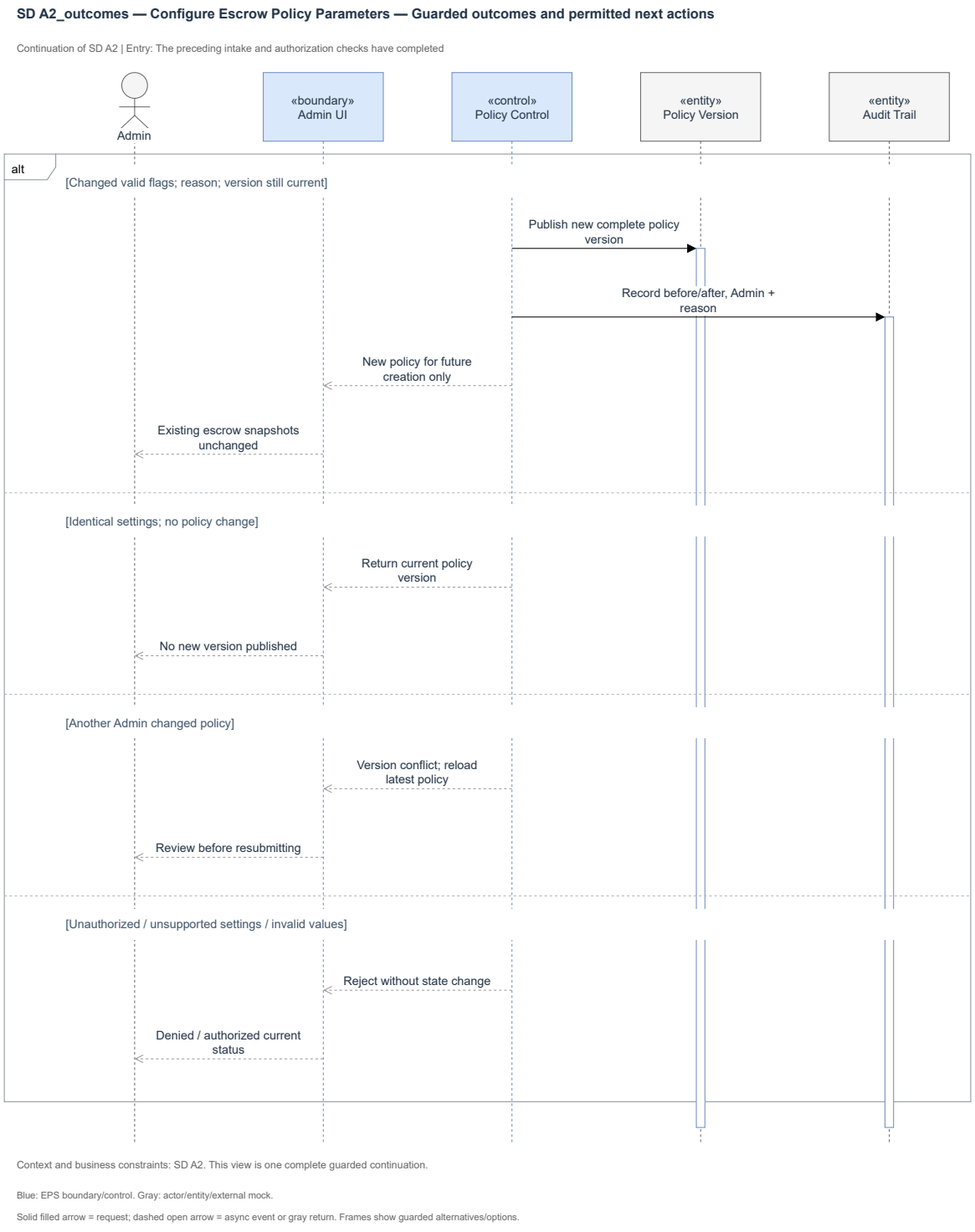


Figure 3.68: A2: Configure Escrow Policy Parameters (continued: A2_outcomes)

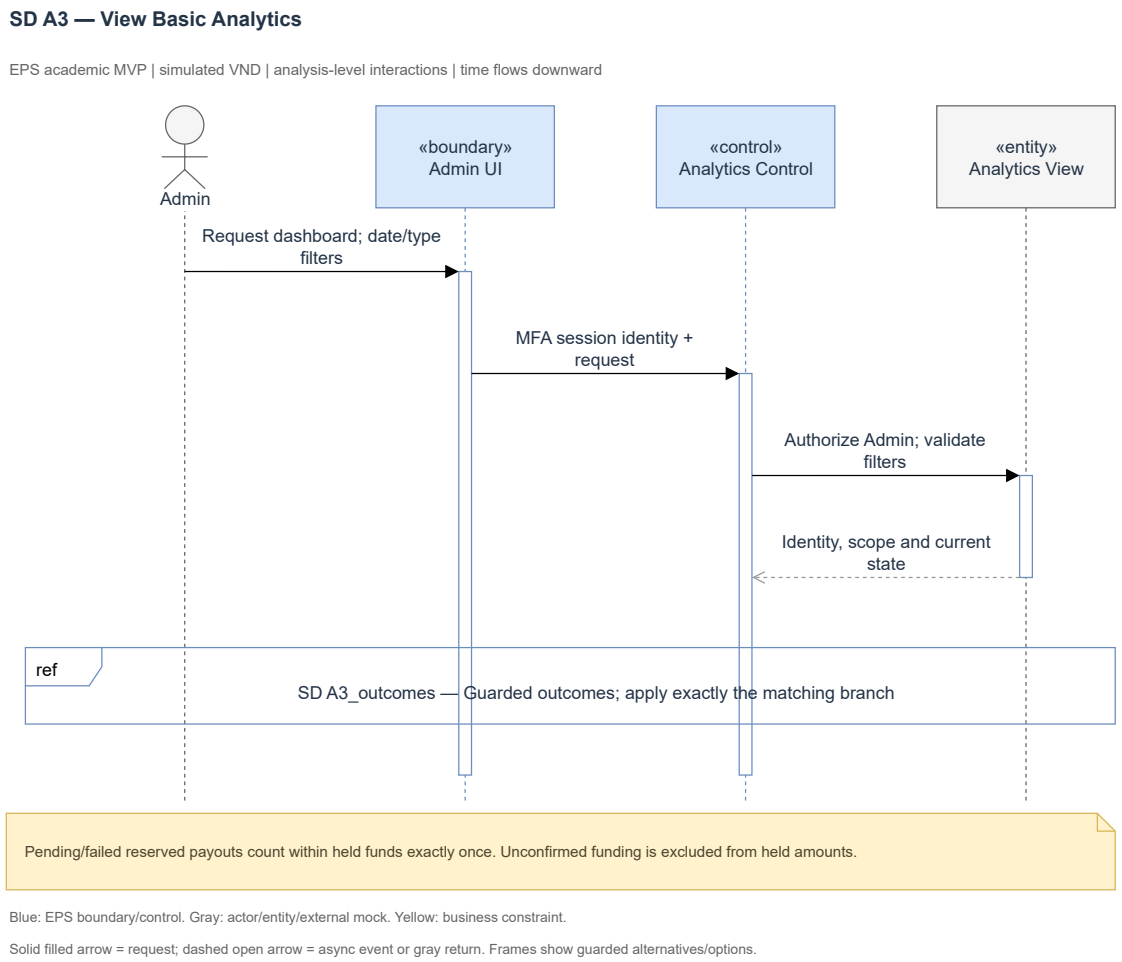


Figure 3.69: A3: View Basic Analytics

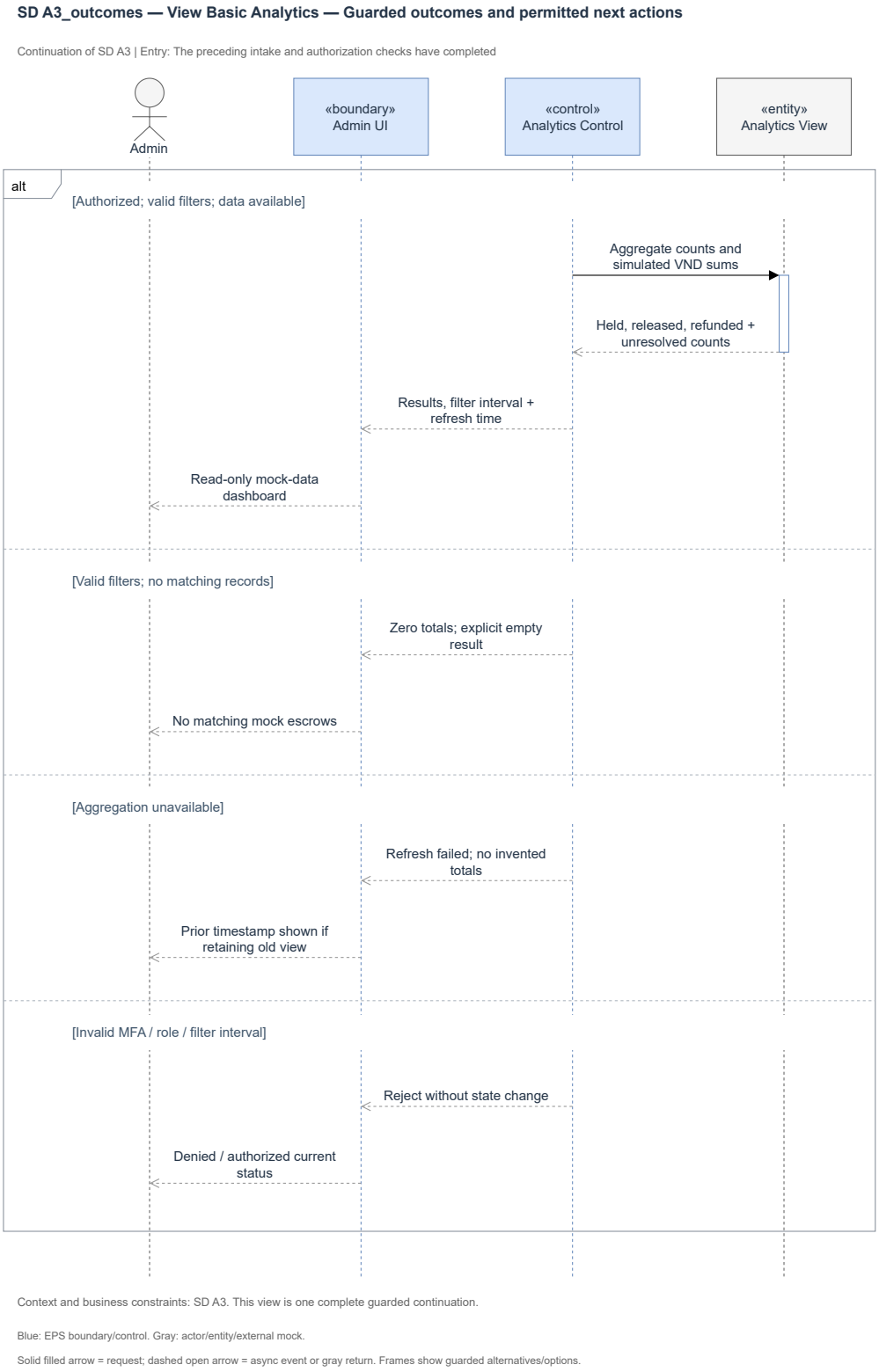


Figure 3.70: A3: View Basic Analytics (continued: A3_outcomes)

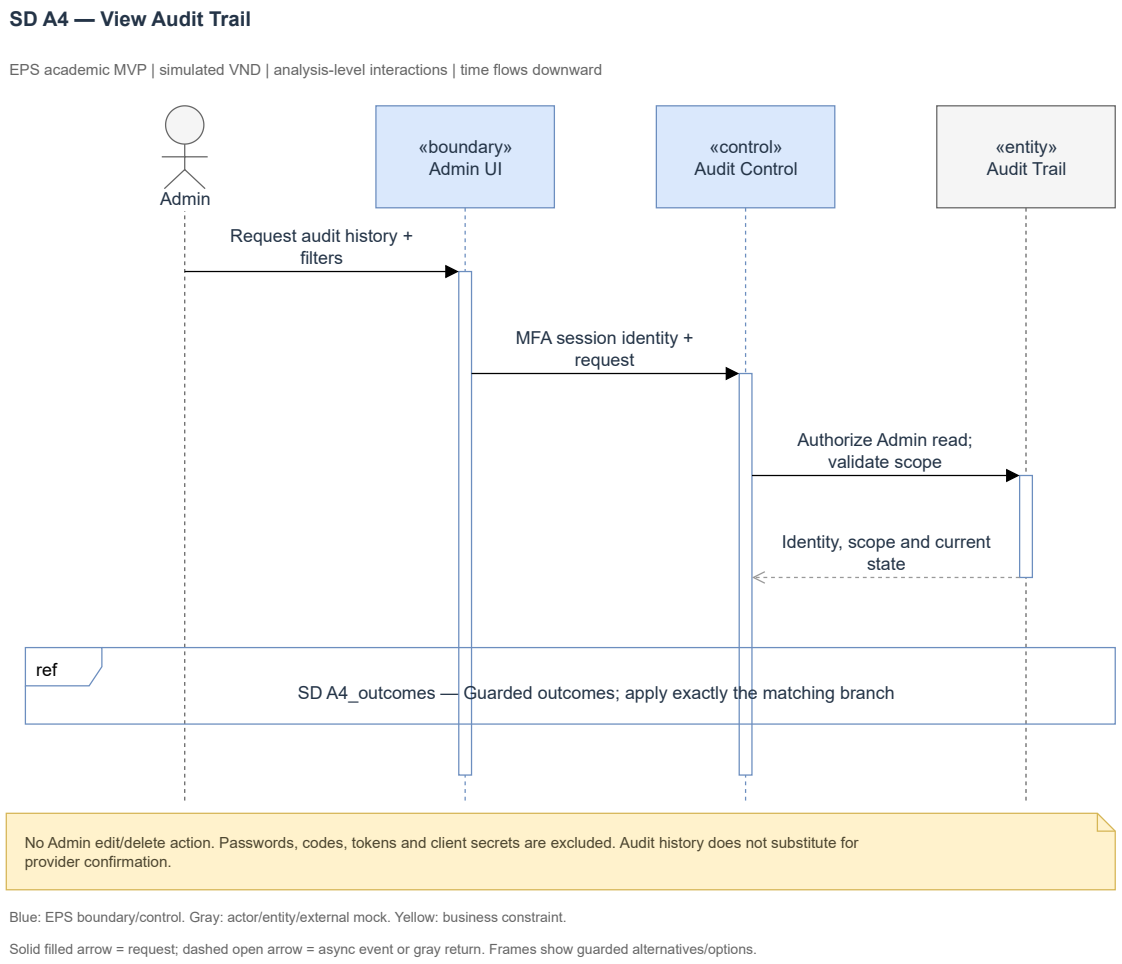


Figure 3.71: A4: View Audit Trail

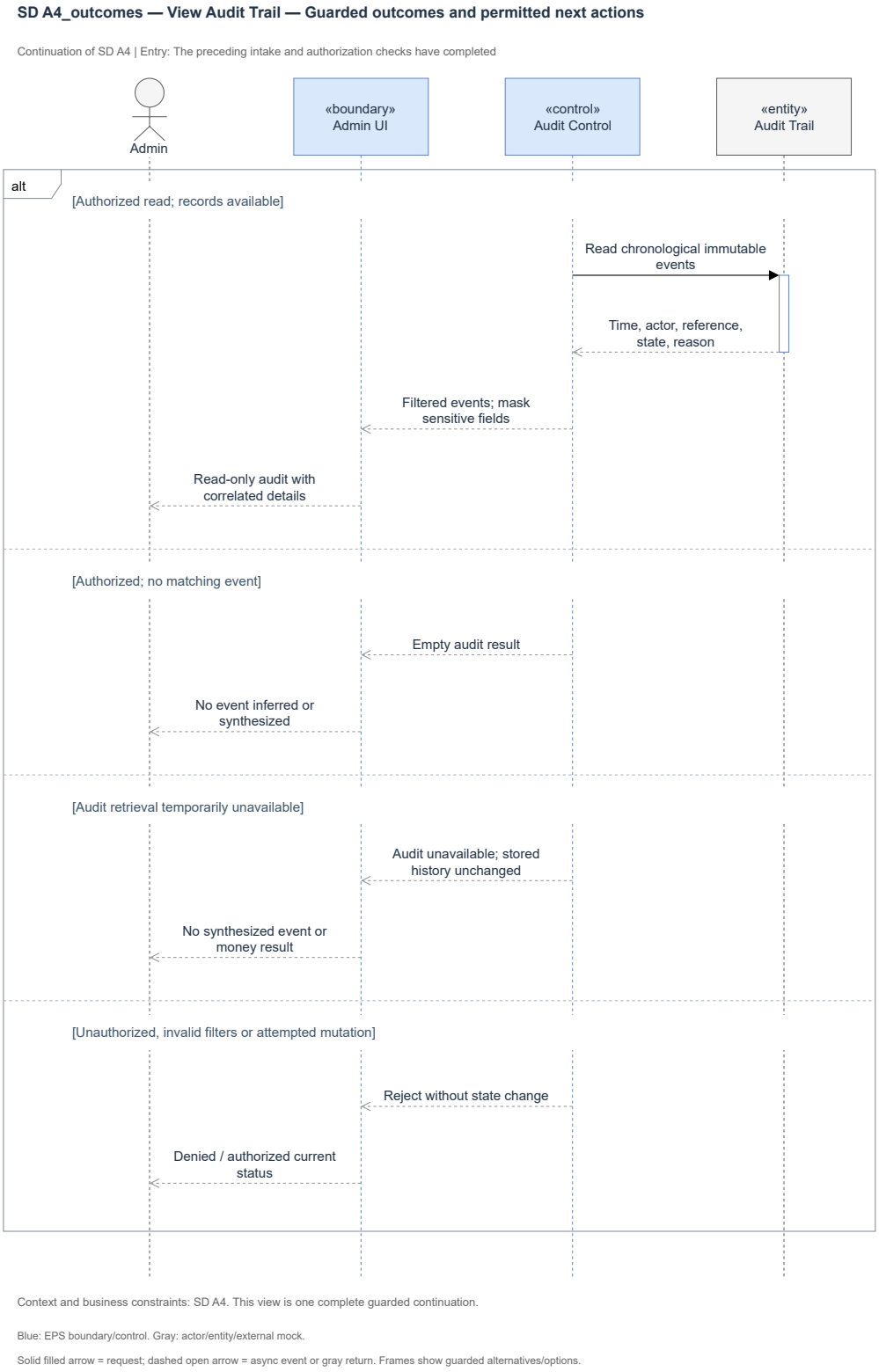


Figure 3.72: A4: View Audit Trail (continued: A4_outcomes)

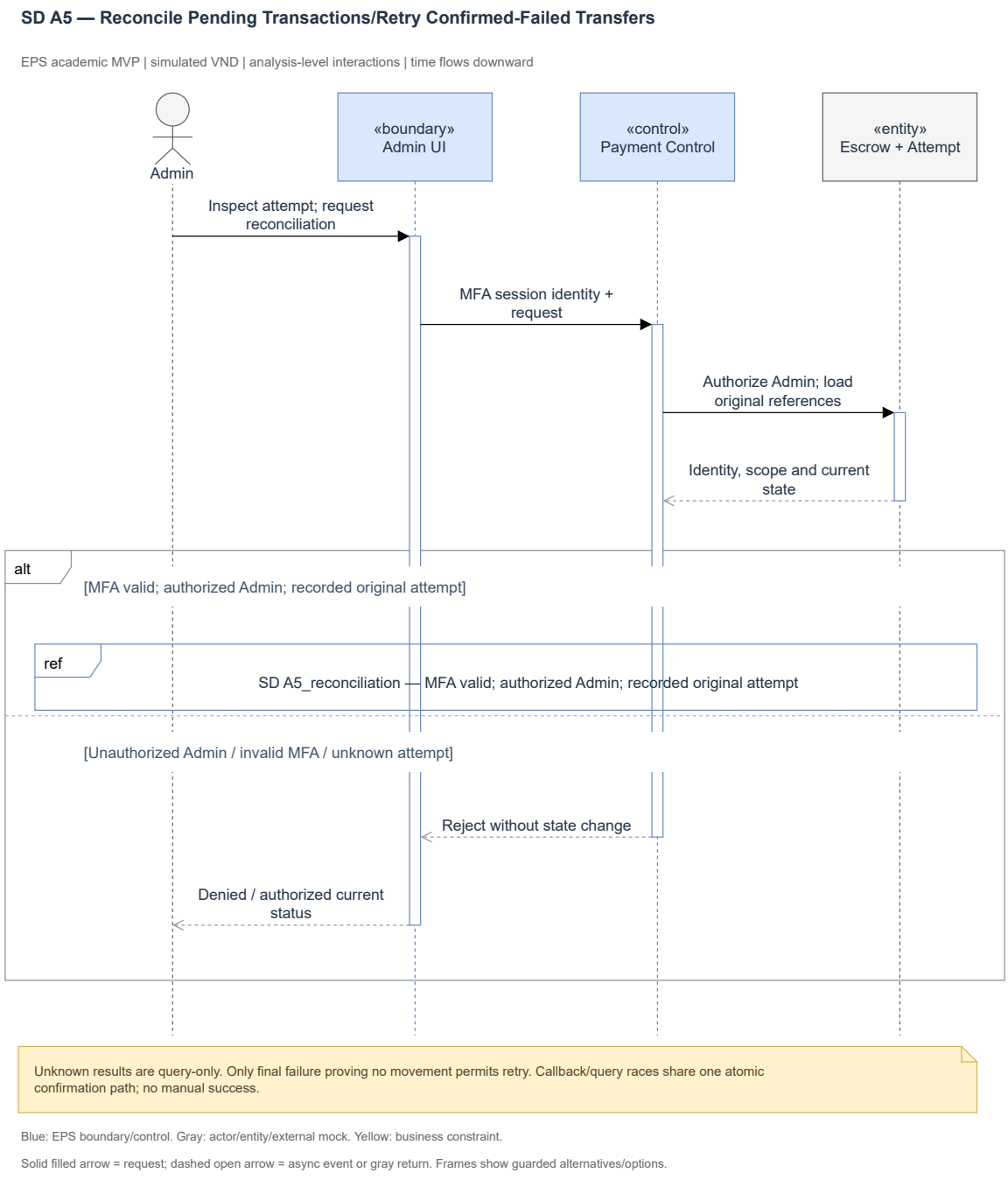


Figure 3.73: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers

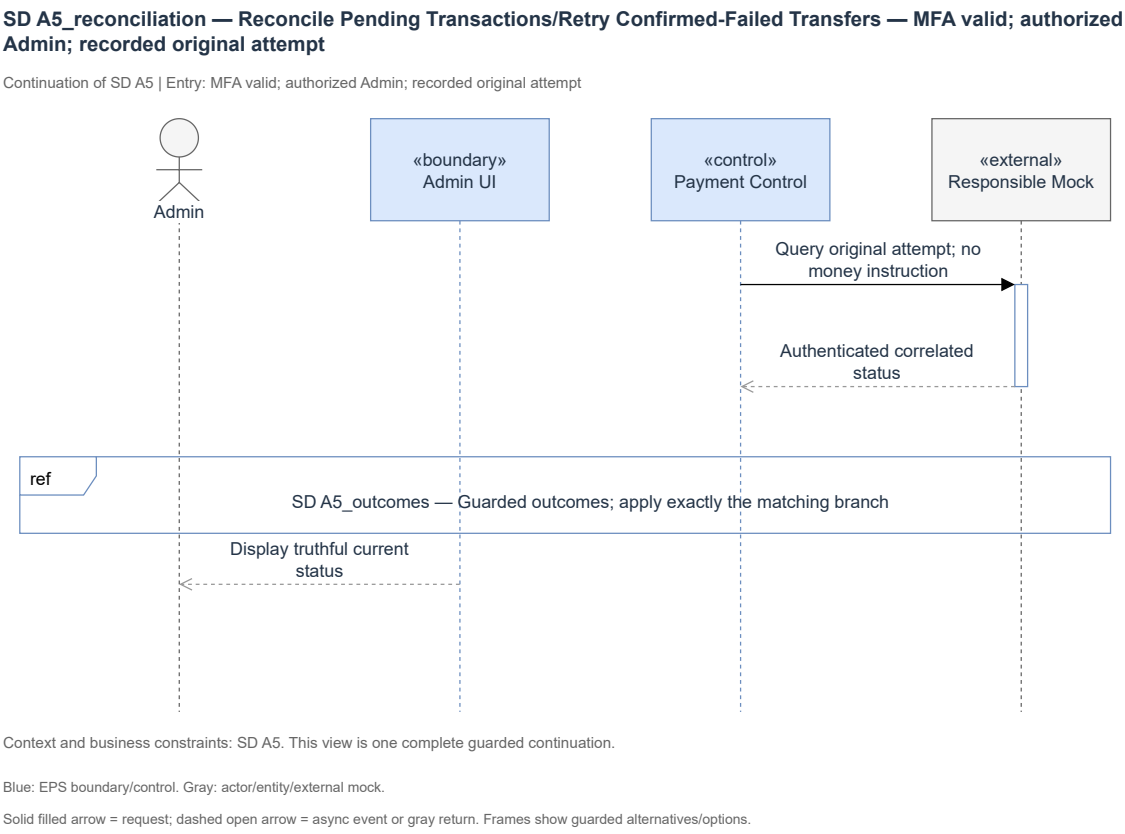


Figure 3.74: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers (continued: A5_reconciliation)

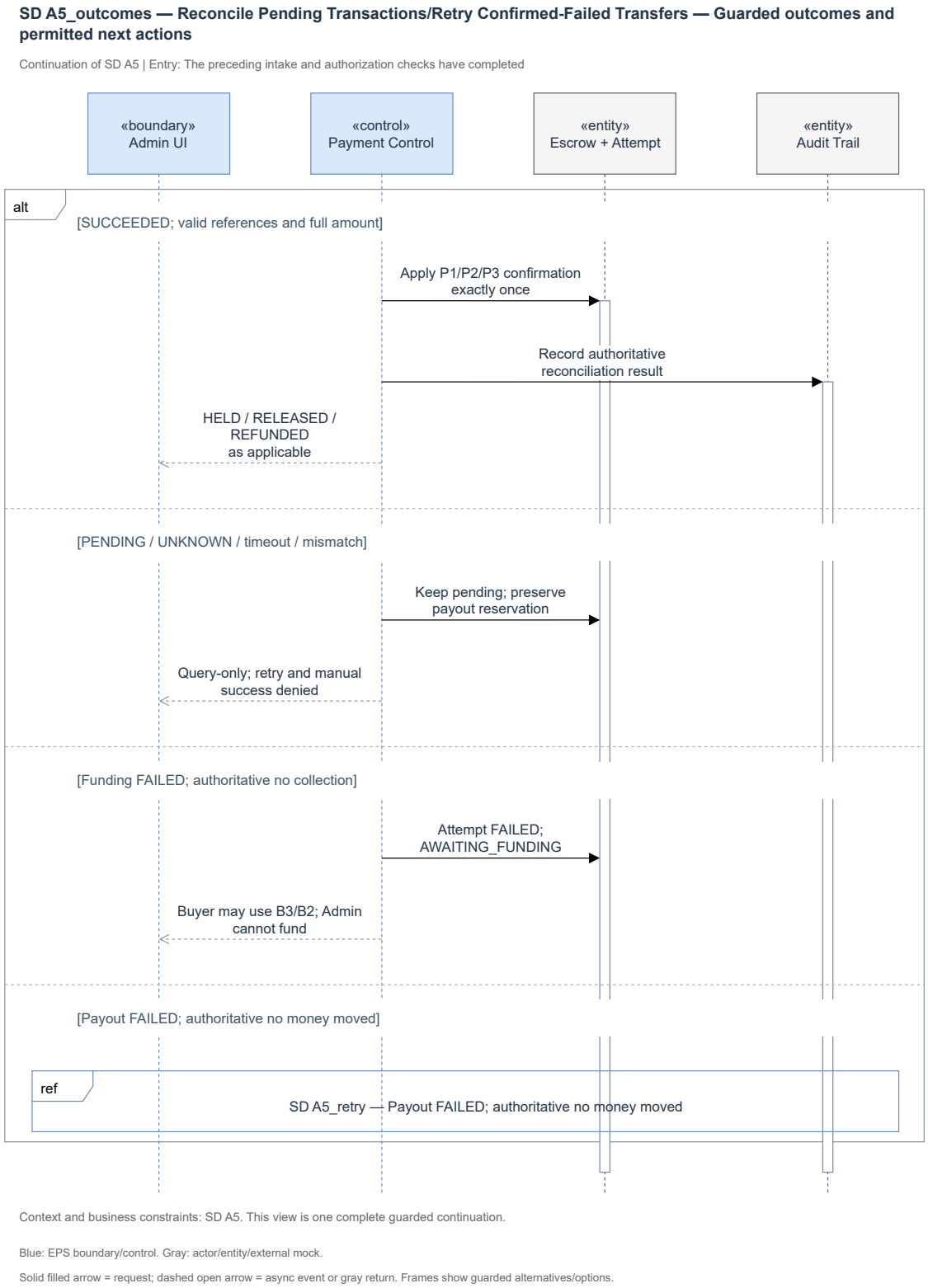


Figure 3.75: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers (continued: A5_outcomes)

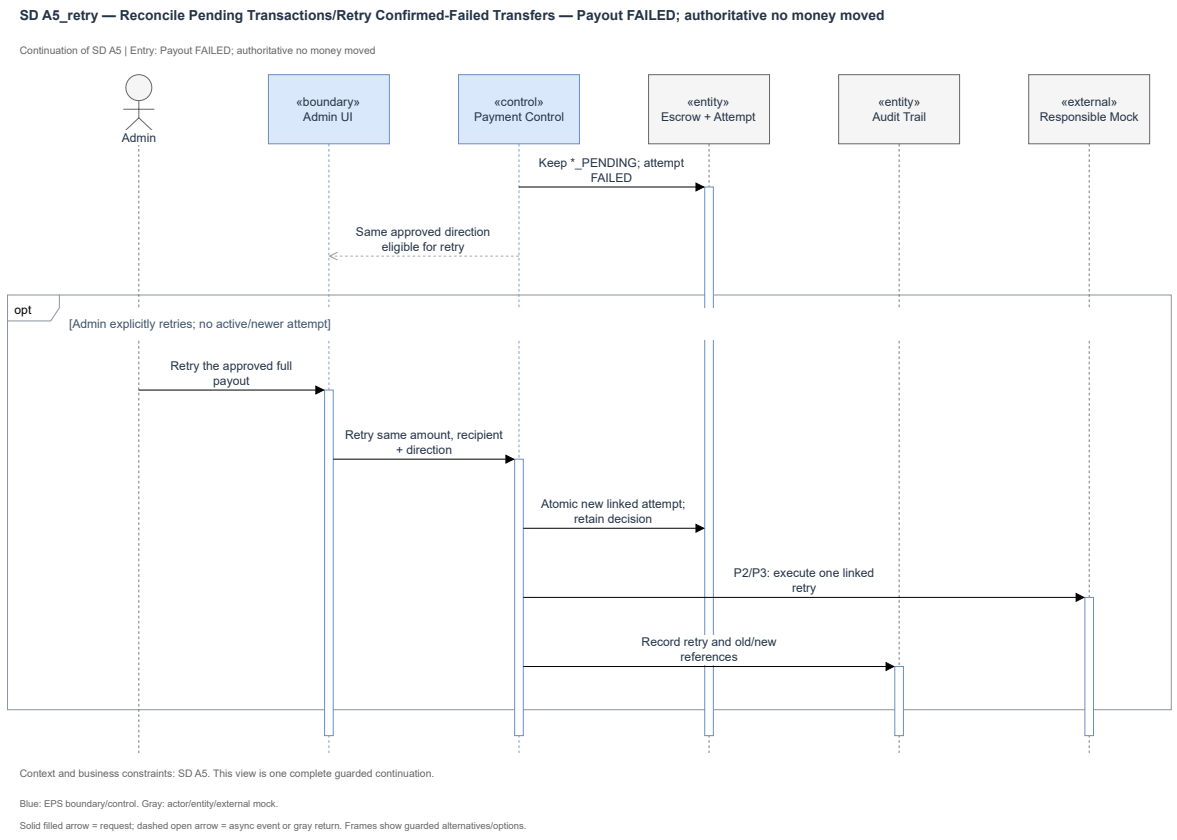


Figure 3.76: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers (continued: A5_retry)

3.4.5 Payment Integration's Use cases

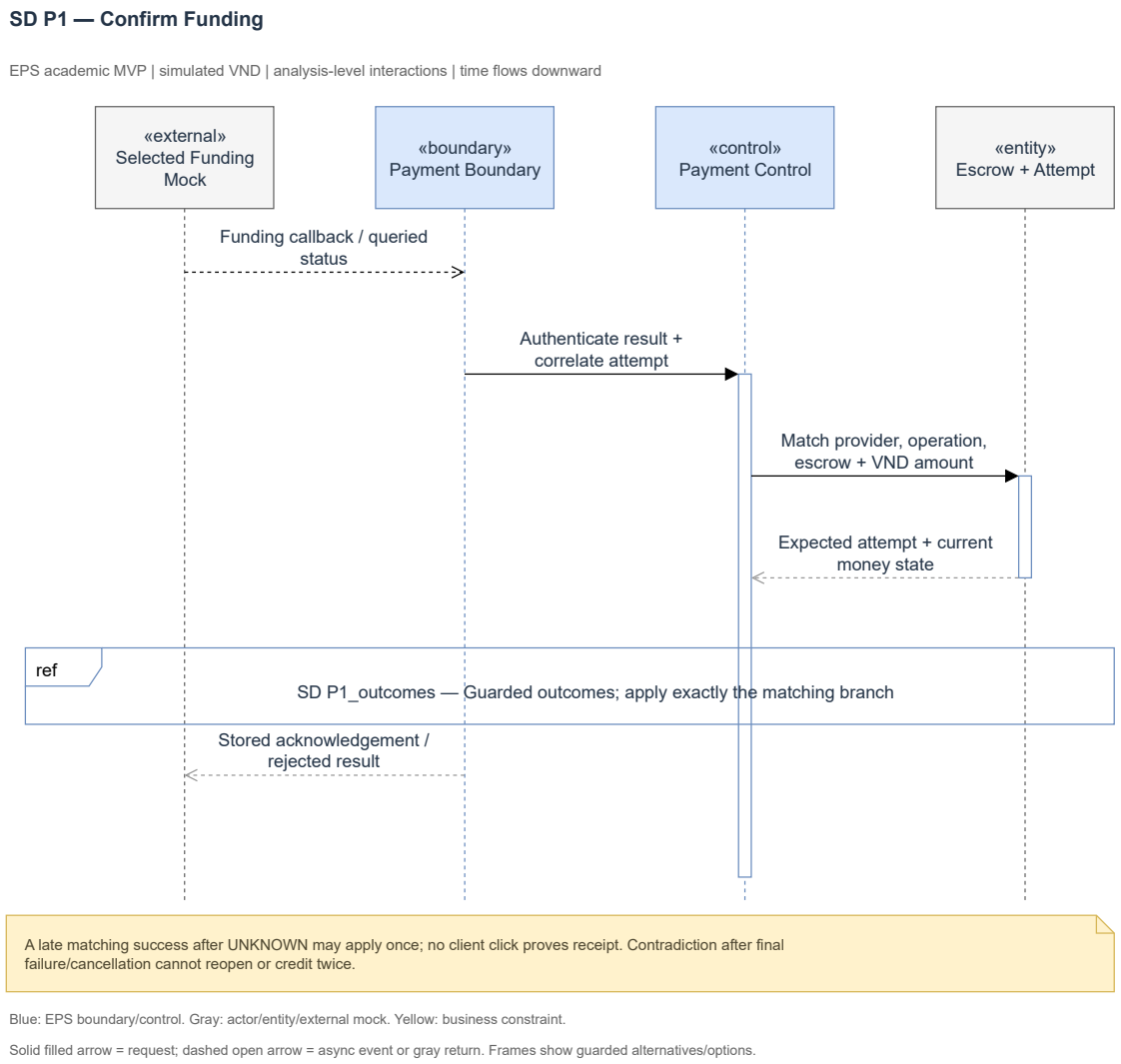


Figure 3.77: P1: Confirm Funding

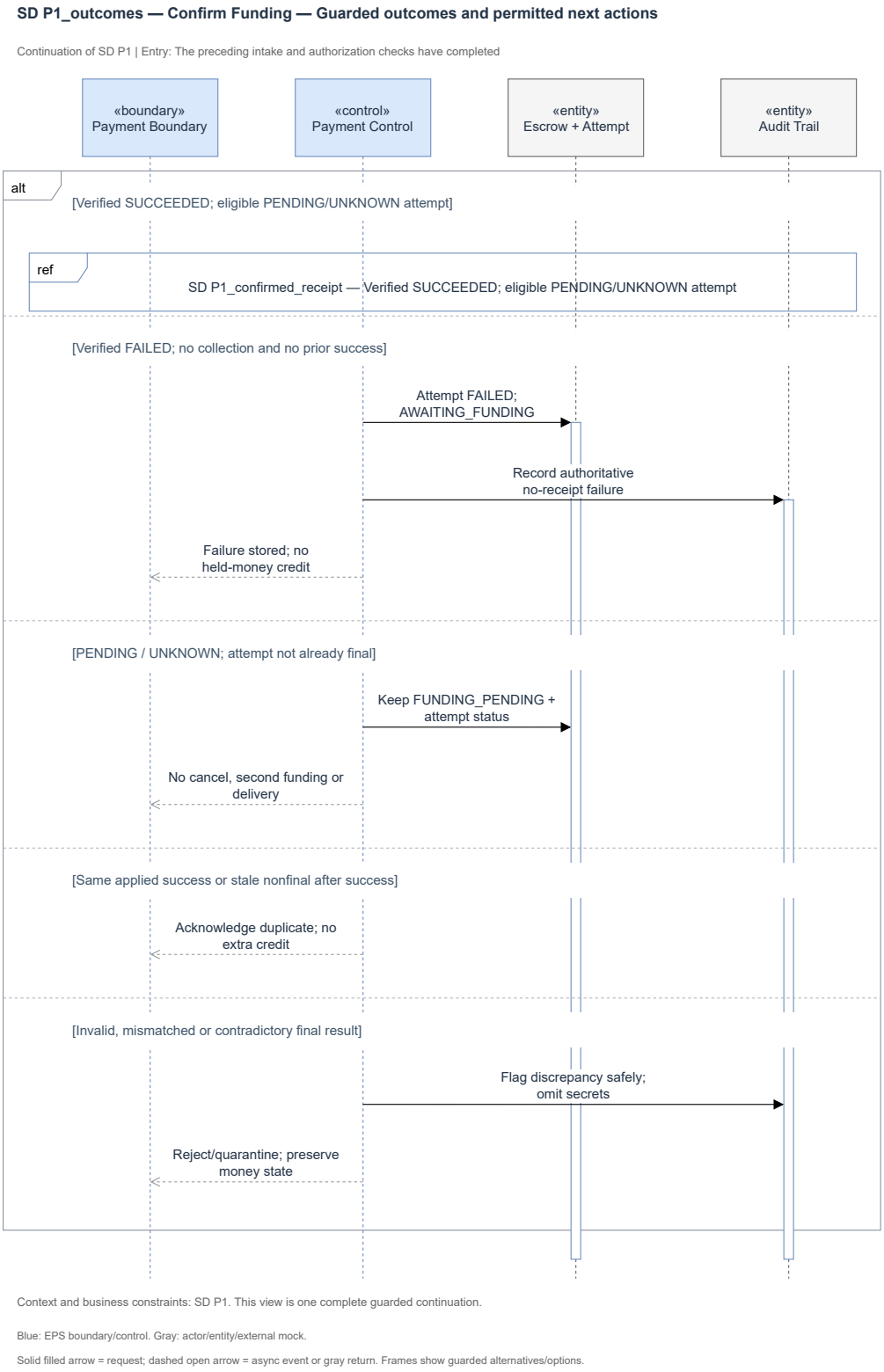


Figure 3.78: P1: Confirm Funding (continued: P1_outcomes)

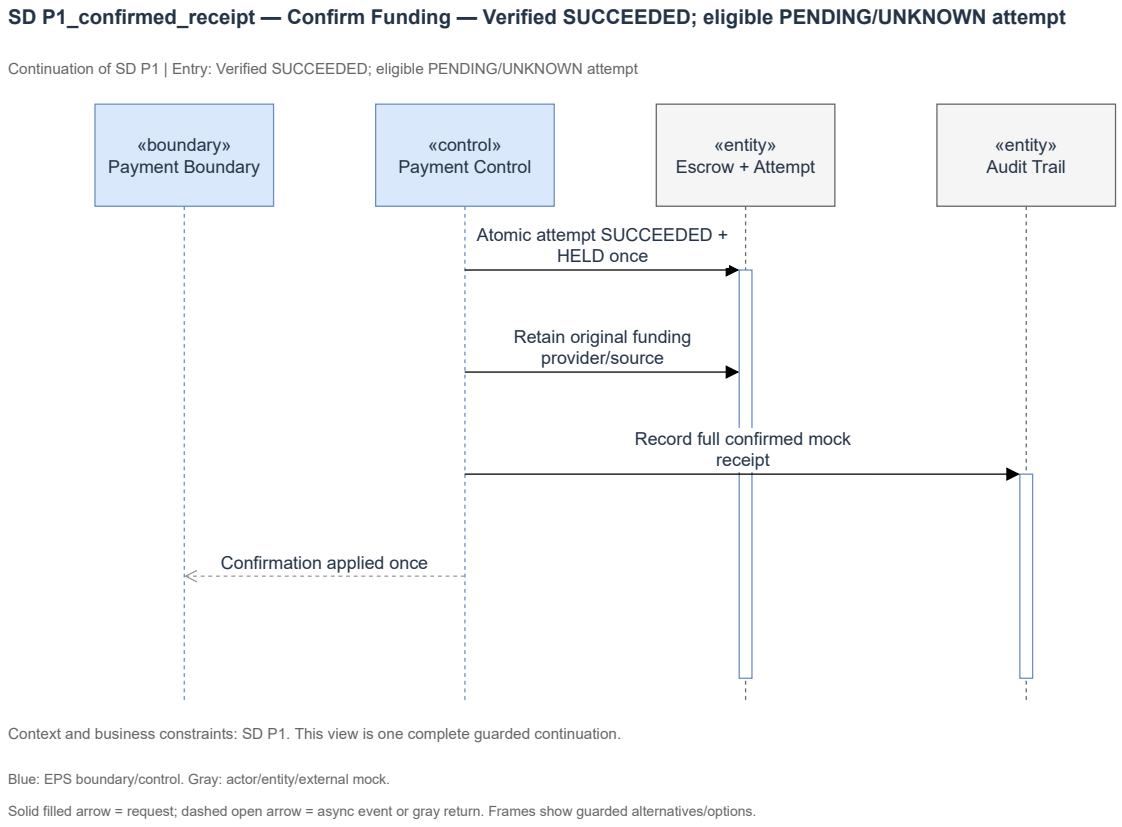


Figure 3.79: P1: Confirm Funding (continued: P1_confirmed_receipt)

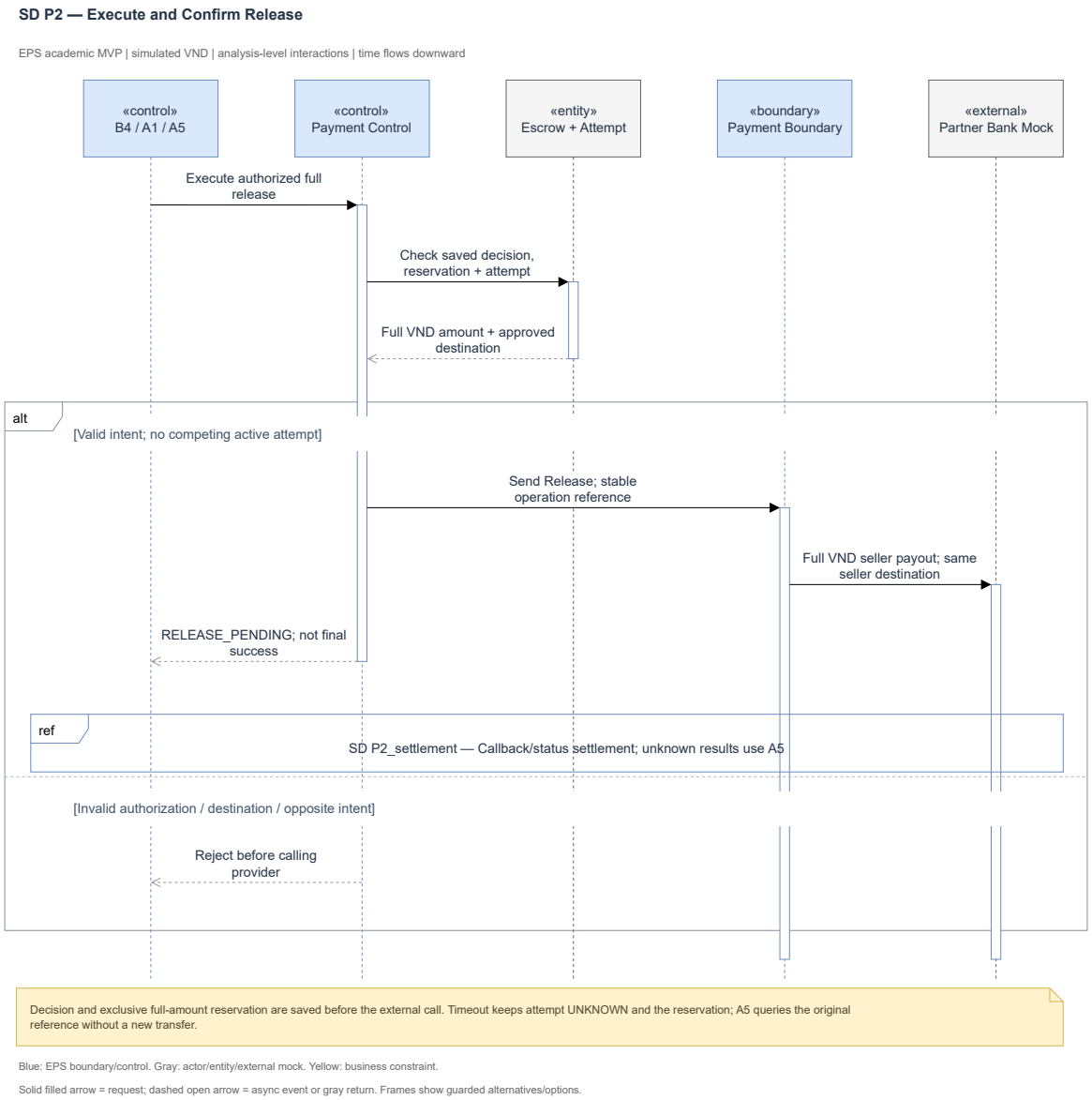


Figure 3.80: P2: Execute and Confirm Release

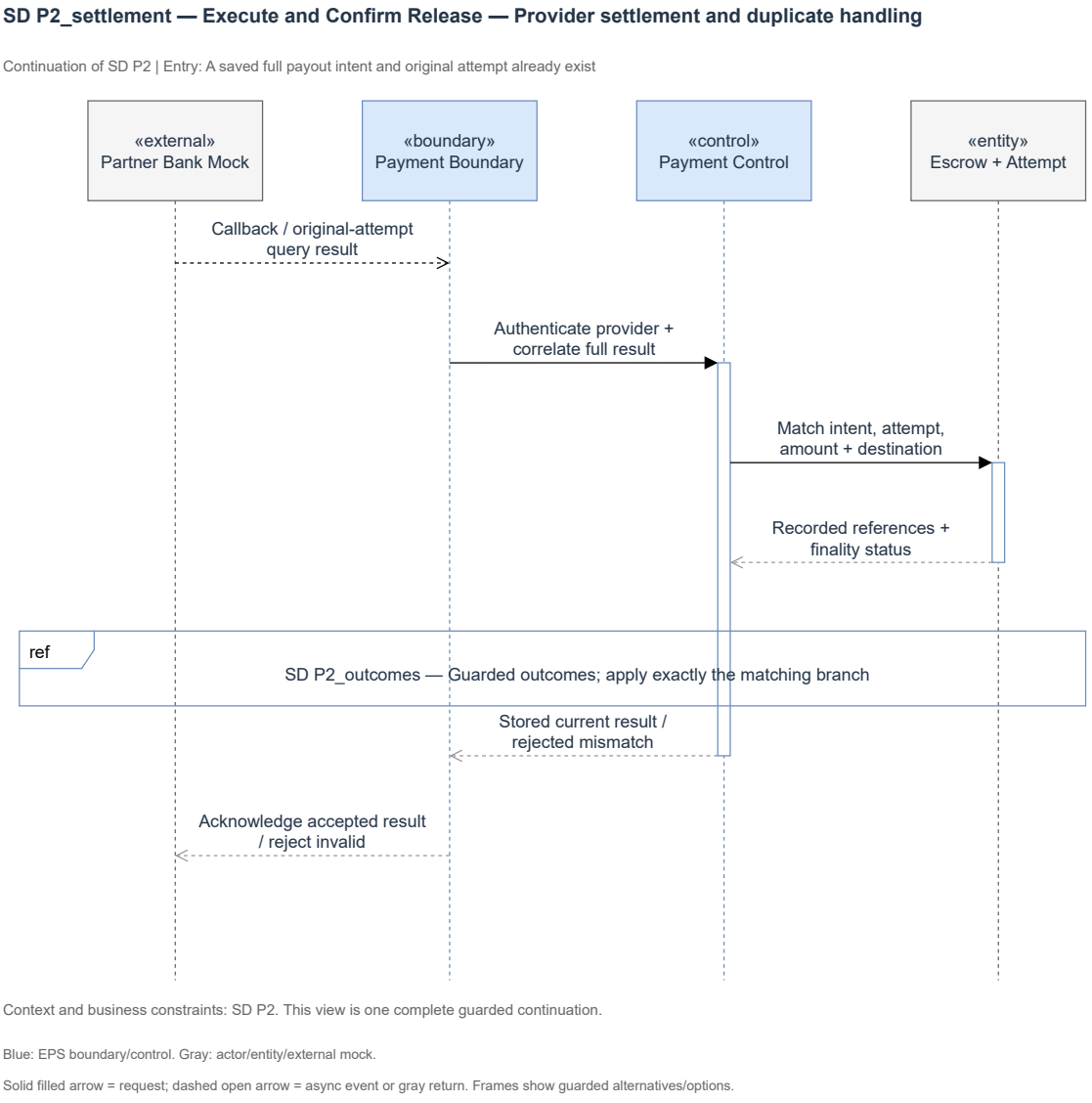


Figure 3.81: P2: Execute and Confirm Release (continued: P2_settlement)

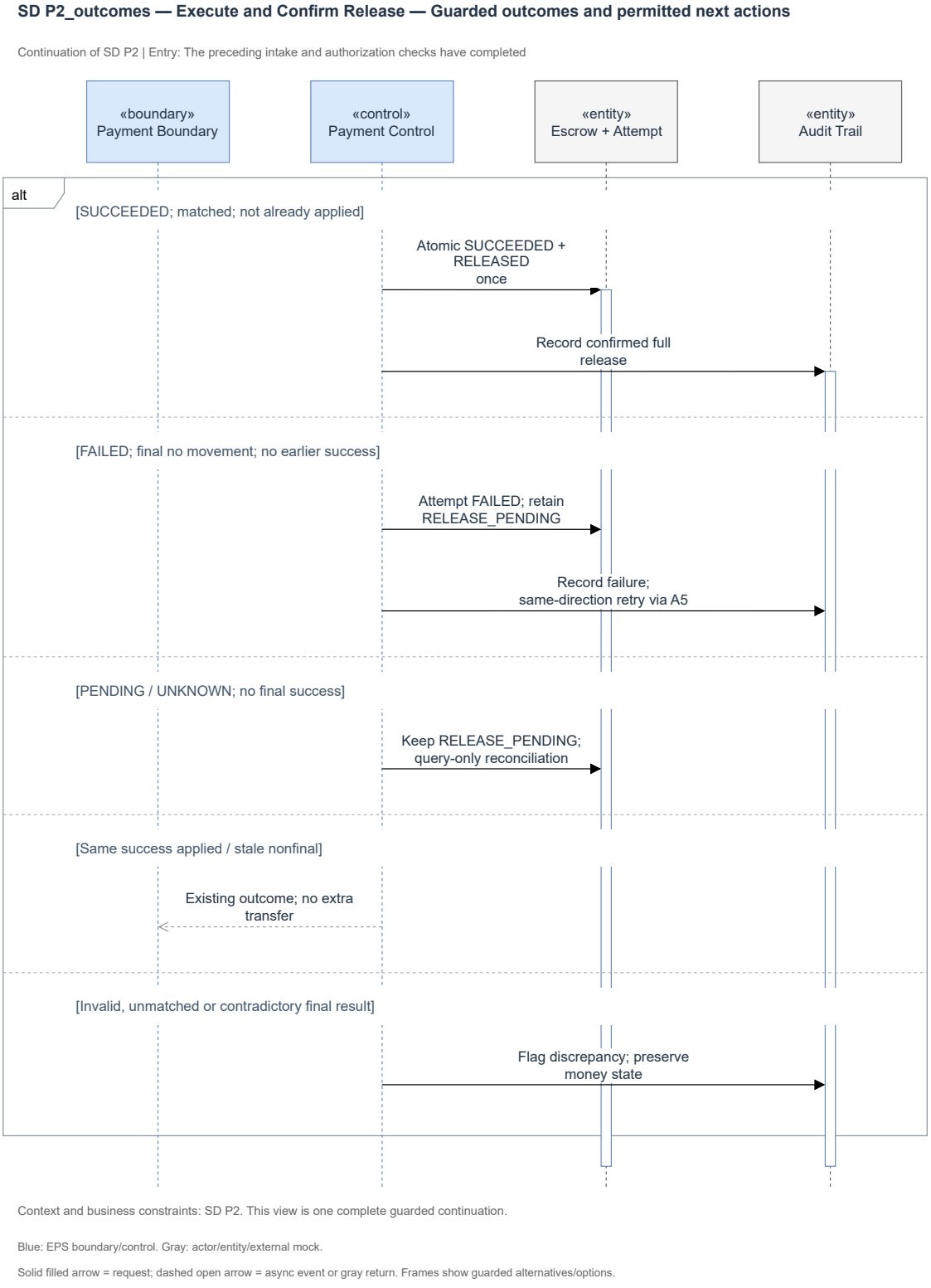


Figure 3.82: P2: Execute and Confirm Release (continued: P2_outcomes)

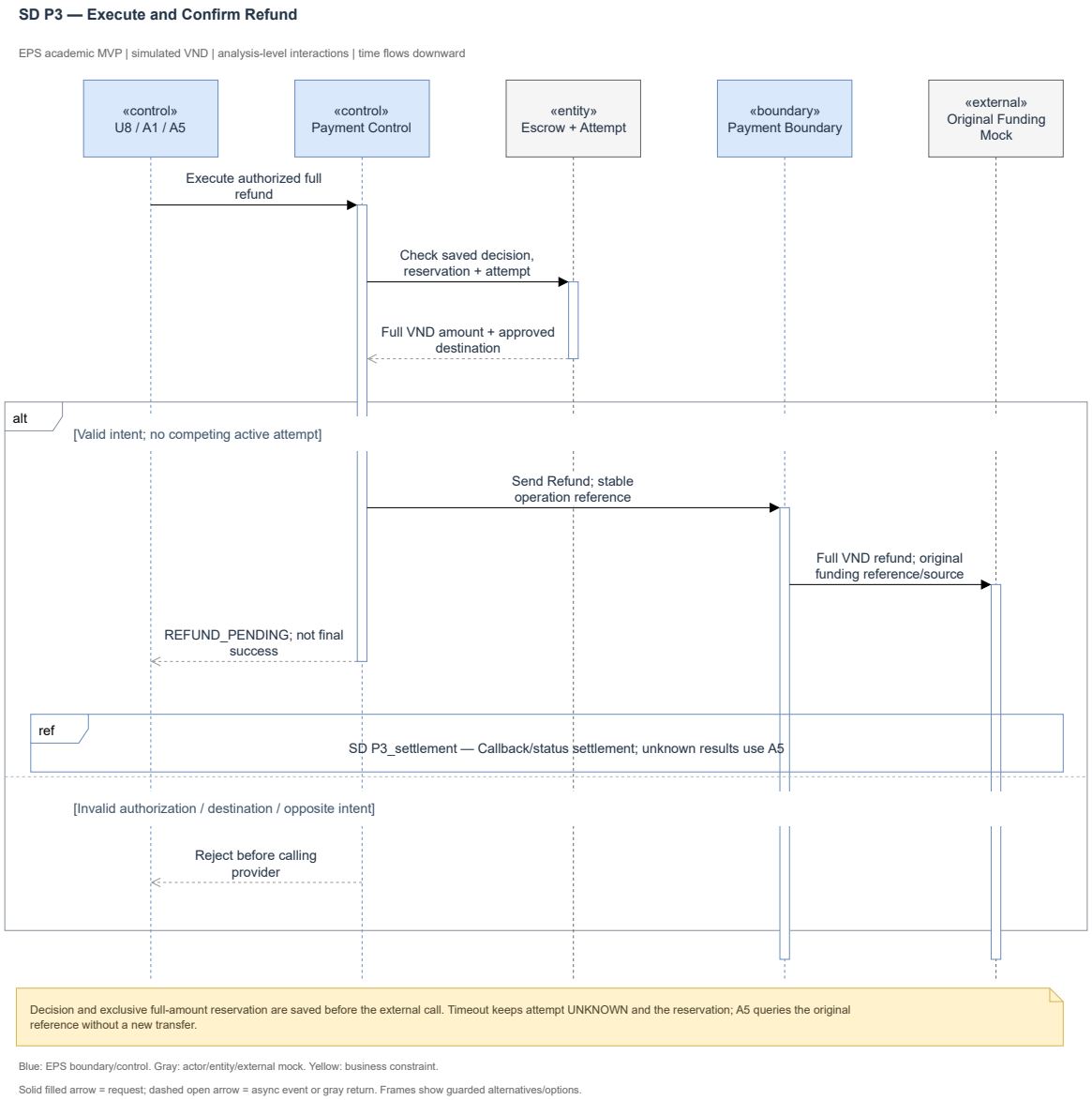


Figure 3.83: P3: Execute and Confirm Refund

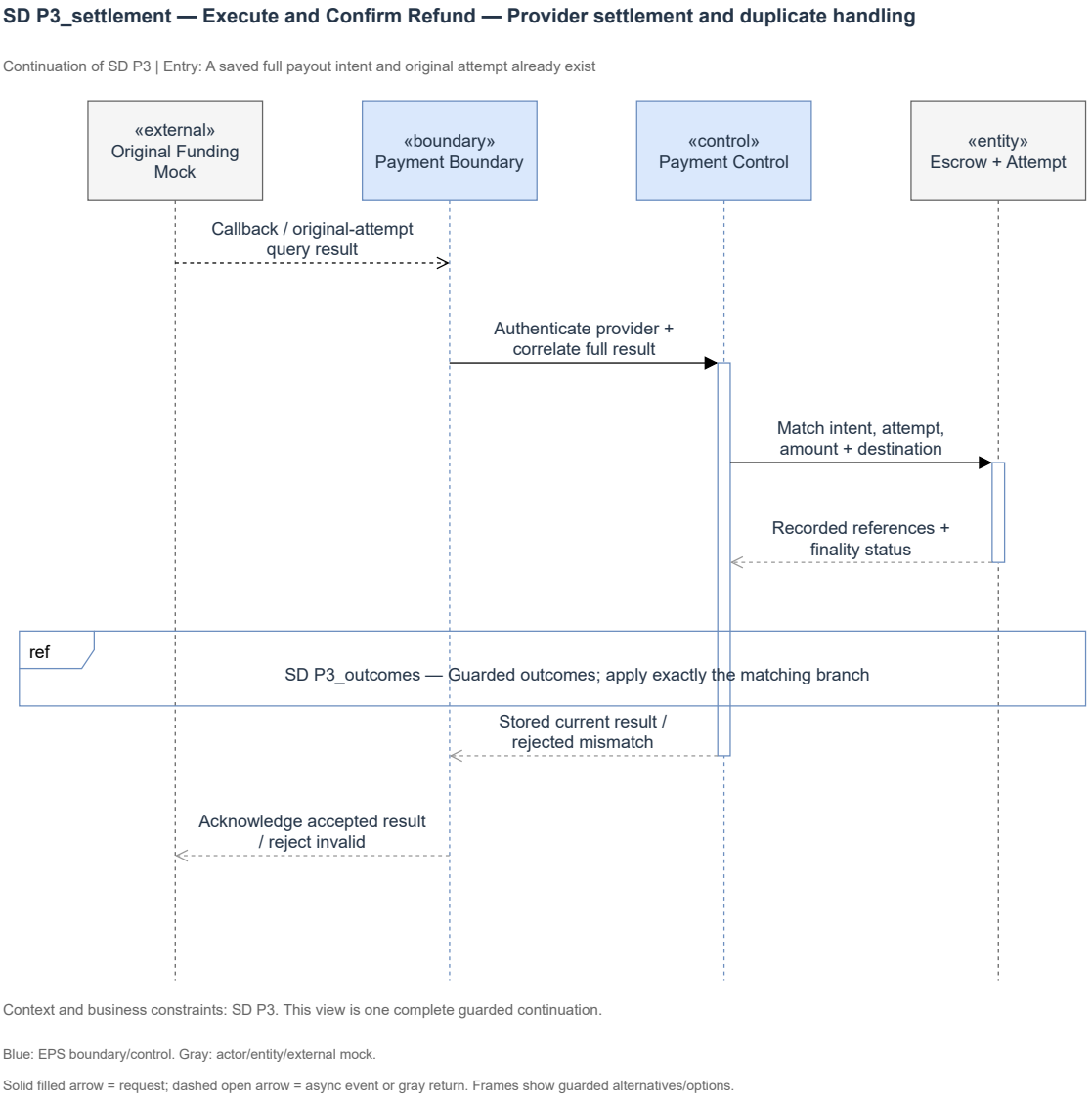


Figure 3.84: P3: Execute and Confirm Refund (continued: P3_settlement)

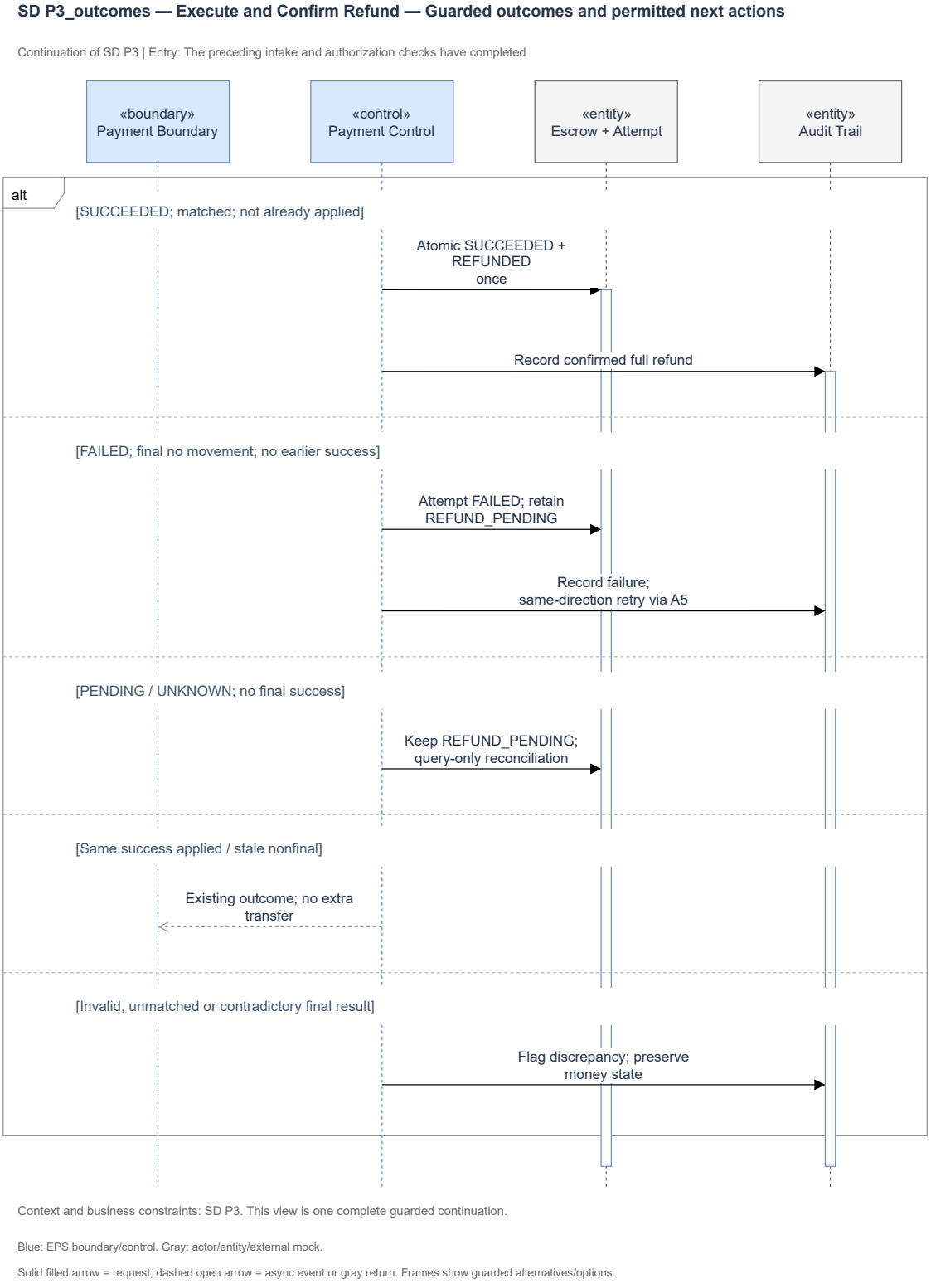


Figure 3.85: P3: Execute and Confirm Refund (continued: P3_outcomes)

3.5 Non-functional Requirements

The system must also meet the following non-functional requirements.

Performance Requirements:

- Target p95 response time is at most 2 seconds for internal create/query operations with 20 concurrent sessions and 1,000 simulated escrows, excluding user/provider waiting. This target has not been measured.
- Accepted operations must return a pending status and query reference when provider settlement is unresolved.

Operational Requirements:

- Preserve pending operations, policy snapshots and provider references after interruption so reconciliation can resume without duplicate transfers.
- Distinguish unknown outcomes from confirmed failures. Retry a payout only after confirmed failure with no money moved, preserving its direction and beneficiary.
- Keep audit records and dashboard counts consistent with recorded escrow states.

Security Requirements:

- Require MFA for human accounts and invalidate the session on logout. Authenticate Merchant clients separately.
- Check role, participant access and Merchant ownership on every protected operation.
- Use TLS and authenticate provider confirmations; reject mismatched amount, currency, destination or operation references.
- Keep the audit trail read-only and exclude passwords, MFA secrets and access tokens from logs.
- Process repeated or concurrent money decisions once, allowing only one final release or refund.

Cultural and Political Requirements:

- Use whole VND amounts and clear delivery, acceptance and evidence terms for both goods and services.
- Use synthetic data, label funds/providers as simulated and restrict personal and dispute evidence to authorised users.
- Legal claims in Planning remain unverified for real deployment; the MVP makes no claim of real-money custody or regulatory approval.